

NABONGO TEACHERS TRAINING COLLEGE

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COURSE PROGRAMME: DIPLOMA PRIMARY TEACHER EDUCATION

SUBJECT COURSE UNIT: CURRICULUM STUDIES NOTES

1.0 Background of Curriculum Studies

1.1 Concepts and Terminologies

1.1.1 Curriculum

The word “curriculum” comes from the Latin word “currere”, which means: “to run or to run a course”. It has its origin in the running/chariot tracks of Greece. It was, literally, a course. In Latin curriculum was a racing chariot; curere was to run. Curriculum is a systematic and intended packaging of competencies (i.e. knowledge, skills and attitudes that are underpinned by values) that learners should acquire through organised learning experiences both in formal and non-formal settings. By curriculum Bishop (1985) is the sum total of all the experiences a pupil undergoes. A document or plan that exists in a school or school system that defines the work of teachers, at least to the extent of identifying the content to be taught to students and the method to be used in the process. (English, 2000).

1. Experiences children have under the guidance of teachers.
2. Curriculum encompasses all learning opportunities provided by school.
3. Curriculum is a plan for all experiences which the learner encounters in school.

The word Curriculum is taken from the Latin word which mean race or race-course.

Oxford Dictionary defined it as a course of study “Curriculum is a disciplining technology that directs how the individual is to act, feel, talk, and ‘see’ the world and the ‘self.’ As such, curriculum is a form of social regulation.”

(a) The Traditional concept;

“It is a body of subjects or subject matter prepared by the teachers for the students to learn”

(b) Static view point

Initially the curriculum was consider the course of study which gives importance only to text book teaching in educational institutions, the objective was to help students to memorize the content of the subject.

(c) Dynamic view point

After 19th century the old concept of the curriculum was replaced, it include now all the activities which are practiced inside or outside school for the well being of the students. It is the totality of learning; classroom Experiences, lab work, play ground and interaction with the community

A curriculum refers to a defined and prescribed course of studies, which students must fulfill in order to pass a certain level of education.

Some influential definitions combining various elements to describe curriculum are as follows

- John Kerr:- According to John Kerr, a curriculum is planned and guided by the school, whether it is carried on in groups, individually inside or outside the school.
- The curriculum is a total learning experience provided by the school. It includes the content of courses (the syllabus), the method employed (strategies) and other aspects like norms and values, which relate to the way schools are organized.
- Thus a curriculum is neither development nor a sequence of experiences. It is a plan for facilitating learning for students.

- This plan starts with where the child is. It enumerates all the aspects and dimensions of learning that are considered necessary. It gives a reason why such learning is considered necessary and what educational aims it would serve.

In a nutshell,

- the curriculum is a means followed by the teachers and students for achieving the set goals and the aims or objectives of education being provided in the school.
- Curriculum, in every sense, is supposed to be used for all experiences. These may be curricular or co-curricular, imparted by the school for the realization of the stipulated aims and objectives of the school education.

1.1.2 Modern Curriculum

The modern curriculum consists of all experiences for learning which are planned and organized by the school. It is composed of the actual experiences and activities of learners inside or outside the classroom under the guidance of the teacher and for which the school accepts responsibility.

Some authors define curriculum as “The total effort of the school to bring about desired out-comes in school and out-of-school situations”

Concepts of Curriculum

The concept of Curriculum is dynamic as the changes that occur in society. In its narrow sense, curriculum is viewed merely as a listing of subject to be taught in school. In a broader sense, it refers to the total learning experiences of individuals not only in schools but in society as well. (Purita P. Bilbo, Ed.D)

A general overall plan of the content or specific materials of instruction that the college should offer the student by way of qualifying him for graduation or certification or for entrance into a professional or vocational field.

A body of prescribed educative experiences under the supervision of an educational institute, designed to provide an individual with the best possible training and experience to fit him for the society of which he is a part or to qualify him for a trade or profession.

(a) Common Concepts of Curriculum

Syllabus Content Outline

Standards Textbooks

Course of study

Planned Experiences

(b) Scope and importance of Curriculum

1. the coverage of the area or subject matter that something deals with or to which it is relevant.
2. The range of one's perceptions, thoughts, or actions “He hasn't the scope to understand this concept.”?
3. The area covered by a given activity or subject

So here the word scope means the range of curriculum and how much area it covers.

Here is the various aspect of the curriculum process;

1. Substantive (real or actual)
2. Technical
3. Social

The important aspect of the substantive is related to goals, aims and objectives and the technical aspect refers to the skill development , methods and subject matter.

The social side is the need and involvement of the society in the process.

(C) Functions of Curriculum

The main objective of the curriculum is the development of the child physical, intellectual, emotional and personality development.

1. Development of individual

Each student learns with his own pace and each individual has his own abilities and potentialities. Some students learn with one method while others learn with another. So it is the function of curriculum to help the student by providing such experience which encounters the need of all students.

Curriculum plays a very important role in the mental, social, emotional, moral and physical development of the student.

2. Produce good Citizens

Good curriculum gives good citizens to a country, it plays a very important role in the character building and providing knowledge about what is wrong and what is right.

3. Patterns of Curriculum

What is the pattern of the curriculum, it is the most important aspect of the curriculum. It is the means and plan of implementation curriculum;

- ☐ Subject based curriculum
- ☐ Activity based curriculum
- ☐ Learner centered curriculum
- ☐ Teacher centered curriculum

4. Develop Basic skills

Develop the basic skills of the students by implementing a good curriculum;

Reading, writing, understanding and speaking

5. Culture preservation and transmission

The main aim of education is to preserve the culture, values and transfer it to the next generation. The Curriculum helps in the process. It inculcates Islamic values, improves mental and physical health of the student, and prepares people for the new era.

(d) Importance of Curriculum

1. The success of a nation is hidden in the formulation of effective curriculum which helps to develop the inner potentialities of an individual.

2. Curriculum provides a plan and structure to the education process inside and outside of school.

3. Curriculum is the reflection of the society, what society needs and wants, how they want to mold the individual capacities.

4. Curriculum guides the society to improvement and brings enrichment.

5. Curriculum gives concrete shape to educational policies of a nation.

6. Curriculum helps in the preservation, improvement and transfer of culture to the next generation.

7. Curriculum helps in producing good and broad minded citizens.

8. It helps in the development of responsible citizens.

9. Curriculum fulfills the future demands of the country.

10. Curriculum gives concrete shape to national requirements.

11. Curriculum helps in the harmonious development of the personality of the people of the society.

12. Curriculum helps in the development of basic skills in the people of the society and prepares them for life.

13. Curriculum gives knowledge.

14. Curriculum helps in the formulation of good attitudes and guides them in the development of desired attitudes in the young ones.

(e) Characteristics of Curriculum

- ☐ Flexibility
- ☐ Integrated & continuous
- ☐ Balance
- ☐ Scope
- ☐ Curriculum consists of religious and moral values.

- ☐ Considering individual conflicts in curriculum
- ☐ Life centered Curriculum
- ☐ Flexibility

The Curriculum has administrative flexibility.

- A good curriculum must be ready to incorporate new changes whenever necessary.
- The curriculum is open to revision and development, to meet the demands of globalization and the digital age.

Integrated & continuous

There should be continuity and integration in curriculum development process for every class.

The curriculum should be balanced in all aspects.

(f) Scope

Cover all aspects;

- ☐ vocational,
- ☐ educational area,
- ☐ social,
- ☐ technical and moral
- ☐ curriculum consist religious and moral values.
- ☐ Cover religious and moral values of a nation
- ☐ Considering individual conflicts in curriculum.
- ☐ Curriculum should have the ability to engage /support all kinds of students. It should be based on individual differences.
- ☐ Life centered Curriculum

1.1.3 School curriculum

A school curriculum is all planned and unplanned teaching and learning experiences to enable the learner acquire and develop the desired knowledge, skills, attitudes and values for personal and society's benefits.

The school curriculum is made up of :

- Objectives
- Learning experience /activities
- Content/subject matter
- Evaluation/assessment procedure

1.1.4 Competence

This is the ability to do something successfully and efficiently.

1.1.5 Core competence

It is the ability to apply appropriate knowledge and skills to successfully perform a function. Is an organizations defining strength, providing the foundation from which education section (nation) will grow, seize upon new opportunities and deliver value to customers. Competency conceptualised as 'the ability to apply appropriate knowledge and skills to successfully perform a function'. They will mainstreamed in every strand and lesson.

CBCs core competencies include:

- Communication and collaboration
- Critical thinking and problem solving ☐ ☐ Imagination and creativity
- Citizenship
- Digital literacy
- Learning to learn
- Self-efficacy

1.1.6 Facilitation

Refers to providing necessary resources, information and support in order for learners to complete a task, rather than teaching.

1.2 TYPES OF CURRICULUM

1.2.1 Curriculum Development

Curriculum development is the process of creating curriculum materials for use by educators and children that are product of curriculum planning. Curriculum development describes all the ways in which a training or teaching organization plans and guides learning. This learning, can take place in groups or with individual learners. It can take place inside a classroom or outside a classroom. It can take place in an institutional setting like a school, college or training centre, or in a village or a field. It is central to the teaching and learning process. (Rogers and Taylor) 1988).

1.2.2 Types of Curriculum

1. Overt

Thus, the overt curriculum is usually confined to those written understandings and directions formally designated and reviewed by administrators, curriculum directors and teachers, often collectively.

2. Explicit

3. written

Is simply that which is written as part of formal instruction of schooling experiences. It may refer to a curriculum document, texts, films, and supportive teaching materials that are overtly chosen to support the intentional instructional agenda of a school.

4. Hidden/covert

That which is implied by the very structure and nature of schools, much of what revolves around daily or established routines. Refers to the unwritten, unofficial, and often unintended lessons, values, and perspectives that students learn in school.

Longstreet and Shane (1993) offer a commonly accepted definition for this term – the “hidden curriculum,” which refers to the kinds of learnings children derive from the very nature and organizational design of the public school, as well as from the behaviors and attitudes of teachers and administrators

It consists of all that is learned during school activities, out of the designated official curriculum.

These includes:

- How learners should sit
- How to greet the teacher
- Role of the teacher and prefect
- How to relate with the teachers, fellow students and other people at school.
- Values such as hard work, respect, obedience, co-operation, value for others, empathy etc.

5. Null curriculum

That which we do not teach, thus giving students the message that these elements are not important in their educational experiences or in our society. Eisner offers some major points as he concludes his discussion of the null curriculum. The major point I have been trying to make thus far is that schools have consequences not only by virtue of what they do teach, but also by virtue of what they neglect to teach. For example, in social studies, the teacher may give a general overview of the history of science while covering the scientific revolution. However, this information is excluded from the formal curriculum.

From Eisner’s perspective the null curriculum is simply that which is not taught in schools. Somehow, somewhere, some people are empowered to make conscious decisions as to what is to be included and what is to be excluded from the overt (written) curriculum. We teach about wars but not peace, we teach about certain select cultures and histories but not about others. Both our choices and our omissions send messages to students.

6. Societal curriculum/social

As defined by Cortes (1981). Cortes defines this curriculum as:...[the] massive, ongoing, informal curriculum of family, peer groups, neighborhoods, churches, organizations, occupations, mass media, and other socializing forces that “educate” all of us throughout our lives.

This type of curricula can now be expanded to include the powerful effects of social media (YouTube; Facebook; Twitter; Pinterest, etc) and how it actively helps create new perspectives, and can help shape both individual and public opinion.

7. Phantom curriculum

The messages prevalent in and through exposure to any type of media. These components and messages play a major part in the enculturation of students into the predominant meta-culture, or in acculturating students into narrower or generational subcultures.

8. Concomitant curriculum

What is taught, or emphasized at home, or those experiences that are part of a family's experiences, or related experiences sanctioned by the family. (This type of curriculum may be received at church, in the context of religious expression, lessons on values, ethics or morals, molded behaviors, or social experiences based on the family's preferences.)

9. Rhetorical curriculum

Elements from the rhetorical curriculum are comprised from ideas offered by policymakers, school officials, administrators, or politicians. This curriculum may also come from those professionals involved in concept formation and content changes; or from those educational initiatives resulting from decisions based on national and state reports, public speeches, or from texts critiquing outdated educational practices.

10. Electronic curriculum

Those lessons learned through searching the Internet for information, or through using e-forms of communication. (Wilson, 2004) These types of curriculum may be either formal or informal, and inherent lessons may be overt or covert, good or bad, correct or incorrect depending on ones' views. Students who use the Internet and electronic media on a regular basis, both for recreational and informational purposes, are bombarded with all types of media and messages.

11. Curriculum in Use

The formal curriculum (written or overt) comprises those things in textbooks, and content and concepts in the district curriculum guides. However, those "formal" elements are frequently not taught. The curriculum-in-use is the actual curriculum that is delivered and presented by each teacher.

12. Received curriculum

Those things that students actually take out of classrooms; those concepts and content that are truly learned and remembered.

13. The internal Curriculum

Processes, content, knowledge combined with the experiences and realities of the learner to create new knowledge. While educators should be aware of this curriculum, they have little control over the internal curriculum since it is unique to each student.

1.3 CBC Curriculum

1.3.1 Concepts of CBC Curriculum

CBC Curriculum offers so many benefits not only to a learner but also the parents and the teachers. There is maximal engagement of all the stake holders involved. These article provides top 15 benefits of embracing CBC Curriculum. Get facts, and be of sound knowledge as we embrace the curriculum for our own good.

CBC Curriculum Kenya was drafted by Kenya Institute of Curriculum Development (KICD). The curriculum was initiated in 2017 after being approved by the education ministry. The education system replaced the 8-4-4 system. CBC Curriculum meaning Competency-Base Curriculum.

The curriculum focuses on the importance of a learner's ability to develop skills and acknowledge and apply those competency skills to solve daily life challenges.

Competency-Based Curriculum applies a 2-6-6-3 system with classes being named as Grades and has the major divisions that include:

Early years Educations

Middle school Education

Senior school Education

1.3.2 15 Benefits of CBC Curriculum

The CBC curriculum is preferred over the 8-4-4 system because the CBC Curriculum's benefits come within shaping the learners' future.

Some of the Importance of the CBC Curriculum includes;

1. Better Decision Making Abilities

CBC aims at letting the learn to have flexibility in exploring content at all angles while teachers have been granted the freedom to carry through the curriculum when classes are on.

2. Learner-Focused System

CBC education system majors focus on the learner as an individual rather than the school performance. As a result, the teachers can familiarize themselves with the student's demands regarding strength and weakness. Learners hence create a good relationship with teachers who help them shape their talents and careers.

3. Balance Education System

The education system provides a distinctive assessment with many areas of the learner's development in classwork and co-curricular activities. According to CBC, the learner's ability is tested all-round.

4. Focus on Competencies

CBC's central focus is around learners discovering their potentials in life. It creates a platform where the learner can positively relate to acquiring knowledge and skills to connect with various disciplines as they advance to their heights.

5. Digitally Based

The teacher-learner alliance creates an opportunity for progressive learner support in the flexible teamwork environment.

6. Synchronous Development

The CBC system encourages corporative learning with learners to assimilate, enhancing positive growth among learners compared to meaningful learning where little positive change is realized.

7. Collaborative Model

The CBC system is a collective responsibility for both stakeholders and curriculum partners to produce local experts who can enhance developments.

8. Community Service Learning

Involving learners in community services exposes them to apply the knowledge acquired during their learning period. Practicing their community development skills creates a ground for the teachers to spot learners' creativity and potential in discovering specific talents.

9. Distinction of Theory and Practical

The learners are tested on the ability to distinguish what they have learned in class and how practically they can follow instructional approaches in applying the skills outside style. In this manner, the teachers can identify every learner's potential and know what works for them.

As a result, it becomes easier to help the learner improve their strength and overcome their weaknesses.

10. Parental Empowerment and Engagement

The CBC form of education requires parents to participate in their children's positive outcomes during their primary education levels. The parents provide a conducive learning environment by being the learners, which motivates the learners to progress in their learning institutions.

Parents need to collaborate with the teachers in providing basic requirements as per the school curriculum demands to enhance the smooth progress of uninterrupted studies.

11. Fosters Excellence

According to the learners' CBC system, excellence is based on pre-eminence and interest areas rather than just completions for excellence in internal and external examinations and ranking.

12. Expands Learner's Opportunities

CBC provides learners with a diver's opportunities to learn from to help them narrow to talent identification and potential exploration according to their needs and capabilities. The system encourages learners to develop a positive attitude towards education by allowing them to access equal education opportunities.

The focus is on discouraging school dropouts as many learners can enjoy exploring their fields of interest.

13. Diversity and Inclusion

CBC system is programmed to help the learners appreciate that everyone is different and unique in terms of brain capacity. They are brought to an understanding that it is ok to be gifted differently and that no gifting is independent.

As a result, all the institutions are compelled to accommodate all the learners regardless of their classwork performance regarding intellectual, physical, emotional challenges. The learners are brought to an understanding in terms of appreciating diversity in the country, such as

- Race
- Language
- Ethnicity
- Culture
- Religion
- Gender

14. Problem-Solving and Critical Thinking

CBC exposes the learners to a learning system that demands them to reason, make a reasonable judgement and use creativity in arriving at solutions. As opposed to the former education system where learners were required to remember what they were taught, CBC enables the understanding.

Every challenge has an explanation attached. They have to find out how to arrive at the solution without simply cramming an answer.

15. Fosters Imagination and Creativity

CBC does not bind the learner to what is entailed to the syllabus but allows the learners to expand their minds to discover new ideas and images and bring them into accomplishment. The learners can analyze and find discrete patterns to generate solutions that can positively affect the phenomenon.

Focusing on learners' ability, the CBC curriculum can be said to be the best education system for learners. Maximum learners potential can be realized for the good of all.

1.4 21st century skills

1.4.1 Definition of terms

(a) 21st century skills

This a broad set of knowledge, skills, work habits and character traits believed by education stakeholders and employers to critically important to succeed in today's world eg technology literacy, critical thinking, leadership, social skills etc

(b) Pedagogy

This refers to method and practice of teaching especially especially an academic subject/thematical concept.

(c) Inquiry based learning

This is a teaching approach that emphasizes the students to generate questions, ideas and observation.

(d) Value based education

Works on values that enhance academic achievement and developing students social and relationship skills that last throughout their lives. The CBC life values include love, responsibility, responsibility, respect, unity, peace, patriotism and integrity.

1.4.2 THE 21ST CENTURY SKILLS

The 21ST Century skills are:

- Critical thinking
- Creativity
- Collaboration
- Communication
- Information literacy
- Media literacy
- Technology literacy
- Flexibility
- Leadership
- Initiative
- Productivity
- Social skills

These skills are intended to help students keep up with the lightning-pace of today's modern markets. Each skill is unique in how it helps students, but they all have one quality in common. They're essential in the age of the Internet.

To start, let's dive into the three categories that 21st Century skills fall into.

1.4.3 The Three 21st Century Skill Categories

Each 21st Century skill is broken into one of three categories:

(a) Learning skills (the four C's) teaches students about the mental processes required to adapt and improve upon a modern work environment. Altogether, these categories cover all 12 21st Century skills that contribute to a student's future career. The four C's are by far the most popular 21st Century skills. These skills are also called learning skills. More educators know about these skills because they're universal needs for any career. They also vary in terms of importance, depending on an individual's career aspirations.

The 4 C's of 21st Century Skills are:

Critical thinking: Finding solutions to problems

Creativity: Thinking outside the box

Collaboration: Working with others

Communication: Talking to others

(i) Critical thinking is the most important quality for someone to have in health sciences.

In business settings, critical thinking is essential to improvement. It's the mechanism that weeds out problems and replaces them with fruitful endeavors.

It's what helps students figure stuff out for themselves when they don't have a teacher at their disposal.

(ii) Creativity is equally important as a means of adaptation. This skill empowers students to see concepts in a different light, which leads to innovation. In any field, innovation is key to the adaptability and overall success of a company. Learning creativity as a skill requires someone to understand that "the way things have always been done" may have been best 10 years ago — but someday, that has to change.

(iii) Collaboration means getting students to work together, achieve compromises, and get the best possible results from solving a problem. Collaboration may be the most difficult concept in the four C's. But once it's mastered, it can bring companies back from the brink of bankruptcy. The key element of collaboration is willingness. All participants have to be willing to sacrifice parts of their own ideas and adopt others to get results for the company.

That means understanding the idea of a "greater good," which in this case tends to be company-wide success.

(iv) Communication is the glue that brings all of these educational qualities together.

Communication is a requirement for any company to maintain profitability. It's crucial for students to learn how to effectively convey ideas among different personality types. That has the potential to eliminate confusion in a workplace, which makes your students valuable parts of their teams, departments, and companies. But when employees communicate poorly, whole projects fall apart. No one can clearly see the objectives they want to achieve. No one can take responsibility because nobody's claimed it.

Without understanding proper communication, students in the 21st Century will lack a pivotal skill to progress their careers.

(b) Literacy skills (IMT) focuses on how students can discern facts, publishing outlets, and the technology behind them. There's a strong focus on determining trustworthy sources and factual information to separate it from the misinformation that floods the Internet.

They're sometimes called IMT skills, and they're each concerned with a different element in digital comprehension. The three 21st Century literacy skills are:

Information literacy: Understanding facts, figures, statistics, and data

Media literacy: Understanding the methods and outlets in which information is published

Technology literacy: Understanding the machines that make the Information Age possible

(i) Information literacy is the foundational skill. It helps students understand facts, especially data points, that they'll encounter online. More importantly, it teaches them how to separate fact from fiction. In an age of chronic misinformation, finding truth online has become a job all on its own. It's crucial that students can identify honesty on their own. Otherwise, they can fall prey to myths, misconceptions, and outright lies.

(ii) Media literacy is the practice of identifying publishing methods, outlets, and sources while distinguishing between the ones that are credible and the ones that aren't. Just like the previous skill, media literacy is helpful for finding truth in a world that's saturated with information. This is how students find trustworthy sources of information in their lives. Without it, anything that looks credible becomes credible.

But with it, they can learn which media outlets or formats to ignore. They also learn which ones to embrace, which is equally important.

(iii) Technology literacy goes another step further to teach students about the machines involved in the Information Age. As computers, cloud programming, and mobile devices become more important to the world, the world needs more people to understand those concepts. Technology literacy gives students the basic information they need to understand what gadgets perform what tasks and why. This understanding removes the intimidating feeling that technology tends to have. After all, if you don't understand how technology works, it might as well be magic. But technology literacy unmask the high-powered tools that run today's world. As a result, students can adapt to the world more effectively. They can play an important role in its evolution.

(c) Life skills (FLIPS) take a look at intangible elements of a student's everyday life. These intangibles focus on both personal and professional qualities. Also called FLIPS, these skills all pertain to someone's personal life, but they also bleed into professional settings.

The five 21st Century life skills are:

(i) Flexibility: Deviating from plans as needed. Flexibility is the expression of someone's ability to adapt to changing circumstances. This is one of the most challenging qualities to learn for students because it's based on two uncomfortable ideas:

1. Your way isn't always the best way
2. You have to know and admit when you're wrong

That's a struggle for a lot of students, especially in an age when you can know any bit of information at the drop of a hat. Flexibility requires them to show humility and accept that they'll always have a lot to learn — even when they're experienced. Still, flexibility is crucial to a student's long-term success in a career. Knowing when to change, how to change, and how to react to change is a skill that'll pay dividends for someone's entire life.

(ii) Leadership: Motivating a team to accomplish a goal

- (iii) Initiative: Starting projects, strategies, and plans on one's own
- (iv) Productivity: Maintaining efficiency in an age of distractions
- (v) Social skills: Meeting and networking with others for mutual benefit.

1.4.4 Need and importance of curriculum development (or construction)

- Curriculum development is a purposeful activity.
- It is undertaken to design or redesign for the realization of certain specific educational objectives.
- The curriculum is the heart of the student's college/school experience.
- The curriculum should be reviewed and revised on a regular basis so that it is able to serve the changing needs of both students and society.

The following points iterate the needs and importance of curriculum development.

1. Clear purpose and goals:- Curriculum construction provide written curricular goals which are nothing but intended student development outcomes. These goals and objectives are specified in considerable detail and in behavioral language.
2. Continuous assessment and improvement of quality:- Valid and reliable assessment of the curriculum is necessary. The curriculum followed by an institution should be reviewed regularly in order to maintain its effectiveness in regards to the changing needs of the society as a whole.
3. A rational sequence:- In a curriculum educational activities are carefully ordered in a developmental sequence. This developmental sequence helps to form a well planned (or coherent) curriculum based on intended goals and outcomes of the curriculum and its constituent courses.
4. Making strategy in teaching and learning:- Curriculum development helps in suggesting suitable teaching-learning strategies, teaching methods, instructional materials, etc. It helps in providing for the proper implementation of the curriculum on the part of teachers and learners.
5. Helps in the selection of learning experiences:- Curriculum development is needed for appropriate selection and organization of learning experiences. It helps in the selection of study matter and other activities so that learners are able to acquire goals and objectives of teaching.

1. The process of curriculum development is needed for conceptualizing a curriculum in terms of the determination of educational objectives for teaching-learning at a particular grade of school education.

2. Helps in continuous and comprehensive education:-

Curriculum development considers the need of providing a scheme of education for CCE of the teaching-learning outcomes. With proper feedback, it helps to bring necessary improvement in the teaching-learning process and environment.

2.0 FOUNDATIONS OF CURRICULUM DEVELOPMENT

This section provides studies on foundations of curriculum. The cardinal principle in foundations of curriculum is that in a democratic society such as Kenya, education should develop in an individual knowledge, attitudes, skills and powers whereby the individual would find his place in the society and be able to use the foundations to shape both himself and the society for noble ends.

You will find in this, statements of the aims of education with specific objectives based.

2.1 Objectives of Foundation of Curriculum

At the end of lectures on foundations of curriculum, you should be able to;

- Define major terminologies used in foundations of curriculum
- Name major foundations of curriculum in development
- Discuss how foundations of curriculum influence Kenyan decision-making and policy on education.
- Identify various ways in which foundations of curriculum are used in Kenya.
- Suggest ways and means of how to develop and utilize Kenya's resources in curriculum development.
- Identify strengths and weakness of the Kenyan 8:4:4 system of education in line with foundations of curriculum.

2.1.1 Purpose of foundations of curriculum

According to Bishop (1985), the main purpose of foundations of curriculum are to:-

- Preserve cultural Heritage
- Control Direction and Size of Education
- Assist in Designing Curriculum
- Use for selecting Curriculum Aims, Goals and Objectives
- Use for identify Curriculum Implementation STRATEGIES
- Guide in the Development of Evaluation Instruments
- Used for identifying innovation Needs
- Used for cultivating Desired Attitudes, knowledge and Skills
- Used in policy and Decision making.

2.2 Types of Foundations of Curriculum

2.2.1 Philosophical foundation of Curriculum

- What is philosophical foundation of curriculum
- How does this foundation influence curriculum practice?
- In what way is philosophical foundation of curriculum the main curriculum source?
- What is the meaning of the following terms as used in philosophical foundation of curriculum; Realism, pragmatism, Idealism, Existentialism, Axiology, Perenniaism, Essentialism, Progressivism and Reconstructionalism?

Philosophical foundation of curriculum refers to the nature of man in respect to his policy to select, design and formulate objectives to develop and evaluate his objectives and evaluate his knowledge, attitudes and skills of man in decision making for situational, society's and learning needs. Man also derives his methods of presentation, expression, skills to develop and formation of desirable attitudes from philosophical foundation of curriculum.

Objectives of the lecture on philosophical foundation of curriculum are to:

- Identify major philosophies of education in ancient and modern periods
- Discuss how the major philosophies of education influence curriculum decision making.
- Establish the fact that philosophies of education influence is the centre of all educational activities.

Decisions are made on; communities, societies and learners needs. Methods of presentation, skills to be developed and desirable attitudes to be formed, require correct decision making denved from philosophy and policy of a country. Philosophy has entered into every sphere of decision making about curriculum teaching. This view caters for approaches to;

Formation of education purpose

- Selection of knowledge
- Organization of knowledge, attitudes and skills
- Formation of basic procedures in education and curriculum
- Selection of education resources such as personnel, materials and equipment
- Identification, selection and development of assessment instruments.

Philosophical foundation of curriculum has contributed to Kenyans Education systems since Independence. Tyler's (1949) view of philosophy in relation to school purposes is based on five criteria for selecting educational purposes

We shall examine four major schools of philosophy, which have guided the writing, organizing and designing of school curriculum. Some of these philosophies are known by names. We shall refer to them as reconstructionism, progressivism, essentialism and perennialism. Reconstructionism is the most liberal of the

four philosophies and perennialism as the most conservative. Most educational practitioners have used a combination of essentialism and progressivism in curriculum planning construction.

Philosophical or value foundations constitute the values and beliefs that make up the philosophies of life and of education and have a permeating influence on the other foundations. Statements of educational philosophy point to methodical efforts by philosophers to:

- i) Examine values in society
- ii) Derive meanings from facts
- iii) Organize experiences in a manner useful to educational practitioners and the pupils in the school system.
- iv) Justify one or some educational beliefs over others.
- v) Develop new proposals of educational practice for translation into action.

In utilizing knowledge of philosophy in curriculum, the curriculum planner is seeking to establish ideas and notions that will indicate a priority of values in selecting experiences for the curriculum. The value or philosophical orientations which the educational system is charged with promoting form an integral part of the culture of a nation.

We shall examine four major schools of philosophy, which have guided the writing, organizing and designing of school curriculum. Some of these philosophies are known by names. We shall refer to them as reconstructionism is the most educational practitioners have used a combination of essentialism and progressivism in curriculum planning construction.

Philosophical Schools

They include:

- i) Traditional school represented by perennialism and Essentialism
- ii) The progressive school represented by progressivists

The three schools have advanced three theories of subject matter as shown below 1. The perennialists

They believe subject matter should be taught for its own sake.

They also believe that permanence of curriculum content and experiences is more important than change.

Subject matter has a value which is inherent in the subject being taught. It's also their belief that educational system should be stable and its purposes steady.

1. Perennialists

The school of perennialist teaches subjects in their customary separate forms, history as history, geography as geography etc. Rather than in the combination as general (social) studies. The teachers and patrons of this school are sure that some subject is too trivial to be included in the curriculum. Only subject matter that is alleged to be hard to learn is admissible. They do not believe in the feelings and emotions of body movement, memory and thinking.

Perennialism can afford education, which is suitable to a small percentage of students who possess high verbal and academic aptitude. The ideal education is not directed to immediate needs, specialization but it is education calculated to develop the mind.

Approaches to subject matter

- i) Emphasis is laid on the classical subjects. These are subjects valuable in their own right and any educated person was expected to have had exposure to them.
- ii) Emphasis is laid on the desire to make children literate and moral to enable them earn a living as well as find a useful place in society.

Principles of Perennialists

- i) Human nature remains the same
- ii) Man's highest attribute is rationality
- iii) Students should be taught certain basic subjects

To the essentialists subject matter should be taught for use. They maintain that there are certain essentials that each student in school ought to know. The essentialists devote their time to:

- i) Re-examining curricula matters
- ii) Distinguishing the essential and the non-essentials in school programmes
- iii) Re-establishing the authority of the teacher in the classroom

To the essentialists and perennialists agree on certain fundamental principles such as:

- i) Learning involves hard work often coupled with unwilling application.
- ii) The teacher's role is to mediate between the adult world and the world of the child.
- iii) The heart of the educational process is the assimilation of prescribed subject matter.
- iv) The school should retain traditional methods of mental discipline.

2. The Progressivists

In the late nineteenth and early twentieth centuries. Progressivism also known as pragmatism, swept through the educational structure of America challenging the time honoured doctrines of essentialism. This movement was led by John Dewey, William H. Kilpatrick, John D. Childs, George S. Counts (faculty members at Teachers College, Columbia University). Boyd Bodde the progressivist maintained that it was time to subordinate subject matter to the learner. Borrowing from time European philosophers like Rousseau who advocated rearing a child in a relaxed environment without forcing learning, the progressivists created the child-centered school. John Dewey formulated progressive beliefs in a series of publications that included among other Democracy and Education, Experiences and Education. How we think and my pedagogic creed progressivists captured the attention of educators when they insisted that the needs and interests of learners bring their bodies; needs and interests or learners bring their bodies, emotions and spirits to school with their minds.

Progressivists view education not as a product to be learned (E.g. facts, and motor skills) but as a process that continues as long as one lives. To their way of thinking a child learns best when actively absorbing presented content. If experiences in school are designed to meet the needs and interests of individual learners, it follows that no single pattern of subject matter can be appropriate to all learners.

At the heart of progressivism thinking is an abiding faith in democracy. Hence the progressivists see little place of authoritarian practices in the classroom and the school pupils and facilitators of learning rather than expounders of the subject matter. Cooperation is fostered in the classroom rather than competition. Individual growth in relationship to one's ability is considered more important than growth in comparison to others.

They believed that subject matter is a medium for teaching life processes and skills.

They take the pragmatist view that change, is the essence of reality.

Educators must be ready to modify methods and policies in light of new knowledge and changes in the environment

To progressivists, a good school is not a place of compulsory instruction but a community of old and young engaged in learning by co-operative experience.

Progressivists' principles

- i) Education should be life itself not a preparation for living
- ii) Learning should be directly related to the interests of the child
- iii) Learning through problem solving should take precedence over the inculcating of subject matter
- iv) Teachers have to advise not direct pupils
- v) The school should encourage co-operation as opposed to competition

Philosophical positions

The three prominent philosophical positions that are closely related to perennialism and progressivism are Idealism, Realism and Pragmatism.

3. Idealism

In considering the influence of philosophy though on curriculum, several classification schemes are possible. The cluster of ideas as organized in idealism are those that often evolve during curriculum development. Plato is the father of idealist/philosophy.

Idealism emphasizes moral and spiritual development reality as the drier explanation of the world. Truth and values are seen as absolute, timeless and universal. The world of mind and ideas is permanent, regular and orderly, it represents a perfect order. The idealist educator prefers the order and pattern or subject matter curriculum that relates ideas and concepts to each other.

It is largely a traditionalist view which is a carry over from Plato's writings

Idealism uses deductive reasoning in its quest for answers to current day problems. They believe in the independence of truth from the individual or the society.

There is over emphasis on the intellectual aspects.

Idealism stresses the role of education in the transmission of the cultural heritage as handed through the ages from the past. Hence it is a preserving function. It allows the concurrent study of liberal and vocational education, as a means to living completely through understanding life.

3. Realism

Aristotle is often linked to the development of realism, which is another school of thought in philosophical foundation of curriculum. The realism views the world in terms of objects and matter. People came to know the world through their serieses and reason. Everything is derived from nature and is subject to its laws. When behavior is relational, it conforms to the laws of nature and is governed by physical and social laws.

Like the idealist, the realist stresses a curriculum consisting or organized, separate subject matter, content and knowledge that classifies objectives. The realist locates the most general and abstract subjects at the top of the curriculum literally and gives particular and transitory subjects a lower order of priority.

They believe that the main aim of education is the disciplining of the mind, the development of the ability of reason and pursuit of truth. Therefore, curriculum should emphasize grammar, rhetoric, logic, classical and modern languages, mathematics and the great books of the truth, which is the same today as it was then and always, shall be. These thinkers add to the curriculum the study of the Bible and Theological writings.

The realist believes in the existence of a real world, divorced from the imaginations of the perceiver.

To the realist, the real world is the physical world of the physical matter of man and has a specific role to play in daily routine and actions.

Here, the problems of man in life are approached through the inductive method by which data is gathered to form a basis for new principles and generalizations. Realism accepts God as the motive cause of all existence.

Realism argues that education should induct learners into their culture and help them to adjust to the natural order of things in order to live in harmony with the universe.

Teachers act as guides: making children aware of the true nature of real world.

Realists advocate a study of physical and social sciences which are instrumental to explaining natural phenomena. Mathematics is also encouraged.

The realism views the world in terms of objects and matter. People come to know the world through their senses and reason. Everything is derived from nature and is subject to its laws. When behavior is relational, it conforms to the laws of nature and is governed by physical and social laws. Advocates of realism include John Amos Comenius, John Locke and John Herbart.

4. Pragmatism

This philosophy is sometimes referred as experimentalism, which is based on change process and relativity while idealism and realism philosophies emphasize subject matter, disciplines and content or ideas, pragmatism construes knowledge as progress in which reality is constantly changing. Learning goes on as the student engages variety of subjects and environment and are constantly changing. To a pragmatism nothing can be viewed intelligently except in relation to a pattern. It is a progressivist position that sees reality as being in a state of flux

or constant change. Pragmatists employ the realist approach in gathering information and facts, and idealist approach in generalizing about the facts gathered.

Pragmatism seeks meaning in the immediate situation according to pragmatists education

- Should enable the learner to experience situations in practice
- Is a means for recreating, controlling and redirecting, experience.
- Should help learners to solve their problems and is to be considered an integral part of life. Systematic sequencing of learning experiences is emphasized by pragmatists.

Teachers should provide an atmosphere in which learners identify the problems and seek solutions to them.

Teachers should also arrange an environment that provides experience for learners. The curriculum should be organized on the basis of the learners' interests and the subject matter selected should help the learner to solve problems.

Sometimes referred as experimentalism, which is based on change process and relativity while idealism and realism philosophies emphasize subject matter, disciplines and content or ideas, pragmatism construes knowledge as progress in which reality is constantly changing. Learning goes on as the student engages in problem solving. The idea of problem solving, is however transferable to a wide variety of subjects and situations, knowing is considered a transaction between the student and environment and are constantly changing.

6. Reconstruction

Reconstructionism is a philosophical school of thought that broke off from the progressive movement because of unresolved problems of democracy when they wanted to rebuild the social order. They see education as a powerful instrument for effecting planned social changes in given society. It is most likely to be favored in times of economic, political and technological turbulence such as has been experienced recently by the newly independent countries of Africa. They believe that new education can make new and better men and women therefore the school should transmit cultural heritage. Also the school is seen as an agent of solving political and social problems.

These schools of thought can be regrouped into two big schools.

a) Progressivism (Reconstructionism)

b) Traditionalism (Perennialism and Essentialism)

c) However, progressivists and traditionalists disagree on many points concerning subject matter which is to be included in the curriculum, such as;

- Which knowledge is most worthy?
- Should we emphasize process or information?
- Should a curriculum be fixed or flexible, constant or differentiated, practical or liberal?

To a large extent one's answers to these questions depends upon one's system of values.

7. Essentialism

Historically, essentialism and progressivism have succeeded in commanding education in the western world. But essentialism is the more powerful than the progressivism. It was only in the early 1950's specifically 1957 the year of sputnik that the progressivism emerged for a short time as the victor among the world philosophies of education. This did not last long. During the 1970's upto now, essentialism has proven that it was not the loser but the leader.

2.2 THE INFLUENCE OF PSYCHOLOGY TO CURRICULUM: FOUR MAJOR FOUNDATIONS

The curriculum is influenced by psychology. Psychology provides information about the teaching and learning process. It also seeks answers as to how a curriculum will be organized to achieve students' learning at the

optimum level and what amount of information they can absorb in learning the various contents of the curriculum.

These are insights gained from psychology which have a bearing on the learning process. Psychologists believe that learning experiences have to be introduced to the learner when such exposure is most effective and most beneficial to him.

The following issues should be taken into account when dealing with learning experiences in the curriculum.

- Capability of the learner
- Maturation level of the learner
- Students' rate of learning as well as how they learn

Psychology as a discipline helps the curriculum designer and the teacher to reach decisions in relation to:

- Sequence – in the stages of development
- Organization – grouping of learning experiences for optimal effect.
- Methodology – dealing with the question of what methods and approaches are likely to promote and guiding learning most effectively.

Psychology also contributes to:

- Formulation of appropriate educational goals
- Decisions regarding the scope of curriculum

The following are some psychological theories in learning that served as major foundations of curriculum development:

1. BEHAVIORISM

Education in the 20th century was dominated by behaviorism. The mastery of the subject matter is given more emphasis. So, learning is organized in a step-by-step process. The use of drills and repetition are common.

For this reason, many educational psychologists viewed it as mechanical and routine. Though many are skeptical about this theory, we can't deny the influences it had on our educational system.

2. COGNITIVISM

Cognitive theorists focus on how individuals process information, monitor and manage their thinking. The basic questions that cognitive psychologists zero in on are:

How do learners process and store information?

How do they retrieve data and generate conclusions?

How much information can they absorb?

With their beliefs, they promote the development of problem-solving and thinking skills and popularize reflective thinking, creative thinking, intuitive thinking, discovery learning, and others.

3. HUMANISM

Humanism is taken from Gestalt's theory, Abraham Maslow's theory, and Carl Rogers' theory. This group of psychologists is concerned with the development of human potential. In this theory, the curriculum is after the process, not the product; focuses on personal needs, not on the subject matter; and clarifying psychological meanings and environmental situations. In short, curriculum views founded on humanism posits that learners are human beings who are affected by their biology, culture, and environment. They are neither machines nor animals. Along this line, curriculum developers must craft a more advanced, more comprehensive curriculum that promotes human potential. Teachers don't only educate the minds, but the hearts as well.

Here is a three-minute video that explains further the humanist's point of view.

4. SOCIOLOGY AND CURRICULUM

Among the major foundations of curriculum development, the sociological theory emphasizes the influence of society to education. It is founded on the belief that there is a mutual and encompassing relationship between society and curriculum because it exists within the societal context. Though schools are formal institutions that

educate the people, other units of society educate or influence the way people think, such as families and friends, and communities.

Since society is dynamic, many developments are difficult to cope with and adjust to. But the schools are made to address and understand the changes not only in one's country but in the world as well.

2.3 HISTORICAL BACKGROUND TO CURRICULUM DEVELOPMENT

Factors that influenced curriculum development in Kenya post-independence

1. Political

Funding plays a fundamental role. Funds help in purchase of materials and approve evaluation systems.

2. Social factors

Gender evaluation and political economy should be factored in.

- Issues at play include:
- Religion
- Culture
- Social groupings
- Race
- Peer culture
- Family
- Drugs, delinquency, sex and crime (should address these needs)

3. Economic factors

A good curriculum should foster global competition in technical fields. It should promote employment, promotions and profits to an individual and the nation.

4. Technological technology

It should utilize multimedia to foster education goals and learning experiences.

5. Environmental issues

It should focus on pollution awareness, recycling and other healthy environmental practices.

6. Child psychology

It should capture changes in personality, cognitive ability and behaviour.

3.0 THEORIES, MODELS AND PRINCIPLES OF CURRICULUM DEVELOPMENT

3.1 CURRICULUM MODELS

3.1.1 What is a model?

It is a three dimensional representation of a person or thing or of a proposed structure, typically on a small scale than the original.

- Curriculum models are based on a body of theory about teaching and learning.
- They are targeted to needs and characteristics of a particular group of learners.
- They outline approaches, methods and procedures for implementation.
- Curriculum development theories and models provide concepts, issues, explanations, proportions and frameworks that give curriculum development directions.

3.1.2 CURRICULUM DEVELOPMENT THEORY

Can be said to be a formal set of ideas that are intended to explain the elements and processes of curriculum development. A curriculum development model is a simple description of a curriculum development system using a diagram. Curriculum development theories and models provide foundations for curriculum development, implement and evaluation. They provide foundations for curriculum implementation review and construction.

3.1.3 RALPH TYLER'S CURRICULUM THEORY AND MODEL

Ralph Tyler was an American scholar and educationist. In 1949, Tyler published his book; Basic Principles of curriculum and instructions.

Tyler identified four basic questions that should be answered by curriculum developers.

i) What educational purposes should the school seek to attain.

– By purposes Tyler was referring to objectives.

– He indicated that curriculum planners should identify general objectives by gathering data from three sources:

a) Subject matter

b) The learner

c) Society

– After identifying these general objectives curriculum planners were to refine them by filtering them through two screens – the philosophy of the school and the psychology of learning.

– What results is instructional objectives

ii) What educational experiences can be provided that are likely to attain these purposes?

– Learning experiences had to take into account both the previous experiences and the perception that the learner brings to a situation.

iii) How can these educational experiences be effectively organized?

– Tyler talked about the organization and sequencing of these experiences.

– He suggested that the ordering of the experiences had to be systematic so as to produce a maximum cumulative effect.

– A curriculum developer should look at the activities that should be given to learners in term I, II, and III and organize them for the realization of the set objectives.

iv) How can we determine whether these purposes are being attained?

– The last question emphasizes the role of curriculum evaluation as an important element, in the curriculum development process.

– A curriculum developer has to consider the means, method and strategies that will be used to determine whether the desired purposes/objectives are being achieved or have been achieved.

Discussion of Tyler's curriculum theory and model.

1. He put more emphasis on the objectives to guide curriculum development.

– According to him the sources of curriculum objectives are:

a) A study of the learners needs in relation to the needs of the society. b) A study of contemporary issues and society's life.

c) Seeking suggestions of subject specialists through consultation.

2. Ralph Tyler's model provided the basis for other curriculum development models. However it is considered simplistic as it ignored the complex process involved in curriculum development.

– It does not offer the relationship between various curriculum elements.

– It was found to deal shallowly with the issue of curriculum evaluation which is an important aspect of curriculum development.

Elements of the curriculum

According to Ralph W. Tyler, a prominent curriculum specialist, the four elements of the curriculum are:

1. Selection of objectives

2. Selection of learning experiences

3. Organization of learning experiences

4. Evaluation process

QUESTIONS

a) Outline four fundamental curriculum questions that Ralph Tyler's curriculum theory emphasized.

b) With the aid of a diagram, discuss Ralph Tyler's curriculum development theory and model

c) Discuss the application of Ralph Tyler's model in secondary school curriculum development.

3.1.4 HILDA TABA'S CURRICULUM THEORY AND MODEL (1962)

Hilda Taba was born in 1902, she was a curriculum theorist, Educator and reformer. Hilda Taba believed that those who teach curriculum should participate in developing it. She believes that teachers should develop curriculum and that curriculum should not be handed down from higher authorities. She advocated what has been termed as the “grass-root approach.”

She advocated that teachers take an inductive approach to curriculum development starting with specific s and building to a general design –as opposed to Ralph Tylers more traditional deductive approach starting from general to specifics.

Hilda Taba noted seven major steps in the process of developing a curriculum.

- i) Diagnosis of needs
- ii) Formulation of objectives
- iii) Selection of contents
- iv) Organization of contents
- v) Selection of learning experiences
- vi) Organization of learning experiences
- vii) Evaluation and means of evaluation

i) Diagnosis of needs

The teacher starts the process by identifying the needs of the students for whom the curriculum is to be planned.

ii) Formulation of objectives.

Objectives formulation should encompass the following areas:

- a) Concepts or ideas to be learned.
- b) Attitudes, sensitivities, and feelings to be developed.
- c) Ways of thinking to be reinforced, strengthened or initiated
- d) Habits and skills to be mastered.

iii) Selection of Content

The objectives selected or created suggest the subject matter or content of the curriculum unit subject.

– Subject matter should be related to the grade level of the student.

iv) Organization of Content

– A teacher cannot just select content, but must organize it in some type of sequences taking into consideration the maturity of the learners, their academic achievement, and their interests.

v) Selection of learning experiences

Content must be presented to pupils or pupils must engage in an interaction with the content. To select learning experiences, Taba suggests that the teacher should ask a number of questions. For example;

- a) Is the experience appropriate for learning the main ideas?
- b) Does the experience promote “Active learning?”
- c) Is the experience appropriate to the students’ maturity level?

Taba also felt that learning experiences should reflect a variety of experiences, including, “reading, writing, observing, doing research, analyzing, discussing, tabulating, painting, constructing and dramatizing.

vi) Organization of learning experiences.

Taba outlines a sequence for organizing learning experiences: –

- Introduction
- Development
- Generalization
- Application or summary

Introduction involves developing student interest and proving diagnostic evidence for the teacher. Development or study consists learning activities that are designed to develop various aspects of the subject and to provide

needed factual material. These activities include “reading, research, analysis of data, committee work and study of various kinds.

Generalization refers to students attempts to put ideas together. E.g comparing and contrasting and exploration of the reasons for similarities and differences.

Application or summary is the stage at which the student applies generalization to a larger framework.

vii) Evaluation and Means of Evaluation

Involves determining whether objectives have been met, diagnosis of the curriculum plan, and assessment of any changes in student behavior.

Revision Question

1. Outline the seven steps that are necessary for the completion of curriculum development according to Hilda Taba.

2. Discuss three differences between Hilda Taba’s and Ralph Tyler’s curriculum models.

3. Using a diagram, identify four curriculum development elements in Hilda Taba’s curriculum model.

3.1.5 OLUOCH'S MODEL OF CURRICULUM DEVELOPMENT (1982).

Oluoch’s model of curriculum development goes a long toward trying to solve the fundamental problem; that much of what is taught in not learned and much of what is presented and tested is not assimilated. Oluoch explains curriculum as all that is planned to enable the learner to acquire and develop desired knowledge, skills, values and attitude.

Oluoch wrote in the context of the Kenya situation. Although most people think of education as the same as going to school, he believed that this was wrong. He defined education in a much broader way: Education is the process of acquiring and developing desired knowledge, skills and attitudes (Oluoch, 1982). This definition is superior to many others as it suggests that education is continuous across the life span and can occur outside school settings.

According to Oluoch, curriculum came from the Roman word racecourse, a tract followed by horses in a race. Curriculum came to be considered a course of study followed by a school or some other teaching institution and also an individualized syllabus in educational institution. He believes that curriculum should be defined as everything that is planned by an educational institution to help the students learn whatever it is that the institution would want them to learn (1982, p. 7). It includes the formal course of study, extra-curricular activities and other informal activities within the school environment. Oluoch developed a conceptual framework of the school curriculum.

Oluoch had the objectives for the learner (what is desired that the learners’ should attain), the learning activities (how the desired objectives are to be accomplished, including the learning activities within the content and the student assessment (evaluation of the learning outcome).

We also note that he includes three different dimensions of school curriculum.

1. Formal dimension: Refers to the learning activities within or outside of classrooms and schools that are normally undertaken by students as formal class work.

2. Nonformal dimension: Refers to learning activities organised in a less rigid manner. The students are not formally grouped as for class work, but group themselves according to interests, aptitude and ages. They may occur in or outside classrooms and schools, and are part of the broad comprehensive education programme made available to learners.

3. Informal dimension: Consists of the guided aspects of learning activities that go on in the school all the time, such as interactions. Although these represent part of the hidden curriculum of schools, it is only the activities that relate to the planned aspects of the environment that count as informal learning activities. The extent of stimulation provided by learning resources within the learning environment is an example of the informal dimension.

Elements of school curriculum according to Oluoch

- i. Goals and objectives of the curriculum
- ii. Learning opportunities through which the goals and the objectives of the curriculum should be achieved
- iii. The evaluation process, to find out if the goals and objectives previously set have been achieved and if they have, to what degree. These elements are appropriate to all dimensions of the school curriculum.

Components of Curriculum Development.

Seven major components of curriculum development can be identified. These are;

1. Needs assessment.
2. Formulation of a curriculum development project.
3. Curriculum designing.
4. Personnel training.
5. Development of curriculum materials and equipment.
6. Implementation.
7. Project evaluation.

1. Needs assessment.

This is a method or technique for identifying educational objectives that most need to be accomplished in a given instructional situation. It is an essential component of curriculum development in the determination as precise as possible of what is that needs improving. The needs to be assessed may include the societal and national needs of education. The component establishes the base of curriculum development.

2. Formulation of curriculum development project.

This component involves statement of goals and objectives of the curriculum development project, approximating the time it would take before completion and how many individuals are required and also the financial amount required. In this way all the necessary resources will be committed for the entire duration of the project.

3. Curriculum designing.

After extraction of information and identification of the learner, designing the curriculum begins. Before designing the content, there is need to be clear objectives. Stating the specific learning outcomes expected that would be in need for figuring out these learning outcomes.

Curriculum designing involves deciding what part in the effort to realize intended learning outcomes. Therefore, is important that detailed curriculum designs be prepared and reviewed from time to time as project proceeds.

4. Personnel training.

For the success of curriculum development procedure, teachers have to be included and they should understand and accept the ideas contained in the curriculum being proposed or implemented. All the categories of personnel should be trained to understand and accept the ideas of the materials forming part of the new curriculum.

5. Development of the curricular materials and equipment.

These materials are designed to help the teacher grasp the knowledge and to implement the ideas contained in the new curriculum. These materials include the syllabuses, books and mass media programs for school teachers and students. Similarity in preparation of these materials should be of importance for the development of relevant curriculum materials.

6. Implementation.

This is the interaction of the learners with the teachers, ideas, materials and equipment with a view to achieving the goals of the new curriculum. Organization of familiarization programs for educators, parents and other members of the society should be important. The implementation component happens after all the above components have been put in place.

7 Project evaluation.

After the new curriculum has been implemented, it should be analyzed if its headed towards the stated goals and objectives and to make appropriate adjustments if otherwise.

These various dimensions are useful for us to remember since they suggest that we need to plan not just for the formal classroom learning activities.

Oluoch described various stages to be used in the process of curriculum development.

1.Information gathering: This is a process of collecting information about the need for and feasibility of the curriculum development process. It should address the issues of what information should be collected, from whom should it be gathered, who should collect it, who should coordinate this process and what should it be used for.

2.Formulating the curriculum project: This is the process of looking if the curriculum project is desirable and also feasible, and if it is then the curriculum development is described within a written project report.

3.Planning the curriculum: This component involves considering each dimension of the school curriculum carefully in reference to:

- Understanding and stating the goals and objectives of the curriculum,
- Determining, organising and stating the necessary learning objectives
- Determining and stating the methods that will be used in assessing the progress and achievement of the learners.

4.Selection and preparation of materials and equipment: In this component the requirements for the materials and equipment to be used in the curriculum are identified and costed. At this point who will prepare the new materials and how they are to be prepared or made are also discussed so that they will be ready for the new curriculum.

5.Curriculum Try-out: In this component the curriculum, the materials and the equipment are piloted. They are used in a few schools to see if the desired results occur. If not, revisions are made before the implementation.

6.Curriculum Implementation: At this point the new curriculum is used in the various schools. The phase may require teacher in-servicing or orienting to the new curriculum, provision of support services and facilities and introduction of possible organisational changes with the schools and districts educational systems. It will be important to inform the general public about the new curriculum and to address their concerns.

7.Curriculum Project Evaluation: In this component the curriculum project is evaluated, and this evaluation focuses on the planning phase, the implementation phase and when it is completed after a specified period of time. This component should be focused on clearly identifying what exactly is to be evaluated, why it is to be evaluated and how it is to be measured.

SHORTCOMINGS OF OLUOCH'S MODEL OF CURRICULUM DEVELOPMENT.

Despite the major positivism towards the success of Oluoch's model of curriculum development, it still faces a number of shortcomings; -

1. It is bulky and contains a lot of processes. The model involves a lot of steps and actions that have to be and undertaken which may be bulky and tiring. This may bring monotony and confusion during the development and implementation of the curriculum.

2. Some of the procedures and guidelines that are suggested in the model are technical and complex and not every teacher can perform.

3. There is a strong tendency in the literature about behavioral objectives in curriculum design to make the assumption that the objectives that can be defined behaviorally, the objectives cannot be easily assessed, and these objectives do not form the important objectives for a school curriculum. The corollary also seems to be accepted. If it seems impossible to detect and assess a specific learning outcome, it probably isn't important.

4. If identification of all worthwhile outcomes in behavioral terms comes to be commonly accepted and expected, then it is inevitable that over time curriculum will tend to emphasize those elements which have been identified. Important outcomes which are detected only with great difficulty and which are translated only rarely into behavioral terms tend to atrophy. They disappear from curriculum because much time is spent on the allotted teaching explicitly for the more readily specifiable learning to which have been directed.

5. The model carries a lot of processes and guidelines through certain people. These processes of research, collection, analysis and implementations which need a lot of financing. Project people, and researchers have to be paid and curriculum implemented.

6. Research, collection of data, analysis, implementation is time consuming and therefore it might take a long time before a curriculum is completely developed. The more the time is taken, the more is the need for more money to finance it.

3.1.5 RELEVANCE TO CURRICULUM DEVELOPMENT IN KENYA.

The model provides guidelines for the contemporary development and improve of the curriculum in Kenya today. The setting up of an organization for curriculum development in Kenya is based on the guidelines provided by the model which have entirely helped for the development of a competent and all round curriculum.

The model has recommended that a procedure to be followed in developing a school curriculum in Kenya which should be based on the principles; - Curriculum development should be based on the outcomes. Should concern itself with all curriculum dimensions and elements. Be a team effort.: all these principles through the Kenya Institute of Curriculum Development have been implemented and has largely help in the development of the major curriculums and organized the basis of well formulated projects making the development simpler and procedural.

The model has made possible reorganization of the formerly Kenya Institute of Education to currently Kenya Institute of Curriculum Development which is the organization in the country with the responsibility of curriculum development in school education. These changes are in the aspect of functions, the control, the panels and the structure of the staff of the institute. This has made the Kenya institute of curriculum development to be more effective in coming up with the competent and reliable curriculums.

Each subject area or curriculum area has subject pane and each of them with a related teacher for educational research and evaluation matters in that subject. These panels interact as much as possible to bring about coordination in education in the country. The panel's main work is to coordinate the content and its relevance in the subject area. The concept of personnel training before the implementation of a curriculum in the model is relevantly visible in the new curriculum. Through seminars and conferences, teachers have been made to understand the changes and ideas in the new curriculum about to be implemented. Each category of teachers has been trained about the content and the teaching and learning activities they are to use for the accomplishment and the success of the new 2.6.3.3 curriculum in Kenya.

3.2 Principle of Curriculum Construction

In curriculum development, we think about the type of learning experiences to be given to a child at various age and grade levels. It needs systematic and sequential planning to widen the sphere of the learning experience at each level by keeping in view the principles of integration and correlation. The curriculum is usually concerned with two questions. What are the principles you will follow when construction curriculum for your children? Principles are guiding rules. They are the rules, which will guide you when constructing, or designing or making a curriculum. The principles include:

(1) Suitability to the age and mental level of the children

- What is to be given to the children in the form of learning experiences at a particular age and grade level should suit their age and mental development
- The capacity for understanding, how children grow with age. The content of the study in any subject should be formed to suit their mental ability.

(2) According to the specific interests of students

- Children will be able to learn better in fields where they have special tastes and inclination of the mind.
- It is also found that at different stages of age groups, children have different interest patterns.
- Interests of children also change according to circumstances and situations.
- Therefore learning experiences should be designed to suit the interests and tastes of the age group of students.

(3) The curriculum should be environmentally centered

- The content of the learning experiences for children should be linked with the needs of the environment in which they live.
- For example, children from rural areas can understand and grasp easily the information which is directly concerned with their experiences in their own rural environment.
- The same thing applies to children in a various environments like urban areas, hilly areas, etc.

(4) The principle of the comprehensive curriculum

- The curriculum must have the necessary details. List of topics to be covered does not solve the purpose.
- Both teachers and students should know clearly what is expected of them, what is the beginning and what is the end of the topic for the particular class.
- Material, aids, activities, life situations etc. should be listed in the curriculum.

(5) Principle of co-relation

- The curriculum should be such that all the subjects are correlated with each other.
- While designing the curriculum, it must be kept in mind that the subject matter of various subjects has some relation to each other so that they help the child eventually.

(7) The principle of practical work

- Children are very active by nature.
- They like new things and can learn more by doing or by activity method.
- Therefore curriculum should be designed in such a way that it provides maximum opportunity to the child for practical work with the help of concrete things.

(8) Principle of flexibility

- Instead of being rigid curriculum should show the sign of flexibility.
- The organization of the curriculum should be on the basis of individual differences as every child is different from the other.
- Apart from these conditions of society go on changing, therefore, the curriculum must be flexible enough to address the needs as aspirations of the society.

(9) Principle of forward-looking

- This principle asks for the inclusion of those topics, content and learning experiences that may prove helpful to the students in leading their future life in a proper way.

(10) The principle of consultation with teachers

- Teachers play a key role in the implementation of the school curriculum of any grade or stage.
- It is therefore quite essential to seek the proper involvement of the teachers in the construction and development of the school curriculum.

(11) The principle of the joint venture

It is necessarily a joint venture where various experts are involved like educational psychologists, educational technologists, curriculum specialists, evaluation specialists, teachers, subject matter experts etc.

(12) Principle of Child Centeredness

When designing a curriculum for your children, you should keep in view the child's abilities, interests and needs.

(13) Principle of actual Participation

The curriculum should provide opportunities for actual participation of children. There should be activities for children to do.

(14) Principle of Co-Ordination and Integration

The curriculum should provide fullness of experience i.e. there should be maximum co-ordination and integration between curricular and co-curricular activities. Curricular activities refer to the activities planned and organized in classroom and co-curricular activities refer to activities done outside the classroom.

(15) Principle of Community Centeredness

Education should prepare a child to fit in his/her community. This is possible only when the curriculum is central on the community of the child that is the content and activities of the curriculum should originate from the culture of the child.

(16) Principle of Conservation

Curriculum should include activities and experiences, which will help in the conservation of past heritage. Heritage means history and culture. Curriculum should preserve what is useful and add what is helpful in solving the life problems.

(17) Principle of Variety and Elasticity

Curriculum should provide a variety of activities because of individual differences and freedom of choice. Children are different in terms of abilities, interests and age. For children to learn best, there should be a variety of activities. Curriculum should also be flexible and not rigid and should accommodate necessary changes.

(18) Principle of Social

Curriculum should meet the needs of a society i.e. should cater for the aspirations of a society. The curriculum should provide knowledge and skills to make children useful in their society.

(19) Principles of Activity Centeredness

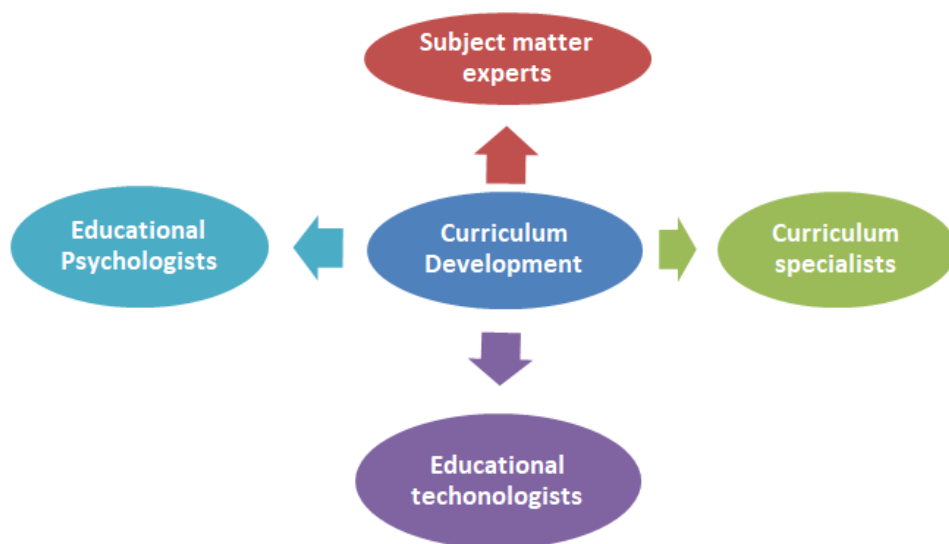
Children learn by doing. Unless they do activities they will never learn. Curriculum should provide a variety of activities for children to do. The activities should be selected according to the abilities, interests and age of children.

(17) Principle of Creativity

Curriculum should promote or develop children's creativity. It should provide activities, which will enable children to exercise their environments according to the needs of the time.

(18) Principle of Utility

Curriculum should have value to children and society or community of the child. It should be of practical use to children. The curriculum should give due emphasis to work experience. It should provide knowledge and skills children need to be useful to themselves, families, communities and to the nation (Kenya).



(19) The principle of availability of time and other resources

To answer these questions we should now discuss the principles of curriculum development.

Curriculum is the means to realize the outcomes of the educational objectives of the school. Implementation of the curriculum is equally important as curriculum construction. While developing curriculum experts should also keep its implementation in mind. They should be aware of the conditions of the schools and possible availability of time and resources available.

4.0 DIMENSIONS OF CURRICULUM

4.1 Types of curriculum: Formal, Informal & Non-formal

4.1.2 What is education?

Education is a gradual process which brings positive changes in human life and behavior. We can also define education as “a process of acquiring knowledge through study or imparting the knowledge by way of instructions or some other practical procedure”.

Education brings a natural and lasting change in an individual’s reasoning and ability to achieve the targeted goal. It facilitates us to investigate our own considerations and thoughts and makes it ready to express it in various shapes.

Education is the main thing that encourages us to distinguish between right and wrong because in the absence of education, we can’t do what we need or we can’t achieve our goal.

Straightforwardly, we can say, “education is the passage to progress”. It is additionally the way to our fate as achievements can only be accomplished when individuals have information, aptitudes, and frame of mind. In this way, education resembles a medium through which we can associate with various individuals and offer our thoughts.

To tackle issues and do inventiveness we first need to gain proficiency with some essential abilities. We require learning and abilities to wind up increasingly imaginative. So education is fundamentally learning of abilities and ideas that can make us increasingly innovative and issue solver. Education is to pick up the capacity to develop and take care of issues in order to achieve their lawful motives.

4.1.3 Dimensions of curriculum

Education goes beyond what takes places within the four walls of the classroom. A child gets the education from his experiences outside the school as well as from those within on the basis of these factors. There are three main types of curriculum, namely, Formal, Informal and Non-formal. Each of these types is discussed below.

1. Formal curriculum

Formal education or formal learning usually takes place in the premises of the school, where a person may learn basic, academic, or trade skills. Small children often attend a nursery or kindergarten but often formal education begins in elementary school and continues with secondary school.

Post-secondary education (or higher education) is usually at a college or university which may grant an academic degree. It is associated with a specific or stage and is provided under a certain set of rules and regulations.

The formal education is given by specially qualified teachers they are supposed to be efficient in the art of instruction. It also observes strict discipline. The student and the teacher both are aware of the facts and engage themselves in the process of education.

Examples of Formal curriculum

- Learning in a classroom
- School grading/certification, college, and university degrees
- Planned education of different subjects having a proper syllabus acquired by attending the institution.

Characteristics of formal education

- Formal education is structured hierarchically.
- It is planned and deliberate.
- Scheduled fees are paid regularly.
- It has a chronological grading system.
- It has a syllabus and subject-oriented. The syllabus has to be covered within a specific time period.

- The child is taught by the teachers

Advantages of Formal education:

- An organized educational model and up to date course contents.
- Students acquire knowledge from trained and professional teachers.
- Structured and systematic learning process.
- Intermediate and final assessments are ensured to advance students to the next learning phase.
- Institutions are managerially and physically organized.
- Leads to a formally recognized certificate.
- Easy access to jobs.

Disadvantages of Formal education:

- Sometimes, brilliant students are bored due to the long wait for the expiry of the academic session to promote to the next stage
- Chance of bad habits' adoption may be alarming due to the presence of both good and bad students in the classroom
- Wastage of time as some lazy students may fail to learn properly in spite of motivation by the professional trainers.
- Some unprofessional and non-standard education system may cause the wastage of time and money of the students which leads to the disappointment from formal education and argue them to go for non-formal education.
- Costly and rigid education as compare to other forms of learning

2. Informal curriculum

Informal education may be a parent teaching a child how to prepare a meal or ride a bicycle. People can also get an informal education by reading many books from a library or educational websites. Informal education is when you are not studying in a school and do not use any particular learning method. In this type of education, conscious efforts are not involved. It is neither pre-planned nor deliberate. It may be learned at some marketplace, hotel or at home.

Unlike formal education, informal education is not imparted by an institution such as school or college. Informal education is not given according to any fixed timetable. There is no set curriculum required. Informal education consists of experiences and actually living in the family or community.

Examples of Informal Education

- Teaching the child some basics such as numeric characters.
- Someone learning his/her mother tongue
- A spontaneous type of learning, "if a person standing in a bank learns about opening and maintaining the account at the bank from someone."

Characteristics of Informal Education

- It is independent of boundary walls.
- It has no definite syllabus.
- It is not pre-planned and has no timetable.
- No fees are required as we get informal education through daily experience and by learning new things.
- It is a lifelong process in a natural way.
- The certificates/degrees are not involved and one has no stress for learning the new things.
- You can get from any source such as media, life experiences, friends, family etc.

Advantages of Informal Education

- More naturally learning process as you can learn at anywhere and at any time from your daily experience.
- It involves activities like individual and personal research on a topic of interest for themselves by utilizing books,

- Utilizes a variety of techniques.
- No specific time span.
- No need to hire experts as most of the professionals may be willing to share their precious knowledge with students/public through social media and the internet.
- Learners can be picked up the requisite information from books, TV, radio or conversations with their friends/family members.

Disadvantages of Informal Education

- Information acquired from the internet, social media, TV, radio or conversations with friends/family members may lead to the disinformation.
- Utilized techniques may not be appropriate.
- No proper schedule/time span.
- Unpredictable results which simply the wastage of time.
- Lack of confidence in the learner.
- Absence of discipline, attitude and good habits.

3. Non-formal curriculum

Non-formal education equivalency preparation includes adult basic education, adult literacy education or school. In nonformal education, someone (who is not in school) can learn literacy, other basic skills or job skills. Home education, individualized instruction (such as programmed learning), distance learning and computer-assisted instruction are other possibilities.

Non-formal education is imparted consciously and deliberately and systematically implemented. It should be organized for a homogeneous group. Non-formal, education should be programmed to serve the needs of the identified group. This will necessitate flexibility in the design of the curriculum and the scheme of evaluation.

Examples of Non-formal Education

- Boy Scouts and Girls Guides develop some sports program such as swimming comes under nonformal education.
- Fitness programs.
- Community-based adult education courses.
- Free courses for adult education developed by some organization.
- Characteristics of Non-formal Education
- The nonformal education is planned and takes place apart from the school system.
- The timetable and syllabus can be adjustable.
- Unlike theoretical formal education, it is practical and vocational education.
- Nonformal education has no age limit.
- Fees or certificates may or may not be necessary.
- It may be full time or part-time learning and one can earn and learn together.
- It involves learning of professional skills.

Advantages of Non-formal Education

- Practiced and vocational training.
- Naturally growing minds that do not wait for the system to amend.
- Literacy with skillfulness growth in which self-learning is appreciated.
- Flexibility in age, curriculum and time.
- Open-ended educational system in which both the public and private sector are involved in the process.
- No need to conduct regular exams.
- Diploma, certificates, and award are not essential to be awarded.

Disadvantages of Non-formal Education

- Attendance of participants is unsteady.
- Sometimes, it's just wastage of time as there is no need to conduct the exam on regular basis and no degree/diploma is awarded at the end of the training session.
- Basic reading and writing skills are crucial to learn.
- No professional and trained teachers.
- Students may not enjoy full confidence as the regular students enjoy.
- Some institutes provide fake certification through online courses just for the sake of earning

6.0 CURRICULUM DEVELOPMENT CYCLES

6.1 History of Curriculum Development In Kenya

6.1.1 Curriculum Changes

(a) The 7-4-2-3 Education System

Introduction of 7-4-2-3 System of Education. According to Ominde (1964), the chairman of the first educational commission in the independent Kenya, “during colonial era, there was no such thing as a nation” only several nations living side by side in the same territory. Education, like society, was stratified along racial lines, there existed an ‘African education’, a ‘European Education’, and an ‘Asian Education’; three separate systems divided by rigid boundaries (Ominde 1964). This stratification was based on the colonialist’s assertion that the mental development of the average African adult was equivalent to that of the average 7-8 year old European boy (Gachathi, 1976).

African education’ therefore tended to be a hybrid, precariously hovering between a European model with a European subject matter, and an education deemed suitable to the place in colonial life considered ‘appropriate’ to the African population (Ominde 1964).

Thus the eve of independence brought with it sweeping reforms in the educational system. With the creation of a single nation came the emergence of a single educational system, no longer stratified along racial lines. Ominde Commission was formed to introduce changes that would reflect the nation's sovereignty. The commission focused on identity and unity, which were critical issues at the time. Changes in the subject content of history and geography were made to reflect national cohesion. Between 1964 and 1985, the 7-4-2-3-system was adopted, seven years of primary, four years of lower secondary (form 1 -4), two years of upper secondary (form 5-6), and three years of university. This does not include the 'pre-primary' schooling provided to children under the age of six. The principle preoccupation for Ominde’s report was introduce an education system that promoted national unity and inculcated in the learners the desire to serve their nation (Simuyu, 2001).

Criticism of 7-4-2-3 System of Education 7-4-2-3 system of education lacked the capacity and flexibility to respond to the changing aspirations of individual Kenyans and the labour market needs, in terms of new skills, new technologies and the attitude to work (Owino, 1997).

According to Simuyu (2001), the 7-4-2-3 policy was criticized in two major areas (2001):

- i) The policy was being too academic and therefore not suitable for direct employment. Thus the policy lacked orientation to employment.
- ii) The policy encouraged elitist and individualistic attitudes among school leavers, something that was considered incompatible to the African socialist milieu. The Gachathi Report (GoK, 1976, p 33-34) raised the issue of unemployment in relation to 7-4-2-3 policy: The report maintained that “One of the largest problems confronting the country is that of unemployment. The problem is aggravated by the annual outputs of school leavers whose number continue to swell following the enormous expansion of the education.

Among those who made calls for change of educational policy were, the Kenya National Assembly’s Select Committee on unemployment (1970) as cited in Maleche (1976, p.13) Therefore, education, which was regarded by Kenyans as a medium for social mobility and national economic development failed to deliver as the number of unemployed school leavers continued to grow system in the first years of independence. ILO also called for a change to the education system in order to help reduce unemployment. The change consisted of increasing the

technical and vocational aspects of the curriculum. The move by ILO towards vocationalising the education system won support from the World Bank.

(b) 8-4-4 Education System

The 8-4-4 system of education was introduced in January 1985, following the Mackay report of 1982. King and McGrath (2002) claims that the 8-4-4 policy arose out of the concerns that a basic academic education might lack the necessary content to promote widespread sustainable (self) employment". Therefore the 8-4-4 policy emanated from the assumption that it would equip pupils with employable skills thereby enabling school dropouts at all levels to be either self-employed or secure employment in the informal sector. King & McGrath, 2002 observes that the new policy would orient youths towards self-employment. The system strongly emphasizes attitudinal and skills preparations for the world of work and especially self-employment. According to Sifuna (1990), there are three events that led to implementation of the 8-4-4 system: the 1966 conference on education at Kericho in Kenya, which stressed the need for integrating rural development; the International Labor Organization mission report entitled "Employment, Incomes and Equality: A Strategy for Increasing Productive Employment of 1972;" and the recommendation of the National Committee on Educational Objectives and Policies of 1975.

In 1979 the Ministry of Education was changed to the Ministry of Basic Education with an introductory nine-year basic education system program. The rationale was that the previous program was too short and not rigorous enough to give graduates enough practical education. It also recommended that the first six years of primary were to concentrate on numeric and literacy skills and the last two years on basic education with practical orientation. This represented a shift from a focus on enrollment to restructuring the program as a means to cater to the influx of unemployed.

(c) The structure of 8-4-4 System

The 8-4-4 System is divided into primary, secondary and university excluding the pre-school years. This sections gives a brief description of the structure.

a) Primary Education

It starts at age of six years and consists of eight years of schooling, out of which first three may be in the mother tongue, while English is invariably the language of instruction from standards 4 to 8. Most schools are public and run on the basis of harambee system, but the number of private schools, though small as yet, is rising very fast. Most schools are co-ed though a few only-girls schools also exist. The primary education ends with an exam for Kenya Certificate of primary education. Enrolment levels in primary education are reasonable by third world standards - partly a result of the making it free.

b) Secondary School

Secondary school consists of four years. Majority of secondary schools are run on the Harambee system. About one fourth are Government schools, but only the most meritorious are able to enter them. The private schools charge high fees and many offer British O-levels, followed by A-levels or the International Baccalaureate. The enrolment in secondary school is far less compared to the primary school, as it is not free, though some plans to do that are being prepared.

c) University and College Education

Since the establishment of University of Nairobi in 1970, the first public University in Kenya, seven more general public Universities and a number private Universities have been introduced, most of them run by religious organizations.

Debate over 8-4-4 System

However, although the 8-4-4 policy has been described a major educational reform in the history of Kenya's education system since independence in 1963, there are views that it is a great devastation

The 8-4-4 system has been the subject of national debate since its inception. It has been criticized for being broad, expensive and burdensome to pupils and parents".

Since its inception, 8-4-4 system was changed to be more accommodative. Several commission have been formed with the view of improving the system: The Wanjigi Report (GoK, 1983); The Kamunge Report (GoK, 1985), The Mungai Report (GoK, 1995); The Ndegwa Report (GoK, 1991) and The Koech Report (GoK, 1999). The Kamunge Report (GoK, 1985) on Education and training which recommended the reduction of examination subjects under the 8-4-4, was implemented in secondary schools but ignored at the primary schools level; The president (Moi) rejected the Koech Report.

Those commissions' reports, that were implemented, made profound contributions to certain aspects of the education system in Kenya, for instance, the Gachathi and Mackay Reports contributed to the change of education policy from 7-4-2-3 to 8-4-4. The Gathathi report noted with great concern the rising rate of unemployment among school leavers and recommended the restructuring of the education system curriculum in order to have more streams of science, mathematics and, technical and vocational subjects (Republic of Kenya, 1976). The Mackay Report influenced the extension of primary education from seven to eight years and thus influenced the restructuring of the education system from 7-4-2-3 to 8-4-4 and hence the introduction of technical and vocational i.e. practical subjects in primary schools. They both commission reports emphasized the integration of technical and vocational education in the entire Kenyan educational system right from primary to tertiary levels (Simiyu, 2001).

(d) 2-6-3-3-3 SYSTEM OF EDUCATION

The new curriculum has been touted as the ultimate remedy to limitations identified in the 8-4-4 system because it is entirely skills-based.

It was piloted this year between May and September across 470 schools – 10 in each county. Learners will not sit exams but they will be evaluated through Continuous Assessment Tests (CATs) on the skills acquired as opposed to cramming for exams as has been the case.

Experts are of the view that it will enable learners to develop beyond academics and also focus on how best they can use their specific talents to make a living. The needs of special needs children have also been incorporated in the curriculum which will integrate ICT at all levels of education.

The 2-6-3-3-3 model places emphasis on formative years of learning where learners will spend a total of eight years - 2 in pre-primary and six in primary. Subjects to be taught in lower primary are Kiswahili, English, literacy, and mother tongue as well as science, social studies and agricultural activities. Upper primary will include Kiswahili, English, Mathematics, Home Science, Agriculture, Science and Technology, Creative Arts (art, craft and music), Moral and Life Skills and Physical and Health Education. Others are social studies (citizenship, geography and history) with an option of a foreign language (French, German, Chinese and Arabic).

Junior Secondary (grades 7, 8 and 9) and Senior Secondary Education (grades 10,11 and 12) will each take three years.

Twelve core subjects will be taught at junior secondary - Mathematics, Kiswahili, English, life skills, health education, social studies, integrated science, business studies, religious education, agriculture, life skills, sports and physical education.)

Learners at this level will also be required to take a minimum of one and a maximum of two optional subjects that suit their career choices, personalities, abilities and interests.

Home science, foreign languages, Kenyan sign language, indigenous languages, visual arts, performing arts, Arabic and computer science will be optional at junior secondary.

Learners at senior secondary (ages 15-17) will focus on three areas of specialization depending on their skills, talents and interests.

These are arts and sports science, social sciences and Science, Technology, Engineering and Mathematics (STEM). Graduates from this level will have the option to join vocational training centres or pursue university education for three years.

6.2 CURRICULUM DEVELOPMENT PROCESS

A process is the flow of an activity from the initial stage to the end stage without repetition. Curriculum process undergoes systematic stages of development which are well ordered one building onto the other in a sequential manner.

6.2.1 Purpose of education and curriculum

The main purpose of education is to socialize an individual among peer groups. Without a community and group, life and socialization of an individual would have very little meaning. From the beginning of human society, the main objective of education has been that of transmitting to the child the accumulated experiences of his people and their culture as well as of training him to fit into the membership of the group. It was through this background that the society has managed to survive and gain increasing knowledge over all forms of human institutions; governance, rituals, survival etc.

Curriculum on the other hand was prompted during the World wars, especially the 1st World War (1918) when Franklin Bobbat published a book called “the Curriculum “ to provide a course of operations on the war and its derived out comes. This period was marked by industrial and technological development. There was need to design programmes of activity or events which had profound effect on the social, economic and political life of people. Curriculum as seen now is to create a situation of social activity- based on development of the people.

6.2.2 Functions of Education and Curriculum

Education provides knowledge desired attitudes and practical skills to individuals.

Education matures and individual for a sense of responsibility, accountability and service to the society.

Education trains an individual for self-discipline, self appraisal and sense of humour for self-reliance.

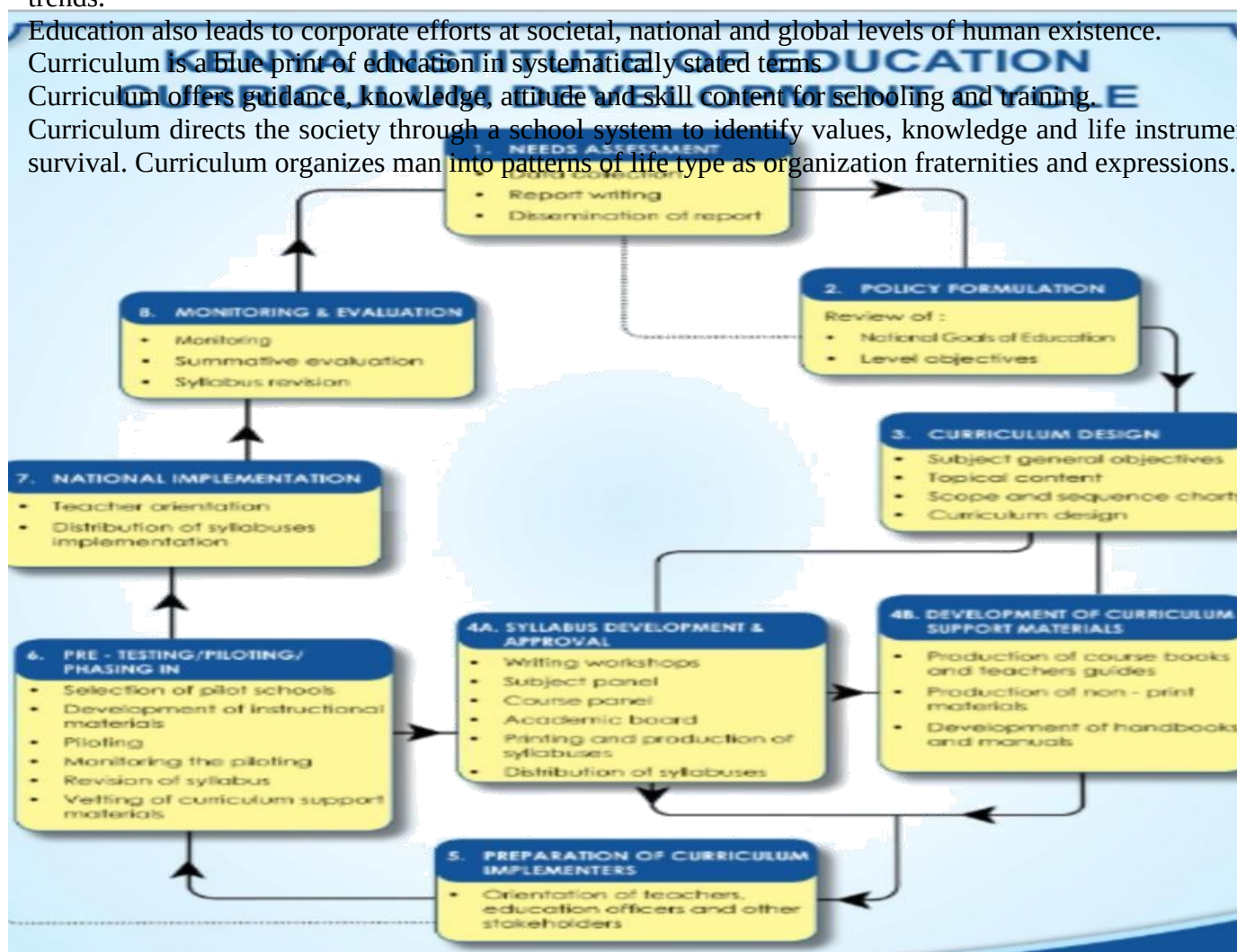
Education provides ways and means of recognizing the environment and its ecological content for use of survival trends.

Education also leads to corporate efforts at societal, national and global levels of human existence.

Curriculum is a blue print of education in systematically stated terms

Curriculum offers guidance, knowledge, attitude and skill content for schooling and training.

Curriculum directs the society through a school system to identify values, knowledge and life instruments for survival. Curriculum organizes man into patterns of life type as organization fraternities and expressions.



The above illustration shows the curriculum cycle used in Kenya by the KICD TAXONOMIES OF LEARNING Benjamin Samuel Bloom (February 21, 1913 – September 13, 1999) was an American educational psychologist who examined and then restructured the way teaching should be approached, to maximise learners' performance. His book, *The Taxonomy of Educational Objectives: The Classification of Educational Goals* (1956), set out a series of learning objectives that became known as Bloom's taxonomy. It continues to impact the way educational curricula are structured to this day. Bloom's taxonomy divided learning into three psychological domains –

i) Cognitive (processing information)

ii) Affective (attitudes and feelings)

iii) Psychomotor (physical skills).

1. The six cognitive levels of Bloom's taxonomy (Revised)

Within those domains, his taxonomy progressed from Lower Order Thinking Skills (LOTS) to Higher Order Thinking Skills (HOTS), through six levels: knowledge, comprehension, application, analysis, synthesis and then evaluation. Essentially, Bloom's model evolved education from being a case of learners just memorising information they had been taught to first remembering it; then understanding it; then applying it (in exercises); then analysing it and, finally, being able to evaluate it at a complex level. Bloom's taxonomy was updated by former students of his in 2001, and the updated version is now widely used in all spheres of education. I go into a deep-dive on Bloom's taxonomy on this page, for those who want some more information. In 2008, a variation of Bloom's was created for use specifically in the modern, digitally-enabled classroom (see Bloom's Digital Taxonomy: A reference guide for teachers).

Bloom's revised taxonomy: what is it, and how is it used?

Bloom's taxonomy is a multi-layered model for encouraging learning by progressing through six levels of increasing complexity. The six levels are:

Cognitive levels Explanation Creating Putting elements together to form a new, coherent and functional whole.

Evaluating Making judgements based on criteria and standards by checking and critiquing. Analysing Breaking information into constituent parts and establishing how they relate to one another. Applying Implementing what was learnt in the Remembering and Understanding phases. Understanding Constructing meaning by interpreting and summarizing information. Remembering Recalling information from long-term memory.

Bloom's taxonomy encourages learners to engage with knowledge at a deeper and more interactive level, working with what they are learning in the real-world sense, rather than just passively taking information on board.

6.2.3 Bloom's taxonomy as an interdependent learning process

While Bloom's taxonomy arranges learning into six cognitive levels in order of hierarchy and complexity, it also sees each level as being interdependent on the other levels, with all levels contributing to the final, holistic learning outcome. This is why, while Bloom's taxonomy is traditionally rendered as a pyramid or even an inverted pyramid, it is also rendered in other forms that place more emphasis on the interdependence of the levels:

Bloom's Taxonomy

The interdependence of Bloom's different learning levels can be articulated through logic:

Before we can understand a concept, we must be able to remember it.

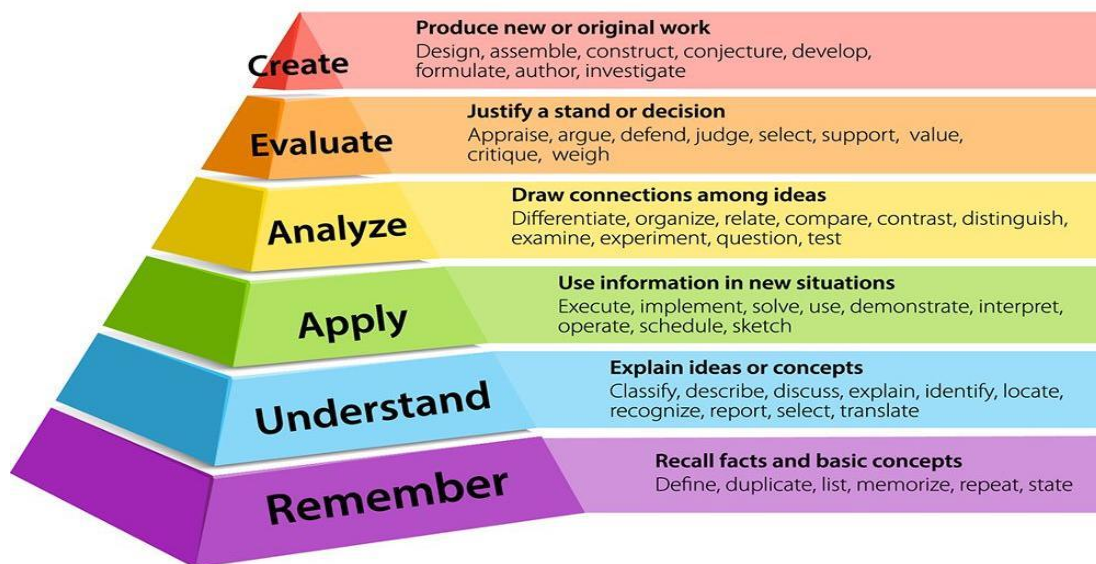
Before we can apply the concept, we must be able to understand it.

Before we analyse it, we must be able to apply it.

Before we can evaluate its impact, we must have analysed it. Before we can create something based on the concept, we must have remembered, understood, applied, analysed and evaluated the concept.

The non-pyramid forms for expressing Bloom's taxonomy indicate that learning may not always progress linearly up through the six levels. Rather, learners might move back and forth between different levels depending on the learning situation, or they might spend more time in some levels of the taxonomy than in others. The human brain, by nature, tends to behave in a lateral manner rather than a set linear manner.

Bloom's Taxonomy



Level one – Remembering

Verbs: Define, Label, List, Name, Recite, Repeat.

Remembering is the act of retrieving knowledge and can be used to produce things like definitions or lists. It is the lowest of the taxonomic levels but is essential for the learning process because learners need to have knowledge in place before they can engage with it at higher cognitive levels. Examples of Remembering include reciting the times' table, naming different parts of the human anatomy, answering true or false questions, recalling critical events on a historical timeline or even naming the six cognitive levels of Bloom's taxonomy. Remembering requires no understanding of the knowledge, only to have it accurately and thoroughly in mind.

Level two – Understanding

Verbs: Explain, Discuss, Examine, Generalize, Group, Order, Paraphrase, Rephrase, Sort.

The next level in the taxonomic structure is Understanding, which is defined as the construction of meaning and the building of relationships. Understanding can be demonstrated by, for example, grouping a list of different animals into the right categories (marine, avian, terrestrial, amphibian); explaining how one event on a historical

timeline impacted on another, discussing the moral of a story or being able to explain why Bloom's has different cognitive levels and the logic behind their hierarchy.

Level three – Applying

Verbs: Compute, Demonstrate, Direct, Dramatise, Formulate, Make, Present.

The third level in Bloom's taxonomy, Applying, marks a fundamental shift from the pre-Bloom's learning era because it involves remembering what has been learnt, having a good understanding of the knowledge, and then being able to apply it to real-world exercises, challenges or situations. Examples of Applying in action could include making repairs to a computer's components; role-playing mediation and conflict resolution between two warring countries; demonstrating the steps that take place in HIV counselling and testing, or presenting a talk on solutions to climate change.

Level four – Analysing

Verbs: Simplify, Criticise, Distinguish, Explain, Illustrate, Inspect, Question.

Analysing is the cognitive level where a learner can take the knowledge they have remembered, understood and applied, then delve into that knowledge to make associations, discernments or comparisons. Analysing would mean a learner can take complex information, and simplify it or summarise it. As other examples, a learner would be able to give reasons why one historic military campaign failed and why another succeeded, or critically examine aspects of Bloom's original taxonomy and explain why his students later updated them.

Level five – Evaluating

Verbs: Decide, Forecast, Judge, Prioritise, Revise, Value, Weigh.

The fifth level in Bloom's Digital Taxonomy is evaluation. This level requires the learner to make criteria-based judgements through the processes of critiquing and checking. Evaluating could involve reading a book and writing a review on its merits; looking at the Universal Declaration of Human Rights and deciding which rights are more or less relevant to a given country; suggesting ways to introduce digital technology into the classroom environment, or making an informed judgement in a role play of court case proceedings.

Level six – Creating

Verbs: Construct, Write, Develop, Design, Invent, Originate, Set up.

The final taxonomic level is concerned with taking various elements and creating a new, coherent product. This level draws on all of the other levels, with the learner remembering, understanding and applying knowledge; analysing and evaluating outcomes and processes, and then constructing the end product, which may be either physical or conceptual. For example, designing and building a house out of wooden segments, and designing a 3D model of a house on a computer would both be examples of Creating. Another example would be a Learner taking the knowledge of Bloom's taxonomy which they have remembered, understood, applied, analysed and evaluated, and creating a brand new model for the tiers of cognitive thinking and learning.

Original Blooms taxonomy

Bloom's Taxonomy's 6 Levels of Learning



2. Affective domain

This domain deals with developing learners feelings and emotions, especially in modifying their attitudes towards things and persons. It also deals with interests, appreciations and modes of adjustment (Ndubisi 1985). Some level of cognitive is involved in development of feelings emotions and attitudes. For example, learners can only develop what they know or perceive, conceive and think. However, unlike the cognitive domain, the affective domain has five levels which comprise the following:

1. Receiving
2. Responding
3. Valuing
4. Organisation
5. Characterisation by a value or value complex

The affective domain was developed by a man called David Krathwohl. The appropriate way to describe it is “Taxonomy of Educational Objectives in the Affective Domain”. Now let me explain to you what each level means:

Receiving: This refers to willingness to attend to or listen to a given stimuli. For example, before one can develop a feeling of or emotion of courage, one may have to be willing to watch a movie or listen to a story of the courageous exploit of someone else. This is the first level in the inculcation of a spirit of courage in a learner;

Responding: This refers to the response that comes from the learner as a result of attending to or receiving stimuli presented. As in the example given, the learner could respond to the story in various ways e.g. staying through the story while it is being told, etc.

Valuing: This refers to the acceptance of the value being taught as ‘courage’, ‘honesty’, by the learner. It is an internalizing process involving accepting the value to one that is contrary to it, and commitment to live the value. For example, having received and responded positively to, say, a story telling lesson on “honesty is the best policy”, the learner may internalize or imbibe the value, preferring it to any other opposed to it, and be ready/willing to commit himself to living the value of honesty;

Organising/Organisation: This is the bringing together of different values and organising them into a value system which eventually leads to character formation. In other words, this level is important in that the learner is able to put together different values of honesty, perseverance, handwork and organize them into a value system that may result in producing in him/her, the character expected of a good citizen;

5. Characterisation by a value or value Complex: This is the last stage and the final or end result of what learners is expected to become as a result of exposure to the learning material in the affective domain. It is expected at this stage that the learner’s behaviour would have been modified or his/her attitude changed for the better, and that the new behaviour learnt becomes a permanent feature or aspect of his/her character.

3. Psychomotor Domain

This domain deals with physical skills training. It involves getting learner to coordinate the hand, eye and ear in such a skillful way as to result in the performance of certain motor skills. Such physical skills may include playing games such as football, tennis etc. Great dribblers in football such as the world renowned Diego Maradona or Nigeria’s Austin “JayJay” Okocha, or people involved in acrobatics, or a racing cars etc. are people

who have developed great manufacturer/physical dexterities Music performers, drummers, saxophonists etc. combine great physical/manual dexterity with aesthetics to produce great music. All these belong to the psychomotor domain of educational objectives, and as you can see from the illustrations, it is a performance objective but combines elements of both the cognitive and affective processes. In other words, both the mental as well as emotional (feelings) processes are involved in psychomotor objectives. For example, the great football dribblers use their intelligence, combined with their love for football in learning various “dribbling” skills, which are put to use on the field of play. This domain was developed by Anita Harrow in 1972. Like the cognitive domain, the psychomotor domain of educational objectives has six levels. These six levels are:

1. Reflex movement
2. Basic fundamental movements
3. Perceptual abilities
4. Physical abilities
5. Skilled movements
6. Non-discursive communication These are briefly explained below:

Reflex Movements: This pertains to movements that are as a result of biological reflex actions, e.g. a person automatically closes his eyes when an insect is flying towards his eyes. This is the lowest level of movement controlled by biological instincts;

Basic Fundamental movements: Performing complicated dance steps, for example, involve certain basic movements which must be learnt first. To play tennis well, involved a combination of various movements which cannot be learnt all at once, but step by step from the basic/fundamental component parts of the complex movements;

Perceptual Abilities: This involves the combination of perception, a mental process, in learning physical skills;

Physical Abilities: To perform physical abilities, physical skills training are necessary, e.g. the development of muscles through certain physical fitness exercise;)

Skilled Movements: This involves learning or combining various basic movements into a pattern that becomes a skill.

Non-Discursive Communication: The body can be used to communicate, i.e. they include movements which are learnt for communication purposes using the body. The body has a language, the body can speak. Actors learn how to use their bodies to communicate feelings or other messages.

7.0 Curriculum monitoring and Evaluation

7.1 Curriculum monitoring

7.1.1 Concepts Related to monitoring of curriculum.

Curriculum monitoring - refers to the process of gathering information for evaluating the effectiveness of the curriculum and ensuring that the intended, implemented and attained curricula are aligned. This process typically focuses on such issues as relevance, consistency, practicality, effectiveness, scaling-up and sustainability, as well as whether learners are achieving the expected learning outcomes. It measures the extent to which the curriculum is commensurate with the diverse needs of all learners.

Scaling - up.

Refers to expanding or replicating innovative pilot or small- scale projects to reach more people and/or broaden the effectiveness of an intervention.

Relevance.

Refers to applicability and appropriateness of a curriculum to the needs, interest, aspirations and expectations of learners and society in general.

Consistency

The concept of consistency means that accounting methods once adopted must be applied consistently in future. It ensures that standards are aligned across grade levels to eliminate instructional gaps between the years. Also same methods and techniques must be used for similar situations.

Practicality.

In assessment practicality means that the test is easy to design, easy to administer and easy to score. No matter how valid or reliable a test is it has to be practical to make, economical to delivering quality and not excessively expensive.

Effectiveness

Curriculum effectiveness is defined as the existent to which a curricular program and its implementation produce positive and curricularly valid outcomes for students.

Sustainability

Sustainable curriculum would be transformative rather than transmissive as its goal is to “equip all people with the knowledge, skills and understanding necessary to make decisions based upon their full environmental, social, cultural and economic implications” rather than transmissive with a goal to provide students with knowledge. It would not be specialised, content-driven units of study on sustainability but integrated across courses and units. Sustainability is a paradigm that requires us all (educators and students) to examine our own values, hidden assumptions, motivations, beliefs and actions. We reflect on how we live and work and how this impacts the environment, economy and others (social and cultural aspects) on local, regional, national and global levels. Sustainable curriculum is based on seven principles.

- transformation and change (not just knowledge),
- education for all and lifelong learning,
- systems thinking (highlighting connections between environmental, economic, social and political systems),
- envisioning a better future (problem solving, critical thinking and reflection,
- participation (engaging groups and individuals.)
- partnerships for change. It would equip students with the skills to learn and develop their own skills to the changing world. It would be innovative. Students would learn to question, think critically from various viewpoints and make well-informed ethical decisions.

Sustainability Education is often referred to as Education for Sustainable Development. Education for Sustainable Development allows every human being to acquire the knowledge, skills, attitudes and values necessary to shape a sustainable future.

7.1.2 Principles of monitoring of curriculum.

- Monitoring and evaluation should refer back to existing baseline data or begin with a baseline study.
- Monitoring and evaluation are closely linked to and should reflect the methodology of the initial project appraisal.
- Monitoring should be linked to the specific programme objectives that were defined through the appraisal and programme planning process.
- Monitoring information should be used in decision-making.
- Monitoring and evaluation should respect principles of participation and involve all programme stakeholders.
- Monitoring and evaluation should respect and protect the rights, welfare and confidentiality of all those involved in the programme.

7.1.3 Process of curriculum monitoring.

Refers to the process of gathering information for evaluating the effectiveness of the curriculum and ensuring that the intended, implemented and attained curricula are aligned. This process typically focuses on such issues as relevance, consistency, practicality, effectiveness, scaling - up and sustainability, as well as whether learners

are achieving the expected learning outcomes. It measures the extent to which the curriculum is commensurate with the diverse needs of all learners.

7.1.4 Purpose of Curriculum Monitoring

Monitoring is important to:

- Determining if the curriculum is still relevant or effective.
- Know if the curriculum is working or not.
- Provide decision on what aspects have to be retained improved or modified.
- Help develop new targets, programs and strategies.
- Stay on track and improve project efficiency and accountability.
- School monitoring and evaluation help in providing a consolidated source of information showing the progress of the school.
- Monitoring and evaluation give a basis for both questioning and testing assumptions. .
- help schools achieve their goal.
- It shows the mistakes and creates paths for both learning and improving.
- Through monitoring and evaluation in schools, good leadership is put into practice and accountability which leads to school improvement. This also helps in figuring out whether the teaching method in the school is delivering to the desired educational results. Through this, the school can be accountable to the stakeholders.
- Monitoring and evaluation systems have enhanced the performance of both the teachers and the students. The teachers can do an assessment and the behavior of the student to identify the areas where the student is failing. It is by that; teachers can align their teaching skills accordingly to improve student performance.
- Monitoring and evaluation help in planning on future of the school.

7.1.5 Importance of setting objectives in curriculum design.

- To help in deciding whether to accept or reject a programme.
- To ascertain the need for the revision of the course content.
- To help in future development of the curriculum material for continuous improvement.
- To improve methods of teaching and instructional techniques.

7.1.6 Aspects of curriculum that should be monitored.

Objectives .

Objectives are statements that describe the end-points or desired outcomes of the curriculum, a unit, a lesson plan, or learning activity. They specify and describe curriculum outcomes in more specific terms than goals or aims do. Objectives are a key component of the curriculum design process, as they help teachers determine learning outcomes (what they should teach and what students should learn), methods of instruction, and objective ways to measure student progress and performance.

Aims are general statements that provide direction or intent to educational action. Aims are usually written in amorphous terms using words like: learn, know, etc. Curriculum goals are general, broad statements that lead towards long-term outcomes.

Content.

Curriculum content simply means the totality of what is to be taught in a school system. The content component of teaching learning situation refers to the important facts, principles and concepts to be taught.

Curriculum content is another main lever of education quality. The knowledge, skills and attitudes imparted by learning areas/subjects, cross-cutting approaches and extra-curricular activities is a main source of systematic and comprehensive learning.

Principles of curriculum content

- Be broad and balanced.
- Be suitable for learners of different ages, abilities and aptitudes.
- Provide for appropriate learner progression.

- Cover every statement of what matters.

Criteria for the Selection of Subject-Matter or Content of the Curriculum

- Self-sufficiency.
- Significance.
- Validity.
- Interest.
- Utility.
- Learnability.
- Feasibility.

Comprehensiveness,

Comprehensive curriculum is a written plan that guides the design of children's goals for learning and development, the experiences children will have to achieve those goals and the way in which adults, both staff and families, will support children's learning to achieve school success.

What makes a curriculum comprehensive?

A comprehensive curriculum centers around four main ideas:

- It is student-centered.
- Values students as individuals.
- It is based on an understanding of widely accepted methods of how students learn.
- Encourages teachers to use methods that promote concepts of learning.

Depth.

What is breadth and depth in curriculum ?

Breadth of learning refers to the full span of knowledge of a subject. Depth of learning refers to the extent to which specific topics are focused upon, amplified and explored. Developing Curriculum Depth' is an informative, research-based course explaining the importance of and ways to deliver a broad, deep, rich curriculum to effectively develop children's knowledge and understanding.

Timeliness,

Curriculum Timelines are broad outlines that give our community information regarding units of study that are included at each grade level and course for year long and term long courses. The timelines are approximate and are intended to provide a general overview of the curriculum.

Ensuring timeliness.

Timely return of feedback is one of the biggest problems the academics have in relation to feedback and assessment and also a great source of student dissatisfaction.

Feedback should be returned in a timely manner and the extent, nature and timing of feedback for each task must be made clear to students in advance so that they can make the best use of the feedback.

Why is feedback important?

- The purpose of feedback is to provide students with guidance as to how to improve their work. If that guidance is not received in time, the feedback cannot be used, resulting in lack of knowledge as to where that improvement should be made.
- Feedback impacts greatly on student satisfaction.

Quality.

A principal objective of a quality curriculum is to enable students to acquire and develop the knowledge, skills and values, and the associated capabilities and competencies to lead meaningful and productive lives.

Curriculum can be organized into three major components: objectives, content or subject matter, and learning experiences.

7.1.7 Tools for monitoring of curriculum.

These include;

- Project records,
- Formal surveys,
- Interviews,
- Direct observation,
- Focus group discussions.

7.1.8 Challenges of Monitoring and Evaluation

- Monitoring and evaluation methods are times consuming to design and implement.
- They require trained staff to ensure success.
- Lack of experience;
- Limited financial and staff resources;
- Gaps in technical knowledge with regard to defining performance indicators, the retrieval, collection, preparation and interpretation of data; and
- Inefficient monitoring and evaluation practices.

Importance of observing ethical issues in curriculum monitoring in cbc.

When most people think of ethics (or morals), they think of rules for distinguishing between right and wrong, such as the Golden Rule ("Do unto others as you would have them do unto you"), a code of professional conduct like the Hippocratic Oath ("First of all, do no harm"), a religious creed like the Ten Commandments ("Thou Shalt not kill..."), or a wise aphorisms like the sayings of Confucius. This is the most common way of defining "ethics"; norms for conduct that distinguish between acceptable and unacceptable behavior.

One may also define ethics as a method, procedure, or perspective for deciding how to act and for analyzing complex problems and issues

Most people learn ethical norms at home, at school, in church, or in other social settings. Although most people acquire their sense of right and wrong during childhood, moral development occurs throughout life and human beings pass through different stages of growth as they mature.

There are several reasons why it is important to adhere to ethical norms in research.

- Norms promote the aims of research, such as knowledge, truth, and avoidance of error. For example, prohibitions against fabricating, falsifying, or misrepresenting research data promote the truth and minimize error.
- Ethical standards promote the values that are essential to collaborative work, such as trust, accountability, mutual respect and fairness. For example, many ethical norms in research, such as guidelines for authorship, copyright and patenting policies, data sharing policies, and confidentiality rules in peer review, are designed to protect intellectual property interests while encouraging collaboration. Most researchers want to receive credit for their contributions and do not want to have their ideas stolen or disclosed prematurely.
- Many of the ethical norms help to ensure that researchers can be held accountable to the public. For instance, federal policies on research misconduct, conflicts of interest, the human subjects protections and animal care and use are necessary in order to make sure that researchers who are funded by public money can be held accountable to the public.
- Ethical norms in research also help to build public support for research. People are more likely to fund a research project if they can trust the quality and integrity of research.
- Norms of research promote a variety of other important moral and social values, such as social responsibility, human rights, animal welfare, compliance with the law and public health and safety. Ethical lapses in research can significantly harm human and animal subjects, students, and the public. For example, a researcher who fabricates data in a clinical trial may harm or even kill patients, and a researcher who fails to abide by regulations and guidelines relating to radiation or biological safety may jeopardize his health and safety or the health and safety of staff and students.

Examples of ethical principles that various codes address.

Honesty.

Strive for honesty in all scientific communications. Do not fabricate, falsify, or misrepresent data. Do not deceive colleagues, research sponsors, or the public.

Objectivity.

Strive to avoid bias in experimental design, data analysis, data interpretation, peer review, personnel decisions, grant writing, expert testimony and other aspects of research where objectivity is expected or required. Avoid or minimize bias or self-deception. Disclose personal or financial interests that may affect research.

Integrity.

Keep your promises and agreements; act with sincerity; strive for consistency of thought and action. Avoid careless errors and negligence; carefully and critically examine your own work and the work of your peers. Keep good records of research activities, such as data collection, research design, and correspondence with agencies or journals.

Openness.

Be open to criticism and new ideas.

Transparency.

Disclose methods, materials, assumptions, analyses, and other information needed to evaluate your research.

Accountability.

Take responsibility for your part in research and be prepared to give an account (i.e. an explanation or justification) of what you did on a research project and why.

Intellectual Property.

Honor patents, copyrights, and other forms of intellectual property. Do not use unpublished data, methods, or results without permission. Give proper acknowledgement or credit for all contributions to research. Never plagiarize.

Confidentiality.

Protect confidential communications, such as papers or grants submitted for publication, personnel records, trade or military secrets, and patient records.

Responsible Publication.

Publish in order to advance research and scholarship, not to advance just your own career. Avoid wasteful and duplicative publication.

Responsible Mentoring.

Help to educate, mentor, and advise students. Promote their welfare and allow them to make their own decisions.

Respect for Colleagues.

Respect your colleagues and treat them fairly.

Social Responsibility.

Strive to promote social good and prevent or mitigate social harms through research, public education, and advocacy.

Non-Discrimination.

Avoid discrimination against colleagues or students on the basis of sex, race, ethnicity, or other factors not related to scientific competence and integrity.

Competence.

Maintain and improve your own professional competence and expertise through lifelong education and learning; take steps to promote competence in science as a whole.

Legality.

Know and obey relevant laws and institutional and governmental policies.

7.2 Curriculum Evaluation

Let us start this lecture by defining the term evaluation. We shall then discuss reasons why evaluation is undertaken. Phases of curriculum evaluation will be examined briefly since they are covered in details in another lecture. You are advised to go through this lecture noting carefully all areas which will enable you to follow and grasp a detailed discussion on evaluation.

Objectives

At the end of the lecture you should be able to;

- Define the word evaluation
- Describe the purpose of evaluation
- Distinguish the differences between measurement, testing, assessment and evaluation.

Evaluation in the content of education is a process used to obtain information from testing, from direct observation of behavior, from essays and from other devices to assess a student's overall progress towards some predetermined goals or subjects. It includes both a qualitative and quantitative description and involves a value judgment of overall student behavior for decision making.

Evaluation and measurement are not the same, although evaluation involves measurement, if we assess a student's knowledge and understanding in a subject by means of an objective or essay type test, this is measurement. If a teacher puts a value on the student's work, talents, attitudes and other characteristics of behavior that is evaluation. Evaluation should in part involve testing that is non-subjective on the part of the teacher; otherwise it is likely to be unreliable.

Assessment is used interchangeably with evaluation. Testing is the process of using an instrument or test to measure achievement. Measurement and testing are thus ways of gathering evaluation and assessment data.

Purposes of Curriculum Evaluation

Evaluation has many purposes; it can be used in the following ways.

(i) Evaluation as a basis for school marks or grades by teachers

Educational systems require that teachers occasionally submit marks or grades on students. These marks or grades can be arrived at through formal examinations, regular tests, assignments, laboratory reports, observational information or combination of these.

(ii) Evaluation as a Means of Informing Parents

Parents of school children have a right to know how their children are progressing in school.

Students' evaluation is the most important way of providing them with this information.

(iii) Evaluation for promotion to higher class

Student evaluation is sometimes used to determine whether a student has made enough progress to be promoted to a higher class or form in the school.

(iv) Evaluation for Student Motivation

Success in tests and examinations as well as sports and other school activities can give great encouragement to students. Similarly, failure to do well can make students work harder or strive to do better. Teachers should try to give their students feedback on the evaluation of all aspects of their learning and behavior so that both who do well and those who do not will be motivated to improve on their performance.

(v) Evaluation for Guidance and Counseling Purposes

All students need to be advised to help them solve their own personal problems, whether academic or emotional. The two types of problems are indeed often connected.

Successful students tend to enjoy school more than those who are not successful. It is generally those who appear to be failures as indicated by the evaluation of the teaching staff, who need the most attention and it is to them that the class teacher must direct herself.

(vi) Evaluation to Assess the Effectiveness of the Teaching Strategy

If a teacher does not in some way assess the students' improved knowledge, understanding and higher cognitive skills as well as their attitudes and psychomotor abilities will not be able to evaluate the success or otherwise of

the teaching strategy she has employed. A higher failure rate in a course is more often due to poor teaching than to the lack of intelligence of the students.

(vii) Evaluation for Employment Purposes

Not all students who pass, through post-primary schools will proceed to University or other Institutions of higher learning. Some students may decide to join a company or business. Employers normally require information on potential employees with reference to academic ability, attitude to work, moral character personality and so on. It is therefore necessary for the teachers to evaluate nearly all aspects of the student's performance while they attend the school.

• Curriculum Evaluation

The primary purpose of curriculum evaluation is of course to determine whether the curriculum goals and objectives are being carried out. These goals and objectives are to be evaluated in the first place to determine if they are the right kind of objective. It also determines whether the curriculum is functioning while in operation and using the best materials and the best methods.

Curriculum evaluation also determines whether the products of our schools are successful in higher education and in jobs, whether they can function in daily life and contribute to our society. Curriculum evaluation also determines whether educational program is cost effective, that is to say whether the people are getting the most of their money.

Phases of Evaluation

There are three phases of evaluation which every teacher needs to know

- Pre-assessment
- Formative Evaluation
- Summative valuation

These terms are technical words to differentiate evaluation that takes place before instruction (pre-assessment), during instruction (formative) and after instruction (summative). Pre-assessment evaluation is provided before instruction that takes place before instruction to determine the starting point on instructional program It identifies need prerequisite skills and causes of learning difficulties and to place students in learning groups. Formative evaluation consists of those techniques of a formal and informal nature, including testing, that are used during the period of instruction. Progress tests given in the classroom are a good illustration of formative evaluation.

Through formative evaluation teachers may diagnose student difficulties and take remedial action to help them overcome their difficulties before they are confronted with the terminal (summative) evaluation enables teachers to monitor their instruction so that they a may keep it on course. It is also used to provide assessment of curriculum quality. It is conducted during the curriculum development process for the additional purpose of providing information that can be used to form better finished product. Thus formative evaluation takes place at a number of intermediate points during curriculum development process.

Summative evaluation is the assessment that takes place at the end of a course or unit. A final examination (post-test) means used for the summative evaluation of instruction. Its major purpose is to find out whether the students have mastered the preceding instruction. A good teacher utilizes results of summative evaluation to revise his or her program and methodology for subsequent groups.

Role of Evaluation in Curriculum

Evaluation entails not a single study but also a series of sub-studies, performed at the various stages of curriculum development process. The main aim of curriculum evaluation is to collect descriptive information about an educational programme, which is then used to modify and improve the program; to compare the program to other programs, and to make judgments to the worth of the program of project.

Evaluation generally answers the questions regarding: selection, adoption, adoption, support change, innovation and worth of an education program. Formative evaluation helps the curriculum team at the planning stage to formulate a new educational policy, to discover new needs, to formulate new goals, to clarify in their definition

of goals, select appropriate goals from competing goals, determine complementary goals, number of new disciplines to be taught provide empirical evidence on the view of teachers, parents, learners, employers, professional on what requires change or towards the envisaged change.

At the project materials development stage, evaluation helps to translate broad educational goals into curriculum objectives for various subjects, redefine general objectives into operational objectives, determine appropriate syllabus and textbooks structure and content, ensure relevance in content scope, sequence, continuity, integration and balance, identify art is that require in-service for teachers and determine appropriate evaluation procedures and process.

At the project field trial (try-out) stage evaluation helps to determine.

- Adequacy of the materials.
- Needed revisions, cost of production, quantity.
- Nature needed.
- Teaching-learning strategies used.
- Evaluation procedures - instrument required.

Evaluation at the school implementation stage check on arrival of teaching-learning resources in schools and their use by teachers and students

- Whether teachers have been adequately prepared.
- Reaction of teachers, parents and other stakeholders on the new program.
- Whether procedures used meet the standard criteria designated by the curriculum developer.
- What is actually being learned-the operational curriculum.
- What requires change?
- Acceptance of the new program by the beneficiaries.

National examinations cater for comparison of schools, districts, pupils, leaders. Grading selection, placement, certification, school academic records. Evaluation of examinations procedures ensures higher standards of education, detects areas of difficulty; determiners new methods of teaching and evaluation; problems teachers and learners have in interpreting course objects.

Principles of Instructional Evaluation

Instructional Evaluation should be used on the following criteria by which worth is determined.

Consistency with Objectives

Evaluation should be used to measure what is indicated in curriculum objectives of a course programme R. Tyler (1950) observed that educational objectives are the criteria by which materials are selected, content is outlined, instructional procedure are developed and tests and examinations are prepared. Evaluation tells us how successful we have been in this effort. The criteria for evaluation and the result so obtained should be underscored and accepted by all those concerned. There is need to develop in pupils the ability to learn further.

Validity and reliability

Evaluation instruments are valid if they measure what they are supposed to measure. A test in CRE for example, should not be expected to elicit scientific knowledge but religious educational concept. A valid test shall relate to objectives of the specific course and appropriate for the level

Reliability refers to the consistency with which an evaluation instrument measures giving the same score of results two different examiners are able to arrive the same score on the test candidates who have gone through the same learning process. A test item that has several answers yet the test constructor required only one correct answer cannot be said to be reliable. A reliable test will also try to elicit the same abilities, skills from the same sample of students.

Continuity

Curriculum evaluation should be an on-going process in order to provide effective feedback, which will lead to course improvement, it should moreover relate to previous, present and future learning experiences and follow

proper sequencing of the course, from easy to complex items. The evaluation system begins with curriculum decision, which results in the identification of the first goals. It continues throughout the planning process into implementation activities, and cycles back to the planning process.

Instructional evaluation should be a continuous process so that the teacher can adequately and effectively assess each student needs in order to select appropriate resources, develop appropriate learning strategies, judge each student merit, and provide effective feedback and motivation. So each student plan group methods, appropriate activities along specified learning objectives.

Through continuous assessment the teacher consistently and systematically provides the educational experience most suited to the educational needs, interests, readiness and ability of each student.

Balance

Balance means that the curriculum developers have weighed the relative importance they have given to each student need and development tasks. Considerations should be given to all student needs. Evaluation should assess all skills weighted against the time allocated to each. Balance also ensures that the various cognitive skills are equally or reasonably weighted. If the test items dwell on higher or lower cognitive skills only, such a test will lack balance. Hence, there should be a balance in the following cognitive levels of skills: knowledge, comprehensive, application, analysis, synthesis and evaluation. There should be also balance theory and practical skills. All subjects have theoretical and practical aspects.

Comprehensiveness

Education aims at the development of the whole person. Hence, all the objectives of the curriculum programme should be evaluated, namely: the cognitive, effective, psychomotor, spiritual and social relating domains. Evaluation instruments should be designed to yield accurate information concerning personal, social adjustment, physical growth, spiritual growth, habits of work, interests and attitudes; special aptitudes, growth in creative ability, home and community backgrounds must also be available if the school is to do the best job possible in fostering the wholesome growth of learners and preparing them for effective citizenship in a democratic multi-part, multi-racial, multi-ethnic, multi-religious society such as obtains in Kenya. Education should prepare the individual to face the vicissitudes of life with constancy, persistency, insistence and courage.

Individuals play different roles at different educational experience and stages of life; Pre-primary, primary, secondary, university; childhood, adolescence, adult middle age, retirement age and old age. Each stage requires definite knowledge, skills, values and are subject to evaluation by society. It is not the mere acquisition of knowledge that matters but how it is utilized. Modern evaluation attempt to obtain as complete a picture as possible of the individual. The evaluation procedure is comprehensive if they utilize a variety of means and techniques in collection evaluation data.

Cooperation

An evaluation system is depended upon the adequacy of the planning which resulted in the selected or creation of curriculum programs activities, procedures, resources and other elements to be evaluated. Evaluation systems are also dependent upon the utility and integrity of the specific data to be gathered, displayed and interpreted (practical skills, work at primary, secondary and university exams) what criteria should be met to ensure validity and reliability?

An adequate evaluations system at least two levels of cooperation, the first concerns the integrity of the relationships established among the planning, implementations and evaluation phase of program building. Evaluation should be inbuilt in the planning and preparation states of curriculum development. The second concerns the comprehensive involvement of all parties, which have legitimate input or from the programs activities or its evaluation systems.

The determination of what constitutes success or failure requires the cooperative involvement of those who implement and are affected by the program and those who evaluate. There should be cooperation among the KIE curriculum developers, the Kenya National Examinations Council and the classroom teachers. There should be

also cooperation among the psychologists, sociologist, philosopher, religious leaders, professors, trade unionists, curriculum developers, employers and teachers and all other stakeholders.

Evaluation instruments should be functional, practical understood and acceptable by all teachers involved. There should be a closer relationship between the examination and the objectives which school education hopes to achieve, therefore, between the style of the

Measurement and evaluation

Confusion usually arises in the use of the terms measurement, testing and evaluation. Let us define each one of these terms.

(a) Measurement and Testing

i) Measurement is the means of determining the degree of achievement of a particular objective or competency.\

ii) Testing on the other hand, is the use of instruments for measuring achievement.

Measurement and testing are ways and tools of collecting information for evaluation and assessment. These are not only ways of gathering evaluation data. There are others that we shall deal with in the following lectures.

(b) Evaluation is the process of giving value, judgment based on the information gathered through measurement and testing for decision making.

(c) Measurement and evaluation are not the same thing in curriculum development process.

Whatever judgment we make regarding the degree to which learners have achieved curriculum objectives will be valid if they are based on empirical data. We can obtain empirical evidence through measurement.

We use measurement to quantify representations of the degree to which a learner reflects certain traits or behavior. Data obtained through measurement is basically descriptive in nature. It is expressed numerical terms. Elements of value connotations are avoided as much as possible by measurement.

□ Purpose for Evaluation

Evaluation has a lot of purposes for both schools and public. Some of those purposes are identified as follows:

Evaluation as a Basis of Student Records by Teachers

Many educational authorities require that teachers submit marks or grade for their students, at the end of the term or year. The grades or makes are kept as records of students performance marks or grades are arrived at through formal examinations, test, regular assignments, oral or written reports or observations.

Promotion to Higher Grades

Schools use examination and test results to determine whether students have made enough progress to be promoted to the next class or form in the school. In many cases a student whose progress is low, is advised to repeat the same class for a year before he can move on-to the next one. The class teacher in consultation with the head-teacher will talk to the parent of the affected student and convince him to see the advantages of the child to repeat. Fear of failing in the final selection examination is cited as a big reason in many schools

Selection and Certification Purpose

Not all students who pass through the primary school systems move automatically to secondary schools or form one. National examinations are used in Kenya to select those who have passed with enough points to go to form one. The National examinations are similarly used to select students to join form five and University-Examinations results are also as a criterion for certification. Thus students who pass national examinations at various levels are entitled to being issued with certificates. For instance, the (Kenya Certificate of Primary Education (KCPE) and Kenya Certificate of Secondary Education (K.C.S.E).

Employment Purposes

Students who complete primary and post-primary school education are provided with certificates which enable employers to make a choice. Some students may want to join a company or training for a profession in public or private sectors. All employers usually require information on prospective trainees or employees. Such information may include academic attainment in specific subjects, attitude towards work, moral behavior,

personality another related data. In many cases, a student's certificate or school leaving¹ testimonial will be important for training and employment purposes.

Student Motivation

Good performance by students on tests and examinations has been proved by educators to be a motivation to them. Failure to do well in examinations may also create a sense of competition with students. They will work hard to improve performance in future examinations.

It has been suggested that teacher feedback will assist students to work harder.

Guidance and Counseling purposes

Students in all schools need to know their progress in academic performances. The only way they can judge themselves and be able to solve their personal problems is by seeing their performance on class tests or examinations. Successful students tend to enjoy school much more than those who do not perform well. Those who seem to be failures need greater attention by the teachers. The teacher must direct attention to such demanding cases in a class.

Assessment of the effectiveness of teaching strategies

Many students fail examinations due to poor teaching methods. Teachers who do not assess students' acquisition of knowledge, skills and understanding in a course may not be able to judge the effectiveness of the approaches in teaching. Assessment of student will therefore assist the teacher to modify the teaching strategy where necessary.

Purposes of informing parents or guardians

Parents want to know how their children are performing in class. It is their right to know the progress of children. Terminal records of students' performance are the most of informing parents about the performance of their children in school. Some parents may use such records to advise their children or recommend to their teachers what steps should be taken to improve the child's performance.

Exercise

What is curriculum evaluation?

Why should parents/guardians be informed?

Difference between Instructional and Curriculum evaluation

Instructional and curriculum evaluation are not the same. Instructional process may be very effective whereas the curriculum may be out of order. Instructional evaluation may reveal that the students are achieving the instructional objectives well. If we do not evaluate and rely through curriculum evaluation that we shall know if we are following items well but they may not be universally acceptable as being correct. The earth is flat, illness is caused by the evil eyes and bad spirits, all developing human beings are corrupt and all children can be doctors and engineers.

The main purpose that curriculum evaluation is undertaken is to determine whether curriculum goals and objectives are being carried out correctly. The following are other questions we may need to provide answers for through curriculum evaluation:

- ☐ We want to know whether the goals and objectives are the right ones.
- ☐ We should be interested to know if the curriculum is functioning while in operation
- ☐ We want to know whether the material we are using is the right one.
- ☐ We also want to know how our products, (graduates) can function in daily life after school, and whether they are contributing to the development of our society.
- ☐ We want to know whether the programme we have launched is a good deal for our investment.

Exercise

How do we obtain data for evaluating a curriculum?

Gathering Evaluation Data

Sources of data for curriculum evaluation are numerous. For the purposes of our study we shall divide into two categories.

- ☐ Teacher-made-tests and examinations.
- ☐ Other informal devices.

(a) Teacher-Made-Tests and Examinations

Established procedures for collecting data for curriculum evaluation are mostly to do with instructional evaluation. In Kenya, teacher-made-test provides the only information that may be used for determining student progress in school. Tests are predominantly used in school for evaluation because:

- a) They are expected to be objective.
- b) They are economically viable for use.
- c) They are easy to administer.
- d) They provide a norm against which individual achievement can be judged.

Tests have their limitations too, e.g.

- ☐ Tests do not test all objectives
- ☐ Non-cognitive abilities are hardly tested.
- ☐ Teacher-made-tests tend to ignore individual learners differences.

However, tests and examinations do provide data for curriculum evaluation that serves a useful function in judging the quality of the whole curriculum.

Informal evaluation devices serve as an important source of evaluation data. Some of the devices used also come under the instructional evaluation. They may include:

- All students' records in school.
- Classroom observation by teachers.
- Students' projects.
- Essays and classroom exercises and other assignments.

The curriculum document itself will be acceptable to serve as a source of curriculum evaluation. Curriculum documents consist of all the elements indicated below which should be useful for providing information for evaluation:-

- ☐ Statement of purposes
- ☐ Curriculum content
- ☐ Learning activities
- ☐ Evaluation procedures indicate.

Teachers form an inherent source of data for curriculum evaluation. Information needed for curriculum evaluation should be obtained from the teachers who are involved in the implementation of the curriculum in schools. Information from teachers can be obtained by means of:

- i) Interview
- ii) Questionnaires
- iii) Oral or written devices.

Teachers' perception of curriculum content, instructional materials, learning activities, relevance and student performance in general yield valuable information about what goes on in our schools.

Students are an important source of curriculum evaluation data. Many curriculum developers forget the fact that students can provide very useful information on the curriculum. Students will provide information freely when they know what it will not be used to determine individual grades. The information obtained from students should be compared with that from teachers.

Materials used for instructions would serve a useful purpose as a source of evaluation data.

These materials include:

- ☐ Text books for teaching and references.
- ☐ Films, slides, periodicals etc.

Some books for instance, may serve no useful purpose as class texts; others may be out-dated while others may provide undesirable information to the students. They may not assist the teacher in attaining objectives.

Follow-up studies of graduates will yield good information on how effective the objectives in the curriculum were achieved. Studies of graduates have been organized and carried out to determine what the youths who complete primary education in this country do. These studies have revealed that a lot of youths who graduate from primary schools cannot be absorbed in employment among the graduates. Crime among youths and lack of relevant skills would be relevant for making decisions on what changes should be taken by curriculum developers.

Society would be mother-important source of evaluation data. How do we get curriculum evaluation data from society whose population consists of people with diverse social and cultural backgrounds? Information about what goes on in schools could be solicited through various ways:-

- i) Letters to prominent members of our society will give us what we expect.
- ii) Parents visit schools and other local personnel who may express concern in the welfare of schools would be acceptable.

Exercise

Examine the process of Curriculum Evaluation in Kenya.

In Kenya the task of evaluating the curriculum in schools is carried out into phases:-

- ☐ Formative Evaluation Phase
- ☐ Summative Evaluation Phase

In all the two phases different personnel are involved. Let us examine the role of each personnel in curriculum evaluation. The ministry of education science and technology has the following branches of personnel who undertake the evaluation tasks in Kenya.

a) The Inspectorate

- i) Subject inspectors at the headquarters are responsible for every subject that is included in the curriculum and taught in our schools. There is a team of subject specialists who inspect and provide information on the effectiveness or weakness of subject-content; methodology and material used for instruction.
- ii) Subject Inspectors at the headquarters co-ordinate activities pertaining to particular subjects for all schools, i.e. secondary, colleges and primary schools.
- iii) Subject-inspectors at the headquarters are assisted by provincial and district schools inspectors.

(b) Curriculum Development K.I.C.D

- i) The section of research and evaluation is responsible for curriculum evaluation at both formative and summative phases. The work of evaluating curriculum is done by curriculum development experts under the auspices of the section of research and evaluation. Whatever evaluation is undertaken by this group is used to supplement what is done by the inspectorate.
- ii) K.I.C.D has a panel which reviews books for schools. Recommendation of the books reviewed is transmitted to the Ministry of Education Science and Technology. The approved books are either bought by the schools Equipment Scheme or sent to schools or head-teachers may be provided with the list of books to enable them to purchase them. The panel may also recommend some books to be withdrawn from Circulation and use by schools if they may find them to be undesirable. You have seen in your schools lists and circulars from the Ministry of Education, Science and Technology showing new books recommended as teachers references, pupils class text etc. the circulars may also inform the teachers if the new books will be set books for literature of fasihiya Kiswahili, etc.

(c) Role of the Kenya National Examinations Council

Instructional evaluation will reveal to us a lot of what goes on in various classrooms and what actually happens in different schools. End of year examinations (National Examinations.) have been used for many years as a yardstick to measure the achievement of curriculum objectives. The role of the Kenya National Examination Council is to facilitate the setting, moderating, marking and grading of all national examinations for various levels of education in this country. In particular, the council is in charge of primary schools, secondary schools, Teachers colleges, Technical Training Institutions and other relevant public examinations taken in the country.

8.0 Curriculum Change and Innovation

8.1 Define the term Curriculum review

Curriculum Review is a critical examination of academic programs for the purpose of optimizing student learning experiences led collaboratively by academic staff who teach within the program.

8.2 Define the term curriculum reform

1. The process of making changes to the curriculum with the intent of making learning and teaching more meaningful and effective. Curriculum reform can be seen as a process that aims to change the objectives of learning and the way learning takes place.

2. Curriculum change means making the curriculum different in some way, to give it a new position or direction. This often means alteration to its philosophy by way of its aims and objectives, reviewing the content included, revising its methods, and re-thinking its evaluator procedures.

Curriculum change is a process not an event which; requires time, energy and resources

Curriculum Innovation

Curriculum innovation is defined as deliberate actions to improve a learning environment by adapting a method of presenting material to students that involves human interaction, hands-on activities and student feedback,.

8.3 Sources of Change and Innovation.

- National imposition.
- Societal needs.
- Stake- holder view
- Due to development of technology. e.g Adaption on the use of computers in education.and using online technology in lessons.

Forms of Changes;

- Substitution
- Alteration
- Addition
- Restructuring

Substitution

e.g new textbooks, new equipment, replacement of teachers and administrators.

Alteration.

Change in existing structures rather than a complete replacement of the whole curriculum, syllabus or course of study.

Addition.

Introduction of a new component without changing old elements or patterns. e.g. audio- visual aids, workshops and equipment.

Restructuring.

Rearrangement of the curriculum to implement desired changes.

Phases of curriculum review.

The four phases of this model are:

- Needs assessment,
- Philosophy and goals,
- Curriculum product.
- Implementation/evaluation.

8.4 Curriculum changes in Kenya since independent.

African Systems of Education before Colonial Period. Education in Kenya has not been new and that African education system played an important role in society development before colonialism period. Kenya inherited a British based colonial system of education that focused on making Kenyans productive labourers and obedient subjects rather than skilled and independent minded workers. The African traditional education system before the colonial period was skilled based (apprenticeship) unlike the 'westernised' model which is mainly theory based.

Education System in Kenya from Independence (1963) to Date (2020). After Kenya attained independence in 1963, a new system of 7-4-2-3 of education was adopted. This means that learners spent seven years in primary, four years in junior secondary education (Ordinary Level), 2 years in high school (Advanced level) and three years in colleges or universities. This system of education was inherited from colonial administration but policy changes happened through various education commissions to ensure that the system education offered was demand driven. The country reformed its education in three pillars over the years.

- i) Manpower requirements.
- ii) Social demand and,
- iii) Cost benefit analysis strategies.

The social demand pillar

It came as a result of the societal pressure to ensure that majority of African children got access to education after independence. The notion was that access to education was a prerequisite of better living standards for majority of Africans who had been denied education by the colonialists. Wanyama and Changach (2013) alluded that the model adopted at independence period was focused on consumption role of education rather than investment aspect. One of the effects of this model was production of many school leavers who were not adequately prepared to meet the demands of the job market. Through the consumption education model, the government introduced free education to primary and secondary schools and this resulted to enrolment increase in schools which was recorded across the country but the quality and effectiveness of education was significantly affected since the available facilities in schools could not sufficiently address the educational need of learners enrolled. With this in mind, the government was faced with a dilemma of having too many graduates who would not fit into the employment sector. This led to rethinking of education system by formation of various education commissions.

Education Reforms since Independent.

i) Through the Gachathi Report of the year 1976, Mackay Report of the year 1981 reformed Kenya's education system of 7-4-2-3 to 8-4-4 education system in the year 1985 with the need to address the problem of educated unemployment by the introduction of vocational and technical subjects in primary education and also to improve the quality of education at all levels. From the time it was enacted in the year 1985 to 2009, the 8-4-4 system had been blamed for focusing much more on children passing in examinations rather than acquiring competent knowledge and skills to enable social and economic development of the country.

ii) Vision 2030 and Education System in Kenya

The Kenya government adopted Vision 2030 in the year 2007 to act as a blue print towards country's attaining an industrialized middle income economy by 2030. During the development of the vision, it was discovered that production of quality human resources to attain the whole vision was pegged on the education sector that prides itself in innovation and discoveries. In the last 20-30 years, the country has relied much on expatriate input to

push various developmental agenda. This situation created the debate for the need to reform the education system to ensure the goals and objectives of the Vision 2030.

iii) The Reforms of Education System from 8 – 4 – 4 to 2-6-3-3-3 With calls for reforms in the education sector, the government of Kenya created a commission of inquiry led by Davy Koech in the year 1999/2000 with recommendations made including the replacement of 8-4-4 system of education with Totally Integrated Quality Education and Training (TIQET), removal of examination between primary and secondary schools and reduction of subjects at secondary schools to ease curriculum.

One of the recommendations led to removal of technical subjects (Arts, Craft, Music, Agriculture, Home Science and Business) from primary school curriculum has been hailed to contribute to the current human resource gaps in the country leading to production of low quality graduates in tertiary education who lack requisite skills to drive the country forward .

iv) In the year 2009 the government conducted a summative evaluation of the 8-4-4 and it was found out that it would be impractical to implement the 8-4-4 system as it did not align with the country's development vision.

v) The promulgation of new constitution in the year 2010, called for the need of review of education system.

vi) Formation of Douglas Odhiambo education commission in the year 2011. In the year 2012, the commission came out with the report that proposed a replacement of 8-4-4 system of education with 2-6-3-3-3 one. This report heralded the introduction of competency based curriculum in the Kenyan system of education that was set to be implemented gradually from pre-primary school up to primary level until the year 2028 when it will be fully functional at all levels of education.

vii) The New Competency Based Curriculum.

The competency based curriculum is aimed at cultivating students' capacities to learn and perform activities to expected standards.

In the previous system of education that was largely theoretical, the expectations by the instructors in classroom were for the learners to learn and pass examinations.

With the introduction of competency based curriculum in Kenya, the shift is to learn and apply the knowledge through activities required by the curriculum. While the previous 8- 4-4 system of education focused more on the summative evaluation, the new 2-6-3-3-3 curriculum mainly relies on the mastery of competencies is what will drive the nation forward other than the certificates that are issued after completion of a certain level of education. Further, the use of summative evaluation has denied many students opportunities to acquire higher education skills as it is the determinant on which courses they are likely to pursue in the future. With this in mind, the government rolled out the programme in the year 2019 after conducting the pilot for two years.

Benefits of Educational Change from Theory to Competency Based Education.

Competency based curriculum education involves many activities that helps to promote student learning in classroom. For instance, the use of formative assessment helps students to correct their mistakes hence provide more learning opportunities to students.

Furthermore, the methods and content of assessment applied using competency based learning approach are more oriented towards application and less on theory.

Unlike the current system (8-4-4), CBC allows learners to progress through classes upon demonstrating mastery of prescribed competencies.

CBC purposes to engage learners in applying knowledge through demonstration as opposed to content overload which has been the case for many years.

The recommended teaching approaches through CBC ensure retention of knowledge and application of it in future by the learners.

Strategies of curriculum change in relation to the 21st Century

“21st century skills are tools that can be universally applied to enhance ways of thinking, learning, working and living in the world. The skills include critical thinking/reasoning, creativity/creative thinking, problem solving, metacognition, collaboration, communication and global citizenship.

Communication, Creativity, Critical Thinking and Collaboration skills can be developed in the following ways;

- Letting students lead the Learning. Learning takes place best in environments where students feel empowered to learn.
- Create an Inquiry-Based Classroom Environment.
- Encourage Collaboration.
- Develop Critical Thinking Skills.
- Encourage Creativity

Based on the Needs Assessment Study carried out by KICD, and the vision and mission of the BECF, there are seven core competencies to be achieved by every learner in basic education. The Framework seeks to develop these competencies so that all Kenyans can thrive in the 21st century. They are;

1. Communication and Collaboration
2. Self-efficacy
3. Critical Thinking and Problem Solving
4. Creativity and Imagination
5. Citizenship
6. Digital Literacy
7. Learning to Learn

Communication and Collaboration.

Communication is the act of transferring information from one place to another, whether vocally, visually, or non-verbally. The discipline of communication focuses on how people use messages to generate meanings within and across various contexts, cultures, channels, and media. Being able to communicate effectively as intended is the most important of all life skills. How well information can be transmitted and received is a measure of how good our communication skills are. Developing communication skills helps in all aspects of an individual's life. Parks (1985) maintains that communicative competency can effectively be measured by determining if and to what degree, the goals of interaction are achieved.

Collaboration.

Collaboration is the process of two or more people or organizations working together to realise shared goals. Strategies for effective communication enhance the attainment of greater collaboration among groups and ultimately increases the success of teams as they engage in collaborative problem solving. Collaborative learning is a system in which two or more people co-operate in a learning experience to share and contribute to each member's understanding of a topic and to complete a given task. Collaborative learning is designed to help learners learn from each other and can be an important aspect of the school curriculum. There are many team building games and activities that can be done in a classroom that force learners to work together to complete a task.

Self-efficacy.

Self-efficacy is a person's belief about his or her capabilities to perform tasks or assignments that can change and transform his or her life. It determines how the person feels, thinks, behaves and motivates themselves. Self-efficacy has the potential to determine four major processes namely cognitive, motivational, affective and selection processes. A strong sense of self-efficacy enhances a learner's accomplishment and personal well-being in many ways. Learners with high assurance in their capabilities approach difficult tasks as challenges to be mastered, rather than as threats to be avoided. Learners set themselves challenging goals and maintain a strong commitment to them. Self-efficacy as a competency enables learners to develop and nurture intra-personal skills and values such as self-awareness, self-esteem, confidence and personal integrity. These competencies enhance

the learner's ability to heighten and sustain efforts in the face of failure and effectively manage stressful situations. . An efficacious learner is aware of the resources at their disposal and takes personal responsibility for the use, care, management, protection and preservation of these resources. A learner with strong self-efficacy is internally motivated to establish and maintain healthy interpersonal relationships. They demonstrate interpersonal relationship skills such as assertiveness, empathy, effective communication, negotiation skills, non-violent conflict resolution skills and peer pressure resistance skills. Creative and critical thinking that leads to effective decision making and problem solving is based on a strong sense of self-efficacy (British Council, 2016). Capacity building of teachers and parental engagement are two crucial factors in the acquisition of self-efficacy. The school provides opportunities for parents to be empowered and engaged in the affairs and welfare of their children's education.

Critical Thinking and Problem Solving.

An important outcome of quality education is teaching learners how to think critically. The British Council (2015) identifies three types of thinking: reasoning, making judgements, and problem solving. When learners are empowered with critical thinking, they avoid being subjective and use logic and evidence to arrive at conclusions. Critical thinking also facilitates exploring new ways of doing things and learner autonomy. Learners learn that for every issue there are multiple perspectives that they can explore, rather than a rigid recall and regurgitation of information. Critical thinking is important for lifelong learning. It helps learners to exercise flexibility and open-mindedness to opinions that may sometimes conflict with their own beliefs and positions. Critical thinking and problem solving are useful for learners of all ages and in all the learning areas and disciplines offered in the basic education curriculum. For example, in the sciences learners need to think critically about observations and patterns to develop ideas on how to solve problems. One way of developing critical thinking and problem solving is through provision of age appropriate activities and implementation of suitable programmes in the school curriculum. For example, at pre-primary school level learners can be asked to come up with the best ways of using and keeping their books, stationery and other personal items safe. At the other end of the basic education spectrum, learners can be asked to come up with the best ways of addressing the challenge of scarce resources such as water in the school and community.

Creativity and Imagination.

Creativity and imagination refers to the ability to form new images and sensations in the mind and to turn them into reality. It is the ability to imagine things that are not real, to form pictures in the mind of things that one has not seen or experienced, and turn those pictures into real things. It also refers to the act or power of forming mental images of things that are not present to the senses, or that are never wholly perceived in reality and creating physical representations of those images. Imagination only exists or happens in the mind, and it remains in the mind. Creativity and imagination is characterized by the ability to perceive the world in new ways, to find hidden patterns, to make connections between seemingly unrelated phenomena and to generate solutions. It is a phenomenon whereby something new and valuable is formed. In educational terms, creativity and imagination refers to the ability of learners and their teachers to form images and ideas in their minds and turn them into real visible creations. Learners who are imaginative and creative are able to make life interesting for themselves and others around them. They are able to use the knowledge, skills and values acquired in the learning process to create new ideas that result in products that add value to their lives and to the lives of others around them. The competence based curriculum endeavours to inspire learners' imagination by presenting knowledge in ways that encourage learners to think as individuals. It creates scenarios that help learners to engage in imagination and encourage them to develop creations steered by the imagination. Their ability to imagine is stretched through exposure to challenging situations that help to expand their thinking and creativity skills. The curriculum also creates room for innovative ways of teaching as well as creating an environment conducive to learning that offers all learners opportunities to explore their full potential in and through creativity and imagination.

Citizenship.

Historically, human beings have always formed communities based on a shared identity. Such identities are forged in response to a variety of human needs which might be economic, political, religious or social. As group identities grow stronger, those who hold them in commonality with others organize themselves into communities, articulate their shared values and build governance structures to support their beliefs. The individuals in these communities identify themselves as citizens. Citizenship is the state of being vested with the rights, privileges and duties of a citizen. It creates a sense of belonging and attachment to one's nation. A sense of citizenship helps to equip young people to deal with situations of conflict and controversy knowledgeably and tolerantly. They are able to understand the consequences of their actions and those of the adults around them. Global citizenship is a way of living which recognizes that our world is an increasingly complex web of connections and interdependencies. It nurtures personal respect and respect for others wherever they live. It encourages individuals to think deeply and critically about what is equitable and just and what minimizes harm to our planet.

Digital Literacy.

Digital literacy can be described as having the knowledge, skills and behaviours which are necessary to effectively and safely use a wide range of digital content and devices. Such devices include mobile phones, smart phones, tablets, laptops and desktops among others. All these fall within the category of network enabled devices. Digital literacy focuses mainly on network enabled devices and should not be confused with computer literacy skills. However, traditional forms of literacy and computer literacy are enhancers in the acquisition of digital literacy skills. Individuals are presumed to be digitally literate if they possess a broad range of digital skills and knowledge and have a basic understanding of the potential uses of computing devices. Digital literacy skills also include being able to use computer communication networks, being able to engage in online communication and social networks, being aware of and adhering to ethical behaviour protocols, being aware of societal issues raised through digital media, and being able to search, evaluate and use information channelled through digital platforms. Digital literacy as a competence therefore encompasses knowledge and skills concerning the appropriate application of a variety of hardware platforms such as computers, tablets and mobile devices, and their software including but not limited to web search or internet application software. Digital literacy is a dynamic competence due to the fast-changing world of information communication technology and the ongoing development of technological devices as well as their related software. This is an area in which there is constant innovation and development as the industry attempts to keep up with a globally increasing demand for efficient and effective communication technologies. Currently, digital literacy is considered as one of the main core competencies for learning and life in the 21st century. It challenges existing thinking and practice while leading to a more innovative, creative and often transformational learning.

Learning to Learn.

Learning is a continuous process that begins at birth and continues until death; it is the process through which we use our experience to deal with new situations and to develop relationships. As a concept, it involves far more than thinking as it incorporates the whole personality – senses, feelings, intuition, beliefs, values and will. If we do not have the will to learn, we will not learn and if we have learned, we are actually changed in some way. If the learning makes no difference it can have very little significance beyond being random ideas that float through our consciousness.

Learning to learn is the ability to pursue and persist in learning, to organise one's own learning by the effective management of time and information, both individually and in groups. This competence includes awareness of one's learning process and needs, identifying available opportunities and the ability to overcome obstacles in order to learn successfully. This competence means gaining, processing and assimilating new knowledge and skills as well as seeking and making use of guidance. Learning to learn helps learners to build on prior learning and life experiences in order to use and apply knowledge and skills in a variety of contexts.

There are four pillars of learning:

- Learning to know,

- learning to do,
- learning to be and
- learning to live together.

8.5 Curriculum innovations in CBC

1. Learner-focused curriculum

There is less focus on the school and the system and more focus on education and the learner. The curriculum is responsive and relevant to learner's needs. It enables teachers - who are designers of learning opportunities- to meet the diverse needs of learners.

2. Focus on Competencies

There is less focus on content since the goal is appropriate application of knowledge and not just acquisition. A curriculum that focuses on competencies enables the creation of meaningful connections within and among learning areas and disciplines.

3. Opportunities for Local Decision Making and Greater Depth of Study.

Prescriptive curricular offer limited flexibility. The goal of CBC is to enable greater flexibility at the local level. Content delivery is flexible and exploratory. Teachers have the autonomy to implement the curriculum while teaching.

4. Balance Between Formative and Summative Assessment

CBC offers a range of assessments that focuses on varied assessment of student learning outcomes and cross - curricular experiences.

5. Digitally Based

The design of the curriculum within a collaborative digital application enables continuous improvement and supports learning with flexible timing and pacing through a range of learning environments.

6. Specific objectives have been changed to specific learning outcomes for each sub-strand in the learning areas and subjects.

7. Developing key inquiry questions that promote inquiry based learning. This is expected to not only help in contextualising learning but to also enhance differentiated instruction. For example, the teacher can vary the inquiry questions for learners of different abilities while teaching the same concept.

8. Developing an assessment rubric for each of the sub-strands. A competency-based curriculum requires competency-oriented learning assessment in which formative assessment (for and as learning) is a key feature to complement summative assessment (of learning).

9. Competency-Based Curriculum applies a 2-6-6-3 system with classes being named as Grades and has the major divisions that include; Early years Educations, Middle school Education and Senior school Education.

10. Learners will have a choice of joining one of three pathways: Arts and Sports; Social Sciences; or STEM after junior secondary education. Each pathway will have specific tracks .

11. Physical Education and Community Service Learning will be compulsory for all learners..Community Service Learning will be compulsory as it integrates classroom learning and community service that enables learners to reflect upon their experience and learn from the community, while at the same time enhancing their citizenship through service. Learners will have opportunities to:

- Apply their knowledge and skills in the community,
- Providing age-appropriate and relevant services.
- Interact with experiencea and learn from the community.
- Develop employability skills including responsibility, flexibilit ,and continuous learning through reflection.
- Promote their personal growth by forging strong and productive relationships.

12. Mainstreaming of Pertinent and Contemporary Issues (PCIs). A major paradigm shift in the curriculum reforms is the recognition that simply teaching subjects or learning areas is not adequate for the achievement of the intended basic education learning outcomes. PCIs are grouped in six broad areas: Global citizenship; Health education; Life skills and values education; Education for sustainable development; Learners support programmes; Community Service Learning and parental engagement.

13. Special needs education. The Framework provides for nurturing the potential of learners with special needs. Education of learners with special needs will start with assessment to determine whether the learner should be placed in a special school, special unit, regular school, or home based programme

8.6 Importance of the curriculum innovations in CBC.

1. The curriculum focuses on the importance of a learner's ability to develop skills and acknowledge and apply those competency skills to solve daily life challenges.

2. Learners have flexibility in exploring content at all angles.

3. CBC education system focus on the learner as an individual rather than the school performance. As a result, the teachers can familiarize themselves with the student's demands regarding strength and weakness. Learners hence create a good relationship with teachers who help them shape their talents and careers.

4. Balance Education System

The education system provides a distinctive assessment with many areas of the learner's development in classwork and co-curricular activities. According to CBC, the learner's ability is tested all-round.

5. Focus on Competencies

CBC's central focus is around learners discovering their potentials in life. It creates a platform where the learner can positively relate to acquiring knowledge and skills to connect with various disciplines as they advance to their heights.

6. Community Service Learning

Involving learners in community services exposes them to apply the knowledge acquired during their learning period. Practicing their community development skills creates a ground for the teachers to spot learners' creativity and potential in discovering specific talents.

7. The learners are tested on the ability to distinguish what they have learned in class and how practically they can follow instructional approaches in applying the skills. In this manner, the teachers can identify every learner's potential and know what works for them. As a result, it becomes easier to help the learner improve their strength and overcome their weaknesses.

8. Parental Empowerment And Engagement

The CBC form of education requires parents to participate in their children's positive outcomes during their primary education levels. The parents provide a conducive learning environment for the learners which motivates them to progress in their learning institutions. Parents need to collaborate with the teachers in providing basic requirements as per the school curriculum demands to enhance the smooth progress of uninterrupted studies.

9. CBC provides learners with a diver's opportunities to learn from to help them narrow to talent identification and potential exploration according to their needs and capabilities. The system encourages learners to develop a positive attitude towards education by allowing them to access equal education opportunities.

10. CBC system is programmed to help the learners appreciate that everyone is different and unique in terms of brain capacity. They are brought to an understanding that it is ok to be gifted differently and that no gifting is independent.

11. CBC exposes the learners to a learning system that demands them to reason, make a reasonable judgement and use creativity in arriving at solutions. As opposed to the former education system where learners were required to remember what they were taught, CBC enables the understanding.

Importance of curriculum change

1. To bring about improvement in the present curriculum. Most times looking at the present curriculum you might feel that there is a need to make slight changes to bring about improvement in the curriculum
2. Curriculum are changed to meet with the international standard of and global trends.
3. Society is always changing, hence the need for curriculum change to meet up with the changes in the society.

8.7 Current Focus and Trends in Education.

It is certain that technology will drive the future of education. The new teaching tools that have come up in the wake of the pandemic have proven to be effective for the new generation of learners and educators. Interesting trends have risen in the months that followed the sudden shift to digital education. These include;

1. Personalized and paced learning

Even though most subjects taught at school aim to increase a student's ability, the road to achieving success can vary from student to student. With the modern learning tools, all students will be able to personalize their learning journey with challenges based on their skill levels, without causing disruption to their peers or educators. This personalization and pace of learning styles can be adopted with ease and help students gain confidence with additional practice and learning methods.

2. Unified learning experiences

With the power of technology, it is now possible to chart a learning path right from school to college. The availability of such details to potential college recruiters also help them make an informed decision on student intakes and the admissions process. This trend of easily available information in the form of a single platform to access any student information across their academic paths is revolutionary and unimaginable pre-pandemic.

3. Vocational learning

Another emerging trend in education is the emphasis to deliver a vocation-based education and techniques. The job opportunities in today's world are adapting to the massive digital audience and remote-work setups, therefore students will have to inculcate a project-based learning mindset to dovetail into the new stream of jobs. The practical solutions that they can apply to different challenges need to be taught to build their skillsets. Since the mandate of the education policy, institutions are working towards bringing in industry leaders and subject matter experts to coach learners on these aspects and give them a rounded education. Soft skills such as team collaboration, time management, and organizational skills can further help students excel in their academic career.

4. Usage of deep technology

A lot of new learning strategies have been introduced to cater to the evolved needs of digital learners. Technologies like gamification through augmented and virtual reality is one of the emerging trends of 2021. Courses designed around these technologies as a base tap into the intrinsic motivation to learn and develop concrete skills outside the setting of a classroom. With appropriate use, lesson concepts are retained better through engagement with the student, and turns them into an active learner over a period of time.

5. Social and emotional learning

Social-emotional learning (SEL) is the process of developing the self-awareness, self-control and interpersonal skills that are vital for school, work and life success. There is increasing focus on this aspect of learning in today's school curriculum due to the pandemic's effects on children's social and emotional lives. Aligning learning outcomes keeping the social and emotional learning benefits in mind is paramount to educators and the presentation of such learning offerings presents the educational institution favorably amongst parents and higher education students.

6. Remote assessments

Experience-based assessments will be used to test acquired knowledge and skill sets. The current trends and issues in education revolve around maximising lesson retention and minimizing assessment bias by incorporating advanced testing methodologies to give examiners a near-accurate picture of progress. Application-based testing methods to gauge critical thinking are what educators will turn to.

7. Digital literacy curriculum

Having a digital literacy curriculum weaved into the mainstream school curriculum will be imperative for students as well as teachers, going forward. Based on the students' developmental stages, appropriately designed lessons which cover the range of learning tools along with potential risks of overuse or misuse are to be covered by the educator.

8. The online platform may also force teachers to change how they teach. They might find it challenging to change how they approach lesson plans to ensure that the students remain engaged even while they cannot see the instructor in person. Platforms may offer opportunities for students to watch lectures live or recorded versions later. The online nature of these courses may also enhance the ability of teachers to offer accommodations for different styles of learning.

9. Soft skills training: a major trend in higher education

According to the Future of Jobs report, some of the most important skills in the workplace include critical thinking, problem-solving, people management, and creativity. Employers want to see emerging professionals who understand how to make hard decisions and showcase their leadership abilities. In an effort to prepare students for their future careers, schools must have the training in place to help students nurture and grow in these skill areas.

10. Facilitating learning versus teaching

As technology has grown, it has also changed how teachers relate to their students and their classrooms. With a wealth of information at their fingertips, students today have the tools they need to uncover a tremendous amount of facts and knowledge independently. In this environment, many students value less of a top-down delivery method. Instead, teachers now function more in a facilitative role. Their job has slowly evolved into a position where they help students understand how to learn, to love learning, and how to uncover and understand the information they find. The best teachers will be those who can help students take ownership of their learning.

11. Life-long learning trend

Professionals who want to remain competitive in their environment will need to constantly re-skill themselves. They cannot assume that an education they earned in the first half of their professional career will be all they need for the rest of their working lives.

Classrooms must leave opportunities for teaching self-learning skills so that students can continue to learn and engage in their chosen fields.

9.0 CURRICULUM FRAMEWORKS

9.1 Curriculum Frameworks from other countries

Specific Learning Outcomes

By the end of the sub strand, the teacher trainee should be able to:

- Examine the concept of curriculum framework in guiding learning
- Analyze curriculum frameworks from other countries for comparison
- Value the need for a framework in curriculum

Suggested Learning Experiences

- Research and write a paper on curriculum framework from around the world (Asia, Europe, America, Oceania)
- Compare and contrast various curriculum frameworks from around the world
- Debate on the need for a curriculum framework in the learning process

Key inquiry questions

Why should we have a curriculum framework

Indicator

- Examine the philosophy, visions and missions of the current frameworks in Kenya
- Discuss the philosophy and pillars that guided the development of Basic Education Curriculum Frameworks Kenya

- Analyze components of current Basic Education Curriculum Framework and Teacher Education Curriculum Framework in Kenya
 - Justify the shift from content to Competency Based Curriculum
 - Analyze the organizational structure of education outlined in BECF and TECF
 - Exploring other national and international Curriculum Frameworks applied in Kenya
- Curriculum frameworks from other countries

A Curriculum framework

Is a supportive structure to help schools to plan and develop their own curriculum.

It comprises a set of interlocking components, including essential learning experiences, generic skills, values, attitudes, and key-learning areas.

Is one of the most important tools in ensuring consistency and quality in a curriculum system.

Is an organized plan or set of standards or learning outcomes that defines to be learned in terms of clear, definable standards of what the student should know and be able to do.

Curriculum is usually a document or a set of documents that sets standards for curriculum and provides the context in which subject specialists develop syllabuses.

Purpose/importance of curriculum framework

To develop qualities that make a child socially effective and happy in various social settings such as friendliness, cooperativeness, self-disciplines, self-control, love for social justice etc.

To develop pre-vocational/vocational skills, willingness to work hard, dignity of manual work and job satisfaction.

For understanding of the environment and its limited resources and the need for conservation of natural resources and energy.

To develop ability to appreciate and discover beauty in various life situations and integrate it into ones own personality.

To develop knowledge of scientific methods of inquiry and its use in solving problems.

To facilitate schools and teachers to make decision about choice of content, pedagogy, teaching, and learning material, evaluation etc, at school level.

To give schools and teachers flexibility and ownership to plan and develop alternative curriculum modes to meet their varied needs.

To help to review school curriculum, learning and teaching strategies and to develop school assessment policy.

To help the teacher in becoming reflective practitioner who learns from their own experience.

To emphasize learning with understanding and learning to learn, and helps children develop their own understanding based on their life experiences.

To set out what students should know, value, and be able to do at the various stages of schooling. We also see few functions of curriculum framework in the field of school education

Is to provide a comprehensive conceptualisation of reforms in basic education; pre-primary education, primary education, secondary education.

The philosophy of current curriculum framework in Kenya

The CBC aims to be a more student-based system, with more focus on the ability of the student to self- learn and develop relevant skills requisite for any engagement carried out. This way of thinking is perhaps informed by the opening up of the world, which has been spurred by the internet explosion.

Visions of the current curriculum framework

The vision of the basic education curriculum reforms is to enable every Kenyan to become an engaged, empowered and ethical citizen.

Mission of the current curriculum framework

To provide innovative learning experiences that ensure each and every child can take their place in the world with confidence and pride as 21st century Kenyans.

Pillars that guide the current curriculum framework

Basic Education Curriculum Framework (BECF) and Teacher Education Curriculum Framework (TECF)

The basic education curriculum framework vision and mission are supported by three

Important pillars; values, theoretical approaches and guiding principles.

VALUES

Value Based Education (VBE) Include;

Love

Responsibility

Respect

Unity

Peace

Patriotism

Social

Justice

Integrity

Theoretical Approaches. This include;

Instructional design theories

Vygotsky Social Theory

Gardner's Multiple Intelligence theory

Piagets Cognitive Theory

Guiding Principles. This include:

Opportunity

Excellence

Diversity

Inclusion

Community source learning

Parental empowerment and engagement

Differentiated curriculum and learning

What are the pillars of the new curriculum?

The review process towards the development of the New Curriculum was centred on five key pillars:

MAIN KEY PILLARS ARE:

The legal and regulatory framework;

Teacher capacity development programme;

Teacher professional standards;

Infrastructure development and.

The centre for education research, innovation and development (CERID).

COMPONENTS OF THE CURRENT BASIC EDUCATION

Quality basic education includes at least these five components:

Healthy and ready to learn children;

Supportive learning environment

Relevant curriculum content

Child-centred teaching and learning

Successful learning outcomes for children

PARADIGM SHIFT FROM CONTENT TO COMPETENCY BASED CURRICULUM

From less	To more
Focus on content knowledge	Focus on pedagogy and professional competencies
Teacher centered	Learner centered pedagogy
Academic and competitive teaching	Reflective teaching and mentorship
No set standards for teacher training	Set standard for teacher training through set competencies
Little focus on skills	Addresses 21 st century skills
Emphasis on summative assessment	Balance between formative and summative assessment, excellence & professionalism

CURRICULUM FRAMEWORKS FROM OTHER COUNTRIES.

Examine the concept of Curriculum Framework in guiding learning.

Curriculum Framework means a comprehensive document developed for a content area consisting of overarching goals, content standards, and performance standards. The curriculum framework serves as a guide to schools as they create a curriculum unique to their needs.

The Framework outlines the vision for the curriculum reforms, including the mission, the pillars of the reforms, the organisation of basic education, core competencies to be achieved in basic education, curriculum approaches adopted in the Framework, general learning outcomes and Subjects in the curriculum.

CURRICULUM FRAMEWORK OF ASIA COUNTRIES.

Japan & Singapore

CURRICULUM FRAMEWORK IN JAPAN.

What is the purpose of education in Japan?

The goals of compulsory education are to enhance students' individual abilities and develop a basis for their social independence and basic abilities as citizens of the nation.

Curriculum Outline

The Japanese school system primarily consists of six-year elementary schools, three-year junior high schools and three-year high schools, followed by a two-or-three-year junior colleges or a four-year colleges.

Compulsory education lasts for 9 years through elementary and junior high school. School exchanges during Japan Educational Travel are mainly implemented in junior high and high schools. For physically or mentally challenged students, there is a system called “Special Needs Education” to support special students to develop their self-reliance and thus enhance their social participation.

School Timetable.

Public schools in Japan have classes five days a week, from Monday to Friday. There are also schools that have classes on Saturday. In junior high and high schools, there are six class periods each day, typically lasting 50 minutes for each. After classes, students clean the classrooms in shifts and then start their club activities.

There are a variety of clubs such as cultural and sports ones.

Academic Calendar.

In principle, the school year begins in April and ends in March of the following year. Most schools adopt a three-semester system, with the first semester from April to August, the second semester from September to December, and the third semester from January to March. There is also a summer break (from the end of July to the end of August), a winter break (from the end of December to the beginning of January), and a spring break (from the end of March to the beginning of April).

School Organization.

Each school has a principal, a vice principal, teachers, a school nurse and other administration staff. As the chief executive, the principal assumes all responsibilities of the school including the courses provided and related administrative work. The vice principal supports the principal to manage administrative affairs of the school and to be in charge of student's educational activities and curriculum as well. Furthermore, in order to ensure school's smooth operation, teachers take on various responsibilities, such as taking care of educational activities, students' school life and employment guidance for students after graduation. Many schools also establish their own committees, for example a International Exchange Promotion Committee, and others.

Regarding the Level of Education.

The level of Japanese education is high even by world standards particularly in science related areas.

Foreign Language Education.

English is a compulsory subject in junior high and high schools. There are also elementary schools that introduce English education from intermediate grade classes. In some high schools, apart from English, students are also allowed to take courses in Chinese, Korean, French, German, etc.

Student Clubs.

Student clubs are a characteristic part in Japan's school education. Under teachers' guidance, students with the same interests in sports, cultural activities, or fields of study voluntarily gather together after classes and on

days off. There are also numerous student clubs revolving around Japanese traditional sports and culture. Club activities provide students with the chance to participate in school exchange and friendly matches.

Japan's primary school curriculum is divided into three main categories: compulsory subjects, moral education, and special activities. Compulsory subjects are Japanese language, Japanese literature, mathematics, social studies, science, music, arts and handicrafts, and physical education.

Reading and writing are perhaps the most important parts of the elementary school curriculum.

What makes Japanese education system special?

The Japanese education system believes that requiring students to clean up after themselves teaches them to work in a team and help each other. Besides, spending their own time and effort sweeping, mopping, and wiping makes kids respect their own work and the work of others.

Why Japan has the best education system?

Children are taught to respect other people and to be gentle to animals and nature. They also learn how to be generous, compassionate and empathetic. Besides this, pupils are taught qualities like grit, self-control and justice.

Education is one of the most important aspects of Japan's national identity and a source of pride for Japanese citizens. Japan ranks highly among wealthy nations in providing equal opportunities to students of all socioeconomic backgrounds.

Japan's primary school curriculum is divided into three main categories: compulsory subjects, moral education and special activities.

CURRICULUM FRAMEWORK OF SINGAPORE

Overview.

At the end of World War II, Singapore implemented the first in a succession of economic development strategies rooted in improved education and training.

Singapore is an extraordinary success story. Since becoming an independent republic in 1965, it has transformed from an impoverished island with no natural resources and a mostly illiterate population to a country of 5.8 million people whose living standards match those of the most highly-developed industrial nations. From the very beginning, Lee Kuan Yew, the prime minister who led Singapore to this achievement, understood that an educated workforce would be essential to fulfilling his ambitious economic goals.

Since the 1990s, the nation has focused on boosting creativity and capacity for innovation in its students.

In 2004, the government developed the “Teach Less, Learn More” initiative, which moved instruction further away from its early focus on rote memorization and repetitive tasks and toward deeper conceptual understanding and problem-based learning. Educators abandoned the practice of funneling students into ability-based tracks and began sorting them into three different “bands” in secondary school based on their ultimate educational goal. Although students take most of their classes within their bands, they can take classes in other bands depending on their aptitude and interest in a given subject. The goal is to achieve full subject-based banding, with students freely mixing and matching classes from different bands, by 2024.

Singapore's current priorities for its education system are reflected in the title of its initiative “Every School a Good School.” This set of reforms aims to ensure that all schools have adequate resources to develop customized programs for their students; raise professional standards for teachers; encourage innovation; and foster partnerships between schools and communities. In addition, Singapore launched the “Learn for Life” initiative in 2018 to promote greater flexibility in teaching, learning and assessment. With more opportunities for self-directed learning in and out of school, Singapore hopes to encourage lifelong learning for all Singaporeans, in ways that bring them satisfaction and meaning.

Governance Structure

Singapore's education system is highly centralized. The Ministry of Education oversees kindergarten (ages four to five) through higher education and lifelong learning. The Ministry allocates funding for all schools, sets course syllabi and national examinations, oversees teacher credentialing, manages the teacher and principal evaluation and promotion system, and hires and assigns principals and teachers to schools. Schools are grouped into geographic clusters, each overseen by a superintendent, to provide local support for the Ministry's policies and initiatives. The cluster superintendents, who are successful former principals, collaborate with principals in their cluster on how to implement the curriculum and which teaching materials to choose from among a set the Ministry approves and strongly encourages teachers to use. The cluster superintendents also facilitate the sharing of resources and best practices between cluster schools.

While the Ministry sets the framework for the educational system, other entities operate within that framework. Independent or semi-autonomous agencies such as the National Institute of Education (teacher training), the Examinations and Assessment Board (national assessments), and the Institute of Technical Education (vocational education) have clearly defined areas of responsibility and work closely with the Ministry.

Planning and Goals.

Singapore articulates clear and comprehensive system-wide goals for education. These goals, which the nation revisits regularly, emerge from widespread discussion with partners in the system and with the public, as well as from extensive benchmarking of other leading education systems. Singapore structures policy initiatives around its education goals and creates benchmarks to measure progress. For example, in 2013, Singapore held a National Conversation to gather input on a vision for the 2030 education system strategic plan.

Goals included;

- Improving character and citizenship education.
- Strengthening digital literacy.
- Building more knowledge and understanding of the history and cultures throughout Asia.
- Expanding supports for disadvantaged students and
- Building more adult education opportunities.

In addition, Singapore's leaders monitor educational research and benchmark best practices from around the world so that the system can continue to match the performance of the world's best.

Education Finance.

The Ministry of Education directly funds all schools based on the number of pupils. In addition, all schools receive a set grant (called an Opportunity Fund) to use for their low-income students and students from ethnic minority groups. Although this supplemental funding is distributed by the Ministry, schools can choose how to spend it. The Ministry also provides funding directly to students from low-income families in the form of subsidies called Financial Assistance Schemes for educational materials and activities, and funds for school meals. In addition, the Ministry in 1970 created the Education Fund, which collects contributions from Singapore residents to both support all students and low-income students through scholarships and by providing textbooks, meals and uniforms for students who need these but do not otherwise qualify for financial assistance.

School Accountability.

Schools in Singapore conduct annual self-evaluations of their practices and outcomes using the Ministry-developed School Excellence Model, which includes nine criteria for performance. Schools then develop improvement plans based on the results. Additionally, external inspectors evaluate each school every five years. The external inspectors, made up of university professors and successful school leaders, provide feedback to the schools and offer coaching and support for improvement.

Improvement efforts are organized through Singapore's school cluster system. Cluster superintendents meet regularly with principals to monitor their improvement efforts. High-performing schools are eligible for

awards. The Ministry annually awards schools that demonstrate outstanding achievement in a single year or over a period of years. The highest award, the School Excellence Award, is given to one school each year.

Teacher Accountability.

Singapore uses the Enhanced Performance Management System (EPMS) to conduct annual teacher evaluations. EPMS assesses teachers' performance based on 16 different competencies, including their work in the classroom and their interaction with the greater school community. Teachers first conduct a self-appraisal and then supervisors evaluate them against the EPMS. These evaluations are qualitative and consist of written feedback rather than numeric scores of specific indicators. Teachers base their professional development plans on EPMS feedback. Principals, alongside the School Staff Developer (in charge of professional learning) and the Cluster Superintendent, co-construct a "Current Estimated Potential" for each teacher using the results of the EPMS. This estimate, or snapshot of the teacher's short-term career trajectory is shared with the teacher and used to help them articulate their career goals. Teachers can earn rewards based on EPMS results including honors and salary bonuses. The Ministry also selects teachers for awards and recognition at the national level.

Supports for School Aged Children.

All students in Singapore receive an Edusave account, to which the government contributes funds so that it can invest in their future. Families can draw on these accounts for any type of educational expense; disadvantaged students receive additional funding. The Ministry of Education also provides financial assistance for students from low-income families. The aid supports school fees and other expenses for students in government or government-aided private schools. Financial aid for independent schools is also available. These supports are available in the context of a broader safety net for children. The government's Ministry of Social and Family Development (MSF) administers ComCare, which provides cash grants to low- and moderate-income families on a sliding scale. It also covers long-term assistance needed for care and school-related expenses for children of disabled parents. The MSF also oversees the National Council of Social Service, an umbrella group of 450 private organizations that provide services to Singapore citizens. Services include school-based social work and support for students at risk of dropping out of school.

Learning System

Preschool

In Singapore, children aged three through six can attend either a public or private kindergarten or a childcare center. But in 2013, the government created the Early Childhood Development Agency (ECDA) to coordinate oversight of all early childhood education. ECDA has developed the Nurturing Early Learners Kindergarten Curriculum Framework as suggested guidance for children ages 4 to 6. The government does not assess learning outcomes for students in kindergarten or childcare; the first nationwide screening of children's literacy and numeracy skills takes place in the first month of primary school.

Primary and Secondary Education.

In Singapore, the system includes six years of primary school followed by four to six years of secondary school and one to three years of postsecondary school. The curriculum for primary schools is common for all students in years one to four. For years five and six, students can take individual courses at the foundation or standard level. Foundational level courses are designed to provide more support for students. As they enter secondary school, students, their parents and their teachers jointly agree on one of three bands or "streams" they will join: Express, Normal (Academic) and Normal (Technical). All streams offer the same course of study but Express is accelerated and Normal (Technical) offers more applied work. In most cases, students' scores on the Primary School Leaving Examination (PSLE) are the primary determinant of the stream they will join but parents and students can advocate for different streams if they demonstrate accelerated learning or need more help. Singapore is piloting and implementing a system under which students choose streams for specific subjects, rather than their overall course of study, a practice known as subject-based banding. Subject-based banding currently exists in all primary schools and the goal is to have full subject-based banding in all secondary schools by 2024.

In addition to these options, Singapore has four specialized schools for students who perform poorly on the PSLE. These schools offer foundational coursework in mathematics and literacy, alongside vocational offerings leading to skill certificates and extensive social supports. There are also specialized independent schools that focus on the arts, sports and mathematics and science. These schools receive public funding and use the MOE curriculum but have more flexibility in their program offerings.

Students who want to apply to university stay in secondary school for an additional two years to take A-level courses, as part of the Integrated Program. Those who do not do that have multiple postsecondary pursue these options: Polytechnics, the Institute of Technical Education (ITE), Junior Colleges, a Polytechnic Foundation program and a small set of Arts Institutions. Students choose their postsecondary school based on their secondary school stream as well as results from the General Certificate of Education (GCE) examinations. Polytechnics offer three-year diploma programs. Graduates may pursue university education after they earn their Polytechnic diploma without taking A-level exams, if they so choose. ITE offers shorter technical or vocational education programs, through National ITE Certificate (Nitec) aligned courses and work-based learning. Students graduate from ITE with a Nitec or Higher Nitec qualification and can then continue their vocational studies at a polytechnic or at university. They can also stay at ITE and earn a technical or work study diploma, which also allows a pathways to select university programs. Junior Colleges offer two- or three-year pre-university education, preparing students for the required examinations to enroll in universities or for entry into Polytechnics.

Standards and Curriculum

The Ministry of Education oversees the development of the national curriculum which includes “Desired Outcomes of Education.” The desired outcomes are student excellence in life skills, knowledge skills and subject discipline knowledge organized into eight core skills and values: character development, self-management skills, social and cooperative skills, literacy and numeracy, communication skills, information skills, thinking skills and creativity, and knowledge application skills.

The primary school curriculum includes ten subject areas: English, mother tongue language (available for Chinese-, Malay- and Tamil-speaking students), mathematics, science, art, music, physical education, social studies and character and citizenship education. A coding class was added to the curriculum in 2019. And in 2021, the Ministry introduced an updated character and citizenship education curriculum which focuses on mental health and cyber-wellness and on the establishment of peer support structures within every school, among other topics. For primary students who qualify as gifted, Singapore offers individualized enriched curriculum opportunities..

Secondary education varies depending on school and program type. Students in the express and Normal (academic program) are required to take English, mother tongue language, mathematics, science, and humanities (geography, history, and English literature). For students in the Normal (Technical) program, compulsory subjects include English, mother tongue language, mathematics, computer applications and social studies. There are electives available for both the Technical and Normal program as well.

The Ministry of Education has been very involved with the implementation of its primary and secondary curriculum. During the shift from rote learning to the current model emphasizing student engagement and creativity, Ministry officials were very “hands-on” in schools. They met regularly with school leaders and developed extensive professional learning opportunities for teachers around the new curriculum. However, in recent years, the Ministry has taken a step back, encouraging schools to consider the curriculum as a framework which they should adapt to their students’ needs. The Ministry also encourages secondary schools to differentiate themselves through theme courses or special programs designed to attract students with shared interests.

Assessment and Qualifications.

Teachers perform continuous assessment of their students at all levels of education. On a day-to-day basis, this assessment is informal and based on student work in and out of the classroom. Previously, all students in primary school took school-based exams throughout the year and at the end of each year but in 2019, the government

dropped the exams for Primary 1 and 2 and in 2021 dropped the mid-year exams for Primary 3 and 5 as well as Secondary 3. By removing these exams, the government hopes to shift focus away from grades and competition and toward learning for its own sake.

At the end of primary school, all students take the Primary School Leaving Examination (PSLE) in four subjects: English, math, science, and mother tongue. Students take exams at one of two levels, based on the level of subjects they took in years five and six. In 2021, the Ministry began to update the PSLE scoring process. Going forward, students will be graded based on individual performance in subjects rather than benchmarked against each other. The Ministry also allows some schools to admit students based on their talents in academic areas, sports or co-curricular activities without factoring in PSLE results, to provide greater diversity in student talents and interests. Since 2018, schools have been able to offer up to 20 percent of their places to students through this process, called direct school admission. The Ministry of Education helps place those students who are not accepted into their schools of choice.

At the secondary level, students take subject-based exams, depending on their band. After four years of study, students take O-level exams in the express and N-level exams in the Normal (Technical) program. Students in the Normal (Academic) program can take the N-level exams after four years of study or the O-level exams after five years. Students who wish to study at university take A-level exams after an additional two years of study.

Special Education

Whenever possible, the government encourages students with special needs to enroll in mainstream schools either initially or after having met certain benchmarks in special education. Currently, about 80 percent of all students with special needs attend mainstream schools. To help facilitate their integration, learning support specialists known as Allied Educators help students with conditions such as dyslexia or autism. As of 2018, there was at least one Allied Educator in every mainstream primary and public school, a 40 percent increase over the previous five years. The Ministry has also provided specialized training in special education to a designated group of general education teachers within each mainstream school, to create a strong support system for students with special needs; about 15 percent of teachers in mainstream schools had completed this training by 2019. In addition, since 2020 the Ministry has provided all teachers in mainstream schools with access to online professional learning focused on supporting students with special needs.

In 2019, the Ministry implemented two peer mentoring interventions to support students with special needs in mainstream schools. Circle of Friends allows students with social, emotional or behavioral difficulties to meet with their teacher or Allied Educator along with a group of six to eight of their peers. Over five to eight sessions, the students work together to find solutions for the student in difficulty. Facing Your Fears is a similar program designed to support students suffering from anxiety. In this intervention, groups of two to four students meet with facilitators and Allied Educators to learn self-management strategies over 10 weekly sessions.

For students who need more intensive or specialized assistance, Singapore has 19 government-funded special education schools run by 12 social service agencies. These schools serve populations with highly specific needs: the deaf, the blind, students with autism, or those with the most severe cognitive challenges. Special needs education is available through the postsecondary level, where students with intellectual disabilities are prepared for the workforce through special training programs.

The Ministry allocates extra funds for special needs students at 150 percent or 300 percent of the base per student cost, depending on whether they attend mainstream or special schools. The Ministry increased spending for special schools by 40 percent from 2015 to 2020 and has pledged to continue this increase. The National Council of Social Services also contributes funding to special schools, specifically for additional social supports.

Teacher Recruitment.

Only one institution—the National Institute of Education (NIE)—is authorized to prepare teachers, and it offers both a master's degree and a bachelor's degree route into teaching. In this way, Singapore limits its teacher

recruitment only to those students qualified for the country's rigorous research universities. Each year, Singapore calculates the number of teachers it will need, and opens only that many spots in the training programs. The selection process is competitive: teaching is a highly-regarded profession in Singapore and students in teacher-education programs receive a stipend during their training. On average, only one out of eight applicants is accepted.

Teaching salaries in Singapore are largely commensurate with those of other professions. Indeed, the Ministry of Education monitors teacher salaries in relation to other professional salaries and adjusts them to ensure they remain competitive.

Teacher Development.

Teachers have access to several types of professional development opportunities. They can improve their practice through courses at NIE or at AST, where offerings range from in-service training to online classes on a variety of subjects related to teaching. Teachers can participate in as many as 100 hours of government-funded professional development per year. The Ministry and NIE also offer scholarship opportunities for teachers seeking master's and doctoral degrees in Singapore or abroad, either full- or part-time.

In addition, the career ladders are a key component of Singapore's teacher development strategy. Career ladders ensure that teachers can stay in the profession while taking on new roles and responsibilities and provide the structure for their peer-to-peer mentoring, collaboration, coaching, and development systems. Further, the career ladder guarantees that curriculum development is led by those who know teaching and learning best.

Principals Recruitment, Preparation and Development

Singapore prioritizes developing skilled principals who can ensure that their schools offer high-quality and equitable learning opportunities. In their third year on the job, in consultation with their principal, teachers can choose the leadership track of Singapore's three-track career ladder. If they continue to demonstrate the right combination of skills and if the role is available to them, they can advance to department head and vice principal. In this way, all principals start out as teachers and then serve in two administrative roles before advancing to school leadership. Department heads and vice principals usually participate in the Management and Leadership in Schools Program run by the NIE, which aims to proactively prepare them for the next stage of the leadership track. Vice principals on the verge of being promoted to principal must participate in a two-day simulation test and interview process during which they respond to real-world scenarios.

Singapore also supports its principals through an international school leader exchange program, Building Educational Bridges, which focuses on building leaders' capacity to innovate by learning more about international education systems' leadership practices. .

CURRICULUM FRAMEWORK FROM EUROPE.

United Kingdom & Switzerland

CURRICULUM FRAMEWORK OF UNITED KINGDOM

The National Curriculum for England and P levels

Introduction

The National Curriculum for England and Wales is the framework used by all state schools in both countries. Children work at different levels according to age and ability. There are 4 key stages with national testing at the end of each (the fourth being GCSE at which point the national curriculum no longer applies). By the end of key stage 1 (age 7), the average child is expected to achieve national curriculum level 2; this rises to level 4 at the end of key stage 2 and level 5 or 6 by the end of key stage 3. Only the most able pupils will achieve a level 8 with an additional exceptional performance category for those who achieve beyond this. Such performance is rare. In exceptional cases, schools can choose to disapply weaker students from some elements of the national curriculum.

Some children with special needs will be working below level one of the national curriculum. They are assessed according to P scales. Pupils aged 5-16 who are working at or towards P levels are likely to need significant help and prompting with tasks and activities. P scales exist for all national curriculum subjects. There are 8 levels of performance with each describing some of the important knowledge, skills and understanding that pupils may gain from the programmes of study of the national curriculum.

Levels P1 to P3 show the earliest levels of general attainment with subject-focused examples. Levels P4 to P8 show subject-related attainment designed with transition to the national curriculum in mind. There is no expectation that a child will achieve a particular P scale by a given age, or that they will work through the P scales at a predetermined rate. However, attainment of a level by a particular age may help identify the likely rate of progress through the P scales.

Children do not undergo any formal assessment or testing and unlike the national curriculum, the awarding of a P level is left to the professional judgment of staff. It is expected that teachers will use their knowledge of the child, consider the contexts in which learning takes place and gather evidence from a variety of sources, to support their decisions and to make a 'best-fit judgment' based on everyday activity and continual monitoring and assessment.

Key Stages and Tests

Children in English state (maintained) schools must follow the national curriculum. This is intended to ensure a broad and balanced education split into 8 levels, with the most able children expected to attain level 8 by the end of KS3. P levels exist for those children unlikely to achieve a level one. The National curriculum subjects are: art and design, citizenship, design technology, English, Geography, history, ICT, mathematics, modern foreign languages, music, PE, science, RE, careers education, work-related learning, PSHE. Not all subjects are studied throughout. There are four key stages and the children are tested at the end of each key stage but only in the core subjects.

Standard Attainment Tests (SATs)

Children take Standard Attainment Tests (SAT's) when they are 7, 11 and 14 and national examinations, often though not exclusively, GCSE's at the age of 16 (end of key stage 4). SATs are statutory. State schools must report their results however it is possible for a child to be disapplied from the tests. Many independent schools take SATs but are not obliged to report the results.

Key Stage One (Y1-Y2)

Covers the 5-7 age group. Schools have flexibility over when and how the tests in reading, writing (including spelling and handwriting) and maths are administered. The average 7 year old is expected to achieve a level 2. This is not a pass/fail situation; the tests are designed to indicate whether a child is working at, above or below the target level. The tests can be taken at a time to suit the school and are not reported separately but are used to inform teacher assessment of the child. Teacher assessment looks at speaking, listening and science skills as well as the areas covered by the tests.

Key Stage Two (Y3-Y6)

This covers children aged 7 to 11. Children are tested in English, maths and science, at age 11 (Y6), prior to the move to senior school,. The results are reported to the DfES and to parents. The average 11 year old is expected to achieve a level 4.

Key Stage Three (Y7-Y9)

Tests are taken in Y9 at senior school, when a child is typically aged 14. Children are tested in English, maths and science and the results are reported to the DfES and to parents. Teachers assess pupils in these areas plus art, citizenship, design technology, geography, history, ICT, MFL, music, PE. The target level at KS3 is level 5, with the most able achieving a level 8. There are wide variations reported in the levels achieved at this stage. Very

few independent schools take key stage 3 tests, most will have assessed the child at 11+ or 13 + via The Common Entrance examination

Key Stage Four/GCSE (Y10-Y11)

Most children embark on a two-year GCSE course (in some schools pupils sit some GCSEs at the end of Y10). Children are usually encouraged to take a broad base of subjects including: English (usually language and literature); maths; science (biology, chemistry, physics - a combination of one, two or all three in a guise of awards: single, dual etc); a modern foreign language (no longer compulsory) - most schools offer French and often a choice of German, or Spanish but always check with a school first; a design technology subject; humanities (history Geography, RE), and the Arts (visual and performing). Often business/vocationally orientated subjects such as business studies, economics, media studies and ICT are offered. Social sciences: psychology, sociology are popular in some schools as are Latin, Greek and Classical studies. Virtually all GCSE's have a compulsory coursework element worth between 20% and 100% of the marks (some independent schools have opted to follow the IGCSE courses in some subjects as there isn't a coursework element). Non-GCSE courses such as PE (which is offered at GCSE), games and Personal Social Health Education (PSHE) or similar, plus citizenship continue to be followed.

Children usually make their GCSE - option choices in Y9 following extensive consultation with the school, their parents and teachers. At this stage the proposed level of study will be discussed. Most subjects offer two tiers of entry, foundation and higher. Foundation level study is aimed at grades C-G and Higher Grades A*-D. Most schools and colleges will expect a student to have achieved at least a grade B in any subject to be studied at A level.

IGCSE

The International GCSE (IGCSE) is becoming an increasingly popular option in independent schools. The IGCSE offers more rigour and depth, than GCSE enabling the more able to be stretched while still meeting the needs of others. IGCSE can be awarded without coursework ie by 100% examination although coursework options exist. Exams are graded in the same way as GCSEs: Foundation targets grades C-G and Higher grades A*-D. Many people consider the IGCSE to be more akin to the old style O level than GCSE. For example maths IGCSE has a strong emphasis on algebra and introduces calculus, in preparation for A level maths.

CURRICULUM FRAMEWORK OF SWITZERLAND.

Switzerland's education system is one of the most advanced in the world. Switzerland is the go-to choice for many students looking to study internationally. The system is well established and functional with delegated accountability and administration.

The administration and regulation of the school system in Switzerland is made by the cantons while having a set framework by the central government. Each canton sets its school calendar, curriculum and criteria. This system is overseen by the State Secretariat for Education, Research and Innovation (SERI). The decentralization has its perks which makes it hard for people to move their children from one canton to another.

A distinctive characteristic of the Swiss education system is taking into account the wishes and the abilities of students. This attribute is done by diversifying the different directions a student can take once they're done with their primary education. This is common to all cantons.

Compulsory Swiss Education.

Education is compulsory for all children and young people from 9 to 11 years. Children start compulsory education when they are about four to six years old until about when they turn 15 years old. The cantons also regulate this.

Even though there are private schools in Switzerland, most students attend public schools in the local area. This brings diversity so children can experience many different backgrounds, including, sometimes, linguistic backgrounds. Public schools are free of charge and kids are not required to wear uniforms.

Primary education

Primary education in most of Switzerland is considered kindergarten and the first learning cycle. Kindergarten is not compulsory in all of the cantons, but primary school is. In kindergarten, children are taught about basic manners and knowledge. That's when they are introduced to other languages and Swiss culture. Kindergarten aims to teach children social skills and basic mannerisms through play-based learning. Through music, games, art, and lots of crafts, they are introduced to basic reading and math. After they have finished kindergarten, around the age of 6, they go to primary school.

In the German-speaking cantons, kindergarten and two years of primary education are combined into the first learning cycle and students aged four to eight years are in the same class. They call this Grundstufe or Basisstufe. In the French-speaking cantons, two years of kindergarten are combined with two other years and make the first cycle of primary education. The Italian-speaking cantons children attend kindergarten from the age of four.

Secondary education.

Secondary education in Switzerland is divided into two stages. The lower secondary education and upper secondary education. Kids start their lower secondary education around the ages of 11 to 12 and it lasts 3 to 4 years. The upper secondary education is equivalent to the high schools in American education systems regarding the age of the students. Still, it is very much different from an American high school.

Lower secondary education

Lower secondary schools in Switzerland are called Gymnasiums or Kantonsschule. This is where kids learn their mother language, a second language spoken in Switzerland and English. Languages are given quite the emphasis in the Swiss education system and therefore most of the people there speak at least a second language.

Natural sciences are also given a lot of importance where math, biology, chemistry, physics and geography are taught as some of the main subjects. Other subjects like history, civic education, music, art and design, physical education and health, home economics, career guidance, and vocational preparation are also taught.

Students are graded with numbers from 6 (being the highest) to 1 (the lowest). They also take tests at the end of the school year, where they get to test their gathered knowledge. At the end of the year, depending on performance, students either continue to the next year or repeat the previous one.

Upper secondary education

High school in Switzerland is not compulsory and its structure is unique. After the students finish their lower secondary education, they can choose what kind of career they want to pursue. They get to choose from a Vocational education and training (VET), Baccalaureate or Upper secondary specialized school.

A vocational school is a preferred choice for most young Swiss students. That is where they get to learn skills in school and see how it is done through the first-hand experience in companies that provide apprenticeships and cross-company courses. They last from 2 to 4 years and allow the students to gain technical and practical knowledge. Through VET school, students get a federal VET diploma or a federal VET certificate that will enable them to work and continue further education.

A Baccalaureate school is the choice of $\frac{1}{3}$ Swiss students. It can be considered a continuation of the lower secondary education, where the subjects are similar to the previous ones in lower secondary education, plus law and economics. The enrollment is based on the student's previous performance, and the grading system is the same as in lower secondary education. These schools usually last from 3 to 6 years depending on the canton.

At the end of the program, students must take a test and present a baccalaureate essay. Depending on their performance, they are granted a certificate called Matura and are then eligible to start cantonal universities, institutes of technologies, and teacher-training universities.

Upper secondary specialized schools provide specialized preparation for a career in healthcare, social work, and education. Students are graded with the same grading system as previously and study the core subjects related to the specific occupation. The grades determine whether the students progress to the next year or not. The program lasts three years and in the end, the students must take a final exam in six of their subjects. If they finish that successfully, they get the upper-secondary specialized school certificate and continue their education in PET colleges.

Tertiary

Tertiary education is a continuation of the higher secondary education a student chooses. Switzerland's education ranking has regularly been among the top 100 worldwide when it comes to higher education.

University is the logical choice for the students with a Matura, while Fachhochschule or Höhere Fachschule is the choice of students with a VET certificate. Students with an upper-secondary specialized school certificate will continue their education in PET colleges.

University

Switzerland has 12 universities in total, 10 of which are run and regulated by the cantons and offer theory-oriented majors. There's a wide range of subjects offered to study at Swiss universities.

Furthermore, universities in Switzerland are known for their BBAs and MBAs and they tend to be some of the best in the world. This has resulted in a considerable number of international students in the country. Many of these students attend top universities in Switzerland for their advanced degrees.

Some of the best-known universities in Switzerland are the federal institutes of tertiary technology and the universities of teacher education.

Fachhochschule or Höhere Fachschule.

There is one private and eight public Universities of Applied Sciences. These higher education institutions offer students degrees in topical areas such as engineering, technology, or business, and they prepare students for the Swiss and global labor markets.

The difference between a University and a Fachhochschule is that the latter generally doesn't award a doctoral degree. Fachhochschule also differs from the traditional university in Switzerland because of its practical-based teaching as opposed to a theory-based orientation.

CURRICULUM FRAMEWORK OF UNITED STATES OF AMERICA (USA).

INTRODUCTION

The United States of America entered the 21st century as the world's sole superpower after the fall of the Soviet Union in late 1991. It is the third most populous country in the world with 327.7 million people; the largest economy; has the most powerful military, among others. The U.S. also has the second largest higher education system (China overtook the U.S. in this regard roughly around 2001) and it is the top destination for globally mobile students. Fewer U.S. students go abroad for their education – either for a degree or short-term credit – though the number of such students has increased steadily over time. The reputation and diversity of its higher education system, along with opportunities to work in the country are among the top reasons that students are drawn to the U.S.

Administration of the U.S. Education System.

The federal governmental system of the U.S.A, along with a historical distrust of a strong central government has resulted in the nation's highly decentralized education system. The country's long-standing belief in free market capitalism and unfettered private competition, as well as limited government intervention has also shaped the education system. The United States Constitution makes no explicit mention of education, but the 10th Amendment to the Constitution states that all powers not specifically delegated to the federal government are delegated to state governments. As a result, the 50 states, as well as the District of Columbia and the territories are directly in charge of their own education systems resulting in tremendous diversity of education nationwide.

Role of the Federal Government

While the federal government is not directly in charge of education at any level nationwide, it does play a role in a variety of ways. The person most directly responsible for education policy at the federal level is the Secretary of Education who is in charge of the Department of Education.

The Role of the States.

Because states are ultimately responsible for education within their jurisdictions, most state constitutions include provisions for how education is to be administered. According to the U.S. Department of Education, state governments are generally responsible for the following:

- i) Developing curriculum guidelines and performance standards.
- ii) Providing technical assistance to school districts and schools.
- iii) Licensing private elementary and secondary schools to operate within their jurisdictions.
- iv) Licensing or certifying school teachers and administrators.
- v) Administering statewide student achievement tests.
- vi) Developing accountability plans and reporting on student performance to the U.S. Department of Education
- vi) Defining minimum requirements for high school graduation.
- vi) Distributing state and federal funding to school districts.
- viii) Establishing the minimum number of school days per year.
- ix) In most states, a state board of education is in charge of educational policy making.
- xi) The state department of education typically implements all education-related policies within the state. The leader of the department may be called "superintendent, commissioner, director or secretary of education," according to the Education Department. Education is one of the largest budget items among all states and territories.

Elementary and secondary (K-12) education at the local level is provided by school districts, which run all public schools within their jurisdictions.

Local school districts are generally responsible for carrying out state education policy at a local level. They are often granted a good deal of autonomy in terms of budgeting and teaching of the curriculum and they are responsible for hiring and supervising teachers and for raising funding, usually through local property taxes.

The head of each individual school at the K-12 level is known as a principal, though the name of this position may vary some among private schools.

The Academic Calendar.

The school year usually runs from mid-August to mid-September to anywhere from mid-May to mid-June at schools and higher education institutions alike. There is typically a one- to three-week break that encompasses the Christmas (December 25) and New Year holidays, usually called winter break. There is usually also a one-week break sometime during the spring, often (but not always) the week before the Easter holiday in March or April, called spring break.

Higher education institutions use different models to sequence their programs. The semester system divides the academic year into two halves known as semesters, which are usually 15 or 16 weeks long. There is a fall semester that runs from the start of the school year to winter break and the spring semester begins after winter break and runs until the end of the school year. Most institutions offer an optional summer semester, which may be shorter and more intensive than the fall or spring semester.

OVERVIEW OF K-12 EDUCATION.

Elementary and secondary education in the U.S. are often collectively referred to as K-12 education, a kind of shorthand that refers to the grades through which students progress. The “K” refers to kindergarten, typically housed in the elementary school system. Thus, K-12 education is 12 years long for most students, plus kindergarten. As part of K-12 education, an elementary school typically enrolls students from kindergarten or sometimes first grade through the fifth or sixth grades. Students then move on to a lower secondary school known usually either as a middle school or a junior high school. The last four or sometimes three years of school are usually known as high school, or sometimes senior high school.

Since education in the U.S. is so highly decentralized, various models of K-12 education exist.

K-12 education is virtually universal. All students are guaranteed a free public education through the 12th grade. Most students in the K-12 system study at public schools; others study at sectarian or parochial institutions (schools affiliated with a faith tradition, such as Catholic schools, Baptist schools, Lutheran schools etc.). The minimum compulsory school age varies by state and ranges from age 5 to age 7, though children generally enter kindergarten at age 5. The maximum mandatory age of education also varies per state, usually from 16 to 18 years of age, or in some cases, students can drop out earlier if a certain grade level is completed. Most states mandate the offering of free public education to students until they reach a certain age (if not complete), with most ranging from ages 19 to 22.

Admission into public schools is usually automatic based on residency. While private schools and some so-called magnet schools for gifted students may require additional admission requirements, there are generally no admissions tests at public secondary schools. High-demand schools may assign seats by lottery. English is the main language of instruction in virtually all schools, with the exception of some territories, notably Puerto Rico, where Spanish predominates. State policies also vary regarding instruction in other languages.

ELEMENTARY EDUCATION.

Elementary schools typically teach students from kindergarten through either grade five or six. Kindergarten is traditionally structured as an early childhood education program, a “pre-primary” program, but it is often housed in elementary schools. Public kindergarten is free. Some kindergartens, however, are stand-alone programs and there are a number of other institutions that provide different forms of preschool education including day care centers and nursery schools. Pupils can enroll in some preschool programs at the age of 3, but typically they start kindergarten at age 5. Programs can be either a full day or a half day.

Beginning with the first grade, all students receive full-time academic instruction. Usually, elementary school students receive instruction in their core subjects from the same teacher for the entire year, but may be instructed by specialist teachers in subjects such as art, music, physical education (PE), and computer science. States determine the curriculum and districts schools and individual teachers decide how to implement it.

Typical core subjects are language arts (reading, writing, grammar and spelling in English), mathematics, science, and social studies (geography, state history, U.S. history, world history). Students may also receive instruction from their regular teacher or specialist teachers in art, music, drama, PE, computer science and other subjects. They often also visit the library on a weekly basis.

SECONDARY EDUCATION; MIDDLE SCHOOL AND HIGH SCHOOL.

Typically, middle school or junior high school (lower secondary education) encompasses sixth through eighth grades, seventh and eighth grades, or occasionally seventh through ninth grades. Sixth grade usually starts around age 11.

High schools mostly run from 9th grade to 12th grade. Some such schools run 10th grade to 12th grade. Ninth grade usually starts around age 14 and students usually graduate around age 18. In some cases, middle school and high school are combined into one school running from sixth or seventh grade until 12th grade. In most middle schools and high schools, students have five or six periods of instruction each day and a different teacher for each subject. Teachers usually stay in their own classrooms and students change rooms for each class subject, moving during short “passing periods.” Student schedules are not uniform in most cases.

The typical curriculum at both the middle school and high school include; PE. Mathematic. High school (9th through 12th grade) curriculum includes the following year-long courses:

- i) Four years of English.
- ii) Four years of history or social studies.
- iii) Three years of mathematics.
- iv) Three years of science.
- v) Two years of foreign language.
- vi) Two years of the arts.
- vii) Four years of technical, vocational or business education.
- viii) One year of computer science.
- ix) Two to three years of others subjects.

Just as with elementary education, states determine the general core curriculum and high school graduation requirements. School districts, schools and teachers then decide on implementation. Either the state or the school districts selects the textbooks.

In many parts of the country, foreign language learning including American Sign Language (ASL) does not start until high school. By far, the most common foreign language to study is Spanish, with almost 70 percent of K-12 students learning this . French, German, Chinese and Latin are the next most commoners. Chinese language in particular is rising in popularity.

During middle school and high school, students can often select one or several electives per year. Some electives may be specialized offerings subjects related to core subjects, particularly at the high school level. For example, in lieu of or in addition to a standard English class, students may be able to take a course in American literature, British literature, or women’s literature. Other electives may focus on student interests and abilities, such as visual art, music, drama, debate or the school newspaper. Some are also meant to teach practical life skills. For example, a once common elective class, home economics teaches students how to cook, sew and perform other household tasks, but has become less common.

In many middle schools and high schools, there is a short class usually known as homeroom, typically at the very beginning of the school day. The purposes of homeroom can also vary, but they are usually designed to record daily school attendance, give daily announcements to students and generally provide a place for students to receive guidance from a specific teacher. Usually, a teacher of any subject can serve as a homeroom teacher. Students may or may not be given a grade for homeroom and it may or may not appear on transcripts or other student academic records.

Extracurricular and co-curricular activities.

Extracurricular and co-curricular activities are usually offered at most middle and high schools and are often make up a large part of students’ school experiences. Extracurricular activities mostly take place outside of regular class time and do not provide credit toward graduation. Sports and theatrical productions (such as plays or musicals) are among the most common. Other extracurricular activities such as chess, science projects or

foreign language study, may be organized by student clubs. During the college admissions process many universities and colleges look favorably at student involvement in out-of-class activities.

Credits.

Most high schools nationwide use a credit system to measure students' progress toward graduation. The main credit system in use is the Carnegie credit system. A Carnegie credit, sometimes also called a unit, represents 120 hours of instruction per year. It generally also represents one hour of class time five days a week for one school year.

High School Graduation Requirements and Specialization Tracks.

The states set minimum graduation requirements for all high schoolers in their jurisdictions. Typically, states require a minimum number of credits or units, usually in the Carnegie system. In most cases, a minimum of six or seven credits earned per year is the standard during high school. Most states specify the number of credits that must be earned in certain areas of study.

The following four core areas of study are used in almost all the states:

- English/language arts
- Social studies
- Science
- Mathematics
- Most states have requirements in other areas as well such as arts, PE, or foreign language.
- In addition to completing stipulated course requirements and earning a certain number of credits, some states require students to pass a high school exit examination.

In most states, high school students are placed into specialization tracks (tracking), usually based on a combination of aptitude (usually determined by statewide standardized exams) and students' post-graduation plans. The three tracks below are the most common;

i) General track: Curriculum is designed for students to meet the minimum state graduation requirements.

ii) Vocational track: Curriculum meets the state graduation requirements and includes courses in specific vocations.

iii) Academic or college-preparatory track: Curriculum exceeds the state graduation requirements and prepares students to meet university admissions requirements, typically those of the state's public universities, through additional coursework in certain areas such as English, mathematics, or foreign language. Advanced Placement (AP) courses, may be part of this track.

Some high schools also offer an honors program, which typically includes more rigorous coursework that follows the academic or college-preparatory track, for "high-achieving students." In some high schools, honors courses are given more weight in the calculation of the overall grade point average (GPA). For example, an institution may equate an A in an honors course to a grade of 5.0, instead of 4.0, thus allowing honors students to achieve higher than a 4.0 GPA. This distinction can benefit them in the university admissions process.

Advanced Placement.

Advanced Placement (AP) is a formal program offered by the College Board in the U.S. at high schools nationwide (and in U.S. high schools around the world). Students take rigorous, college-level courses usually at their own high schools that count both towards high school graduation and potentially for university or college credit. At the end of the school year, students take the AP examination in the subject they studied. Students must pay for the exams. In most examinations, students are scored on a 5-point scale. A score of 5 is generally considered equivalent to a grade of A, a score of 4 is equivalent to a B, and so forth. A score of 3 is usually considered the minimum passing score.

U.S. universities and colleges vary in their policies on accepting AP coursework, but many do, usually with a score of 3 or higher. Most AP courses satisfy general education requirements or sometimes foundational courses in certain fields of study.

There are 40 courses and examinations available, in areas such as the arts, English, history and social studies, math and computer science, the sciences, and world languages and cultures. All of them are listed on the College Board's website.

SPECIAL EDUCATION AT ALL LEVELS.

Education provided to children with disabilities—including physical, mental, emotional and developmental disabilities—is known in the U.S. as special education and is guaranteed by federal law to children in the K-12 public school system. Students can receive special education in both school and non-school settings, such as medical institutions. The primary federal law guaranteeing special education is the “Individuals with Disabilities Education Act (IDEA)”.

IDEA denotes 14 categories of disabilities for which students may receive special education.

- Autism
- Deaf-Blindness
- Deafness
- Developmental Delay
- Emotional Disturbance
- Hearing Impairment
- Mental Retardation
- Multiple Disabilities
- Orthopedic Impairment
- Other Health Impairment
- Specific Learning Disability
- Speech or Language Impairment
- Traumatic Brain Injury
- Visual Impairment Including Blindness

Most special education students are schooled in regular public schools, as opposed to special institutions, and most learn in mainstream classrooms, often with classroom aids. Some see special education teachers during at least part of their day. In some cases, at the secondary level, special education for some students may be its own track.

HOMESCHOOLING.

Homeschooling is legally recognized nationwide and regulated by the states. In the majority of cases, parents do the teaching, but sometimes a teacher or tutor is hired. There are a variety of reasons why parents choose to homeschool. Among the most common reasons are concerns about the quality, safety, or social environment of local public schools; as well as the desire to provide instruction at a pace appropriate for the student and often the desire to provide a religious-based instruction.

Many states allow homeschooled students to attend some public school classes or to participate in co-curricular or extracurricular activities, such as sports. An increasing number are participating in online education. Few states, if any, award accredited high school diplomas to homeschool graduates. However, a parent's certifying that the student has completed the program is usually recognized as being formally equivalent to high school graduation. It might, however, be more difficult for home-schooled children without an official high school diploma to pass muster with employers, enlist in the U.S. military or gain admission to higher education programs.

Secondary Grading Scales.

There are no mandatory national grading scales in the U.S. As a result, there is considerable variation between grading scales used by high schools throughout the country, even though most use a 100-point scale. However, it should be noted that many institutions also use pluses and minuses or different numerical ranges, including varying minimum passing grades. Most schools also calculate a student's GPA, which allows the school to measure overall achievement for each student. GPAs are frequently reviewed in the higher education admissions process.

HIGHER EDUCATION

Types of Institutions

With more than 4,500 accredited degree-granting higher education institutions nationwide, the U.S. higher education system is extremely diverse. There are many different types of higher education institutions (HEIs) in the country. There are no nationally standardized definitions of "university" or "college," and the name of an institution alone may not indicate exactly what type of institution it is. A university, at minimum offers bachelor's programs and at least some master's programs.

Colleges, on the other hand, can vary quite a bit. Community and junior colleges mostly focus on associate degrees, while baccalaureate colleges award bachelor's degrees. However, institutions with the title "college" may also be university-level institutions that offer graduate programs. For example, Boston College is a full-fledged research university awarding doctoral degrees. It has retained its name most likely because of its prominent neighboring institution, Boston University.

In addition, there are various big, multi-disciplinary state universities and a large number of private universities that offer master's and doctoral programs. There are also mono-disciplinary, special focus institutions that offer programs in professional, vocational, and arts fields at different levels of education.

The following describe some of the many types of institutions in the U.S;

- Community colleges (also called junior colleges)
- Liberal arts colleges
- Historically black colleges and universities (HBCUs)
- Tribal colleges
- Hispanic-serving institutions
- Master's colleges and universities
- Doctoral-granting universities
- Special-focus HEIs

By international standards, the number of private institutions in the U.S. is high.

The Role of Community Colleges.

Community Colleges (or junior colleges) play an important role in the U.S. education system and have greatly expanded access to education since their inception at the beginning of the 20th century. Hailed as "people's colleges," community colleges were initially designed to provide a low-cost, publicly-funded higher education alternative for broader segments of society while simultaneously freeing up traditional universities from providing general education, allowing them to concentrate more on advanced study and research. While this clear-cut distinction never fully materialized, two-year community colleges certainly have proven to be a highly successful model and today enroll a large portion of U.S. students.

Community colleges are usually open admission or open access—applicants need only have a high school diploma or its equivalent to be admitted—and charge, on average, much less tuition than universities. This makes them an important vehicle for educational and social mobility for lower-income groups, minorities, and

immigrant populations. Community colleges are also a popular option for older adults who want to upgrade their education and skills.

University and College Governance.

As with K-12 education, states are in charge of overseeing all post-secondary education within their jurisdictions. States usually set the general framework for higher education within the state, particularly in regard to public institutions, and license or charter institutions to operate. However, universities and colleges have, in general, a high degree of autonomy and self-governance.

Various models of governance in higher education have emerged over time. In some states, major public universities such as the University of Michigan function with almost full autonomy. Others are more subject to supervision from the state government. Virtually all post-secondary institutions, public and private, have a board of trustees or board of regents who govern the institution. For public institutions, this board is usually appointed by the governor or state legislature, though in some states, they are popularly elected. At private institutions, board members are appointed by the governing board. Community colleges are more likely to be locally run, featuring a board of local members that is either appointed or elected.

Most institutions have a chief executive typically called the president or sometimes chancellor who is in charge of running the institution. This individual is usually hired by the board. The board usually assigns the day-to-day operations of the institution to the president, who usually hires other senior administrators. In some states, particularly those that are larger, public universities may be part of a university system. This usually produces another layer of governance between the state government and the individual institution. The university system typically has a board appointed by the state authorities and a chancellor in charge of running the system. As the American Council on Education (ACE) notes, system officials “oversee campus budgets, set policies such as admissions standards, coordinate degree programs, and facilitate credit transfer and articulation between the state’s public colleges and universities.” They also represent their interests to the state and sometimes federal government. Still, even within a university system, universities still have a considerable degree of autonomy.

Funding.

Funding of U.S. higher education institutions is highly complex and many institutions make use of a variety of sources of revenue. However, there are some key differences between public and private institutions. Additionally, there is one major trend that has emerged over the last several decades among all types of institutions, but particularly public universities: the growing dependence on student tuition dollars as a source of revenue.

Public institutions, which are under the authority of the states, have traditionally received most of their funding from state governments. Private institutions, which receive little or no direct government funding, have long relied on student tuition and fees for revenue. Nonprofit private institutions, relative to public institutions, have tended to enroll students from higher income households. For-profit institutions often enroll lower income students who rely heavily on federal student aid. However, many institutions, particularly private nonprofits, increasingly offer “tuition discounting,” a common term for institutional financial aid, to students who cannot or who are unwilling to pay the full price of tuition.

Both public and private (particularly nonprofit) institutions have developed various other forms of revenue aside from student fees (and, by extension, federal and state student aid) and state appropriations (for public institutions). Other common sources of revenue are private gifts (often from alumni), investments and holdings, and research grants from government, nonprofit, and industry sources.

Accreditation and Quality Assurance.

States are responsible for authorizing universities and colleges to operate within their jurisdictions. Accreditation, however, is a voluntary process for institutions. Unlike in other countries, where accreditation is

often provided by governments or state-authorized organizations, accreditation bodies in the U.S. are private nonprofit organizations.

TEACHER EDUCATION AND TRAINING.

Like most professions, the teaching profession is regulated at the state rather than the national level, but there are some common practices. There are two types of teacher certification; traditional and alternative.

Most teachers are certified using the traditional method, within which are two main routes:

i) Earn a bachelor's degree in education (particularly for elementary education) or in a subject content area (for example, English, mathematics) with a focus on education, depending on the state's requirements. This tends to be a faster method and usually works best when students know fairly early on in their postsecondary education that they want to become teachers.

ii) Earn a bachelor's degree in another subject and then complete a post-bachelor's program, either a certification program or a full master's degree. This is the typical method for individuals who discover late in their undergraduate program or later that they wish to teach.

Both routes are sometimes known as Teacher Preparation Programs (TPPs). Most states require that students pass an entrance examination in reading, writing, and math in order to enter a post-bachelor's TPP, regardless of the intended subject or level of teaching. The most common of the examinations required is Praxis, developed by the Educational Testing Service (ETS).

Many states impose other, specific additional requirements—either in the form of coursework (typically included in TPPs) or examinations—that future teachers must complete in order to be eligible for a teaching certificate. Once they meet all the requirements are met, future teachers can apply to the state agency for their certification in the state where they wish to teach.

Student teaching is a common part of TPPs. Regulations vary by state regarding the length and type of student teaching required of prospective teachers. According to a study by the National Council on Teacher Quality (NCTQ), about half of all states require at least 10 weeks of student teaching with a “cooperating teacher,” an ill-defined term for a current teacher, either explicitly required or assumed to be experienced, under whom the student teacher works. Individual TPPs then set their own requirements and parameters for student teaching in compliance with state regulations. There are alternative routes by which individuals can be educated, trained and certified to be teachers. One method, depending on state regulations, is to immediately start teaching and earn certification in the process. This uncommon method is usually used in schools and subjects that urgently need teachers, where teachers who have been certified via the traditional methods are in short supply.

Once certified and teaching, teachers are generally required or expected to continue professional development, and many go on to eventually earn master's degrees. After three years of teaching experience, teachers may work toward the completely optional National Board Certification (from the National Board for Professional Teaching Standards), which can help teachers advance in their careers, earn higher salaries, and more easily transfer to work in another state.

TECHNICAL AND VOCATIONAL EDUCATION AND TRAINING (TVET).

TVET in the U.S. is usually referred to as Career and Technical Education (CTE). It is not standardized compared with vocational training systems in other countries. CTE is delivered in various different forms and by various different institutions. For example, there are thousands of small proprietary career colleges, most of them offering sub-degree, certificate-level programs in specific vocations. In addition, CTE is delivered by community colleges or through work-based apprenticeship programs. Private institutions constitute the majority of providers in the CTE sector.

At the secondary level, CTE can be offered in specialized high school courses, or in programs at secondary-level career academies that combine career training with college preparatory curricula.

At the post-secondary level, community colleges offer a multitude of applied, career-oriented two-year programs leading to degrees like the Associate of Applied Science, Associate of Applied Arts or Associate of Applied Technology, as well as sub-degree certificate programs. Four-year institutions offer some of these programs as well. Though most programs include some general education coursework, they are primarily focused on applied technical training in specific vocations and may include an internship component.

In addition, there are programs that straddle secondary and post-secondary education. So-called Tech-Prep programs, for instance, combine two or more years of secondary education with two years of post-secondary education, concluding with the award of an applied associate degree as well as a high school diploma.

Furthermore, apprenticeship programs provide CTE through part-time classroom instruction that augments practical training at a workplace. More than 500,000 trainees were enrolled in formal apprenticeship programs registered with the federal government in 2017. Formal apprenticeship programs last between one and six years. CTE in the U.S. is highly decentralized. Licensing and training requirements vary widely between states. Different states have different education and licensing requirements for different occupations, so that training and certification obtained in one state may be difficult to transfer to another.

CURRICULUM FRAMEWORK OF BRAZIL.

The School System

Certain levels of education are compulsory for all children living in Brazil while others such as pre-school and upper secondary school remain optional.

Pre-school education.

Pre-school education is entirely optional and for pre-school children there is a choice between:

- Maternal or state- for children aged two to five
- Jardim, for children from three to six years of age

The Maternal is basically a playgroup whereas Jardim has a more academic focus for small children. The school meets with the parents and child before enrollment and depending on the academic and social skills of the child the school will advise parents on where it feels the child will best fit in. All state pre-schools and nurseries are free of charge.

Primary and lower secondary education.

In Brazil it is mandatory for children to go to school from age 6 to 14. Children under the age of six may be enrolled as long as they turn six in the first semester.

These compulsory nine years of education are known as Fundamental Education (Ensino Fundamental) and are divided into two levels:

i) Ensino Fundamental I.

ii) Ensino Fundamental II.

A core curriculum is set by the Education Council. During the Ensino Fundamental I, children study mathematics, Portuguese, science, arts, history, geography and physical education. During Ensino Fundamental II, pupils also study at least one other compulsory language.

The normal practice in Brazilian schools, both public and private is to mix all academic levels together in the same class.

Under the state system, an exam is given to all pupils at the end of each academic year to determine whether the child will move on to the next year or whether they will be held back to repeat a year. It's not unusual to hold a child back, which means that the age mixture of classes is often quite varied.

Depending upon the pedagogic philosophy of the private school, children may be required to return and repeat a year, although this is not a very common practice these days. If it becomes obvious that a child is not managing

to keep up with the advance in learning levels of their classmates, the school will speak to the parents about options involving other academic systems.

There are some privately-run schools, which divide a year into levels of academic ability for classes, in line with the British state model, for instance. This is however becoming less common as the private schools adopt more tightly controlled pedagogic models.

Upper secondary education.

Upper secondary education (Ensino Médio) is for young people aged 15 to 18. On top of the core curriculum subjects studied during Ensino Fundamental, students will also study philosophy and sociology.

The courses provided during this period are essentially designed to allow a young person to enter into a (usually public) university. The courses can be conducted either at the private school the child has been attending up until this point or at specific colleges which train youngsters to take the specific entrance exam set by a specific university. Students can also choose to take professional training at the same time.

Once a student has successfully completed secondary education, they may continue their studies at a public or private university. To enter a public university, students must sit an entrance exam known as vestibular. Entrance exams to a private university are often little more than a formality and as a consequence, public university degrees are valued much more highly than those from private institutions.

Home schooling early years

Brazilian law demands that a child start their compulsory education at six years old. Should the parents feel that the schooling options available where they live are insufficient for their child's needs, they may apply to the government for the right to provide their own education at home. This is a lengthy process, involving the Ministry of Justice and Ministry of Education.

Private Schools.

Within the private school system, numerous pedagogies can be found from which to choose a child's educational direction. These systems are largely based on existing European models and essentially focus on the degree of attention placed on the arts and humanities versus the scientific and technical subjects, while attitudes towards disciplinary measures and the rigidity of teaching methods also varies.

CURRICULUM FRAMEWORK OF OCEANIA COUNTRIES.

Australia and New Zealand

CURRICULUM FRAMEWORK OF AUSTRALIA.

Structure of education system

Pre-higher education:

Duration of compulsory education:

- Age of entry: 6
- Age of exit: 15

Structure of school system:

i) Primary

- Type of school providing this education: Government and Non-Government Primary Schools
- Length of program in years: 6
- Age level from: 6 to 12

ii) Junior Secondary

- Type of school providing this education: Government or Non-Government Co-Educational Comprehensive/Multi-Purpose High School
- Length of program in years: 4
- Age level from: 12 to 16

- Certificate/diploma awarded: Junior Secondary Certificate of Education (Year 10 Certificate)
- iii) Senior Secondary
- Type of school providing this education: Government./Non-Government Co-Educational Comprehensive/Multi-Purpose High School/Senior College
 - Length of program in years: 2
 - Age level from: 16 to 18
 - Certificate/diploma awarded: Senior Secondary Certificate of Education (Year 12 Certificate)

School education.

Primary and Secondary education in Australia is delivered through both government and non-government providers. State and Territory governments have major responsibility for government school education and contribute substantially to funds for non-government schools. Most students are enrolled in government schools which operate under the direct responsibility of the State or Territory Education Minister, while the rest choose non-government schooling. The Federal Government's role with respect to schooling includes the provision of significant supplementary financial support to government and non-government school authorities to support agreed priorities and strategies.

Primary education

Schooling is compulsory from age 6 to 15 in all States except Tasmania, where it extends to 16. However, in most States, children start primary school at the age of 5 when they enrol in preparatory or kindergarten year, after which primary education continues for either six or seven years depending on the State.

Secondary education

Secondary education is available for either five or six years, depending on the State and the length of primary education. Students usually commence their secondary schooling when aged 12 or 13, reaching year 12 at 17 or 18. Links between the education and training sectors have been strengthened, partially through the introduction of the Australian Qualifications Framework which is a unified system of twelve national qualifications in schools, vocational education and training (Technical And Further Education Colleges and private providers) and the higher education sector (universities). The Framework links together all these qualifications and is a highly visible, quality-assured national system of educational recognition, which promotes lifelong learning and a seamless and diverse education and training system.

Higher education.

The main purposes of Australian higher education are to enable individuals to develop their capabilities for effective participation in the workforce, for constructive contributions to society and for personal growth and fulfilment; to advance knowledge and understanding; aid the application of knowledge and understanding to the benefit of the economy and the society; enable individuals to adapt and learn, consistent with the needs of an adaptable knowledge-based economy at local, regional and national levels; and contribute to a democratic, civilised society. Australian universities are autonomous self-accredited institutions established by Federal, State or Territory legislation.

Non-university level post-secondary studies (technical/vocational type).

Vocational education and training (VET) is the term used in Australia to describe the sector which prepares Australians of all ages for employment and to improve the knowledge and skills of those already in the workforce. Since the late 1990s VET has been available in all three sectors of the education system—school, postsecondary and higher education—but the most important of these is at the postsecondary level.

The postsecondary vocational education and training sector is diverse and includes government-funded Technical and Further Education (TAFE) colleges and other Government providers (public registered training organisations), private registered training organisations, and community-based registered training organisations. These training organisations provide a wide range of vocational training for trade, technical, professional and para-professional occupations as well as a range of adult education, leisure and general enrichment programmes.

University level studies:

University level first stage: Undergraduate Studies.

Access to undergraduate education is through the Senior Secondary Certificate of Education. The main stage of university education leads to the Bachelor's Degree. Undergraduate studies last between three years (Arts, Science, Commerce), four years (Education, Engineering), five years (Veterinary Science, Dentistry, Architecture) and six years (Medicine and Surgery) full-time.

University level second stage: Postgraduate Studies.

A graduate with a Bachelor's Degree can proceed to a one- to two-year postgraduate course leading to a Postgraduate Diploma in a particular subject. A student who has qualified for a Bachelor's Degree (Honours) at a sufficiently high level may proceed to study for the Master's Degree which constitutes the second stage of higher education.

University level third stage: Higher Doctorate.

Many faculties award a Higher Doctorate in Science (DSsc) or Humanities (DLitt) upon submission of published work representing a substantial contribution to knowledge in a particular field.

Teacher education:

Training of pre-primary and primary/basic school teachers

There are no national requirements for the teaching profession in Australia. States and Territories have responsibility for those universities within their jurisdictions that offer teacher education, although the Federal Government is the main source of funding for those institutions. The responsibility for school systems and the employment of teachers lies with both government and non-government education authorities in each State and Territory. Programmes for preparatory or pre-primary teachers are often referred to as early childhood education/teaching programmes. They provide education in the care and teaching of children from birth to 8 years of age, or from age 3 to 8 years. Programmes may be at the undergraduate or postgraduate level. The most common awards are a Bachelor of Education (Early Childhood Education) or a Bachelor of Early Childhood Education, both of which require 4 years of full-time study. Alternatively, students who have already completed a 3-year Bachelor degree in a discipline other than education may complete a 1- or 2-year Graduate Diploma of Education (Early Childhood Education). Primary teacher education programmes provide education to teach children from ages 5 to 12 in years 1 to 6 or 7 or primary school. The programmes generally offer a balance of professional and curriculum studies to foster the intellectual, physical and social development of children. Students follow a four-year course leading to a Bachelor of Education (Primary). Alternatively, students who have already completed a Bachelor degree in a field other than education may complete a one- or two-year Graduate Diploma in Education (Primary) or a two-year postgraduate Bachelor of Education (Primary). These programmes provide a combination of professional studies in education and methodology and supervised teaching practice in primary curriculum subjects. There are also a number of four-year combined degrees such as the Bachelor of Arts/Bachelor of Education (Primary) and Bachelor of Applied Science/Bachelor of Education (Primary) which combine primary education studies with studies in a discipline related to teaching. There are also some programmes which prepare teachers to teach at both primary and secondary school - for example, the Bachelor of Education (Upper Primary/Lower Secondary) and Bachelor of Education (Primary and Secondary). Training of secondary school teachers.

Secondary teacher education programmes provide education to teach pupils of around 12 to 18 in Years 6 or 7 to Year 12 of secondary school. Students follow programmes that provide a balance of integrated professional studies and curriculum studies in one or two disciplines in key learning areas appropriate for both the lower and upper years of secondary schooling. Programmes may be at the undergraduate or graduate levels. The most common undergraduate award is the Bachelor of Education (Secondary) which requires four years of full-time study. Students who have already completed a Bachelor degree in a relevant curriculum area may complete a two-year full-time Graduate Diploma of Education (Secondary) or a two-year full-time Bachelor of Education (Secondary). These programmes provide a combination of professional studies in education and methodology and supervised teaching practice in subjects appropriate to the first degree. There are also a number of combined degrees such as the Bachelor of Arts/Bachelor of Education which combine secondary teaching education studies with study in a discipline relevant to secondary school.

Teachers of Vocational Education and Training (VET) courses in schools are expected to be competent to conduct education and training programmes and to assess skills based on the industry-endorsed competency standards in the Training Packages developed by Australian industry bodies and endorsed under the National Training Framework (NTF). Teachers usually have qualifications and substantial employment experience in a specialised field covered by the NTF and hold the minimum of an Australian Quality Framework Certificate IV in Assessment and Workplace Training. Educational authorities also require teaching qualifications up to Bachelor degree level.

Distance higher education

Distance education has been a feature of higher education in Australia for at least 80 years. The geographic size of Australia and the relatively few large population centres in early years meant that for many distance education provided the only opportunity they had to gain a university degree. Traditionally, distance education was text-based, but Australia has been an early adopter of new education technologies and while text is still important, universities use a range of delivery modes, including on-line courses, email, telephone, video-conferencing, web TV, radio and television.

CURRICULUM FRAMEWORK OF NEW ZEALAND.

Early learning. (0–6 years old)

Early learning helps children to be confident and curious about the world. It also helps them do better when they go to school or kura.

Education system reflects our unique and diverse society, which welcomes different abilities, religious beliefs, ethnic groups, income levels and ideas about teaching and learning.

Education in New Zealand is a student-centred pathway providing continuous learning progression and choice so that students progress every year and their learning at one level sets the foundation for the next steps along a chosen pathway.

New Zealand's education system has three levels:

- Early childhood education: from birth to school entry age
- Primary and secondary education: from 5–19 years of age
- Further education: higher and vocational education.

Early childhood education

Early learning helps children to be confident and curious about the world. It helps them to do better when they go to school or kura, and it helps them develop important skills to become strong, happy and successful in later life.

Early Childhood Education (ECE) isn't compulsory but around 96.8% of children attend ECE.

There are different types of ECE services and all learning that children experience at an ECE service or Kōhanga Reo is guided by the Te Whāriki curriculum framework.

The government subsidises all children who attend ECE for up to six hours a day (a total of 30 hours per week). The 20 Hours ECE is a higher funding subsidy available for all children aged 3–5 years who attend ECE.

Primary and secondary education.

Primary and secondary schools are the second level of education.

Education is free between the ages of 5 and 19 at state schools (schools that are government owned and funded) if they're a New Zealand citizen or a permanent resident.

Schooling is compulsory from age 6–16. In the majority of schools, children can start school on the day they turn five years old (they don't have to wait until the start of a new school year). However, some schools have a policy of starting children at school together as a group at the start of each term (cohort entry). Most children stay at school until they're around 17 years old.

The education system for schools is made up of 13 Year levels. Primary education starts at Year 1 and goes to Year 8 (around 5–12 years of age). Secondary education starts from Year 9 to Year 13 (around 13–17 years of age).

Types of schools.

Local schools

Many children go to a school close to where they live. Many schools have an enrolment scheme called zoning. If you live in an area close to a school (the school's zone), your child is guaranteed to get a place at that school. If you want your child to go to a school outside the area where you live, you may have to apply and a place isn't guaranteed.

Depending on the schools in your area, you may have the choice to send your child to a single-sex or co-educational school.

State integrated and private schools

Most schools in New Zealand are owned and funded by the state (state schools). They teach the national curriculum and are secular (non-religious).

State-integrated schools are schools with a special character. They are funded by the government and teach the national curriculum. They'll have their own sets of aims and objectives to reflect their own particular values and are set within a specific philosophy or religion.

Private schools get some government funding but are mostly funded through charging parents school fees. They develop their own learning programmes and don't have to follow the national curriculum.

Māori-medium education (Kura Kaupapa Māori)

Māori-medium education is where students are taught all or some curriculum subjects in the Māori language for at least 51% of the time (Māori Language Immersion Levels 1-2).

Māori language in English-medium is where students are learning te reo Māori as a language subject, or are taught curriculum subjects in the Māori language for up to 50% of the time (Māori Language Immersion levels 3-5).

National curriculum

The national curriculum covers subjects that are taught at primary and secondary schools and the standards students should reach in each subject. Primary education focus on foundation learning across a range of subjects and competencies but especially in literacy and numeracy. At secondary school learners learn a broad and balanced curriculum, with some specialisation possible in Years 11–13.

English-medium schools use the New Zealand Curriculum. Māori-medium schools (Kura Kaupapa Māori) use Te Marautanga o Aotearoa (a curriculum based on Māori philosophies).

National Certificate of Educational Achievement (NCEA).

The National Certificate of Educational Achievement (NCEA) is the national senior secondary school qualification.

Assessment is during their last three years at school (Years 11–13). They can achieve NCEA at three levels in a wide range of courses and subjects.

Learning support.

Additional learning support is available in every local early childhood centre or school. It's planned to support students, educators, families and whānau in a range of different ways depending on individual needs. They include;

i) Home and distance learning

If attending a school isn't the best option your child can learn with New Zealand's correspondence school, Te Aho o Te Kura Pounamu (Te Kura).

Te Kura teaches early childhood, primary, secondary and special needs students using multimedia and online learning. Your child can also study one or two courses if a subject they want to study isn't available at their school.

Further education.

i) Technical and vocational education

At a senior secondary school level, students may begin to specialise in vocational learning. They can get help into work or further education from a number of programmes and institutions.

ii) Youth Guarantee

Youth Guarantee courses provide students aged 16–19 with an opportunity to study towards NCEA Level 1-3 or Level 1-3 certificates at tertiary providers free of charge. These courses are usually vocationally focused.

iii) Trades academies

Trades academies teach trades and technology programmes to students in Years 11–13 (ages 15–18). They are run through schools and other providers.

iv) Institutes of technology

Institutes of technology and polytechnics teach professional and vocational education and training from introductory studies to degrees.

Industry training organisations represent particular industries (for example, agriculture, building and construction, motor trade). They offer training and qualifications for those sectors. They funded by the government and industry.

v) Private training

Private training establishments offer specific vocational courses at certificate and diploma level (for example, travel and tourism).

vi) State-owned Māori teaching and research institutions (Wānanga)

New Zealand has three wānanga (state-owned Māori teaching and research institutions). They teach according to āhuatanga Māori (Māori tradition) and tikanga Māori (Māori custom). They offer certificates, diplomas and degrees. Some teach in specialised areas up to doctorate level.

vii) Universities

New Zealand has eight state-funded universities. Each university offers degrees in a large choice of subjects and has strengths in specialised professional degrees.

All of the universities are well recognised internationally. They work with universities in other countries on research and teaching programmes and with the business community in New Zealand and overseas on research and development.

9.2 BASIC EDUCATION CURRICULUM FRAMEWORK.

Learning Outcomes

Analyse the Basic Education Curriculum Framework in Kenya

Apply the Basic Education Curriculum Framework to the learning process in Kenya

Appreciate the need for Basic Education Curriculum Framework.

The Basic Education Curriculum Framework (BECF).

- Is the outcome of extensive stakeholder engagement, a national needs assessment study, deliberations from a national curriculum reform conference and several benchmarking studies.

-The Framework is aligned to the Constitution of Kenya, Vision 2030, the East African Community Curriculum Harmonization Structures and Framework, and other policy documents that express the aspirations of the country.

-The BECF is based on pillars which are anchored on values and underpinned by theoretical approaches and guiding principles.

-The BECF has been developed to actualise the curriculum reforms. Its purpose is to provide a comprehensive conceptualisation of reforms in basic education; pre-primary education, primary education, secondary education, and inclusive education

Meaning of Competency Based Curriculum

A competence based curriculum emphasizes what learners are expected to do rather than what they are expected to know

The curriculum is learner centered and adaptive to changing need of the learner, teachers and society.

CBC focuses on acquisition of competencies that enable the learner to cope up with life challenges.

B. E.F (BASIC EDUCATION FRAME WORK)

RATIONALE FOR CURRICULUM REFORMS

Reforms of Basic Education Curriculum were based on

constitution of Kenya 2010

State that every child has a right to free and compulsory basic education (EFA)

National values and principles of governance(article 10)

Chapter six -leadership and integrity

Promoting English as official language and Kiswahili as a national and official language

Encouraging teaching and learning of (KISL)

Development and promoting use of indigenous language

Emphasis on communication formats and technologies accessible to persons with disabilities

Kenya vision 2030 recommended

Integrating early childhood into primary education

Reforming secondary curriculum

Modernizing teacher training

Developing programs for learners with special needs among others

The task force report 2012 proposed

Major reforms to align the curriculum with constitution of 2010 and to address the defects of vision 2030

Inclusion of community service programmes in the school curriculum

Sessional paper (No 7 of 2019)

A policy framework for reforming education

Training and research for sustainable development recommended

Reforming the education sector to provide for the holistic development of learners

A competency based curriculum

Early identification and nurturing of talents

Introduction to values
Introduction to three pathways at senior secondary schools
Arts , sports and science 3 years
Social science
Stem
Recommended commencing of CBC

Summative evaluation of the 8-4-4 system of education 2009

Noted that the following with regard to the education system
It was more academic and examination oriented with emphasis on summative assessment as opposed
Graduator at secondary school level did not acquire adequate entrepreneurial skills for self reliance
It did not provide flexible education pathways for identifying and nurturing attitudes and talents. interest of
learners early enough in order to prepare them for work, core career progression and sustainable development
Identify the skill gaps as follows

Agricultural Entrepreneurial
Vocational technical skills
Innovation and creativity of ICT
It recommended a needs assessment survey

KICD Needs assessment survey 2016

Proposed
A curriculum reform
A wide curriculum to cater for various talent areas to enable learners achieve their potential and be better
equipped to develop themselves and the nation
A functional holistic curriculum that will produce an individual with morals, spiritual and social aspect or with
ability that will help him live amicably with others
Development and acquisition of competences at every level of learning
A flexible and comprehensive educational structure and system
East African community protocol made the following recommendations on
National goals of education
Structure for basic education
Key competence
Learning areas or subjects
Learning outcomes for ECDE, primary and secondary
Guidelines on assessment

Global standards

Such as
Internal bureau of education i.e. IBC recommends that the curriculum be renewed every five years
Sustainable for goal No. 4 consuming quality education 21st learning skills and approaches
21st century learners should be
Self directed learner
Innovator
Financially and economically literate
Information and media literate
Collaborators
Communicator
Critical thinkers

Civically engaged

Globally aware

A problem solver

Paradigm shift in the reformed education

From (less)	To (more)
Content focused	Competence focused
Rigid and prescriptive with limited flexibility	More flexible with opportunities for specialization
Primarily focused on	
Summative assessment and competition	More balanced formative and summative assessment of excellence
Emphasis on schooling	Emphasis on Education
From teaching	To learning

Vision

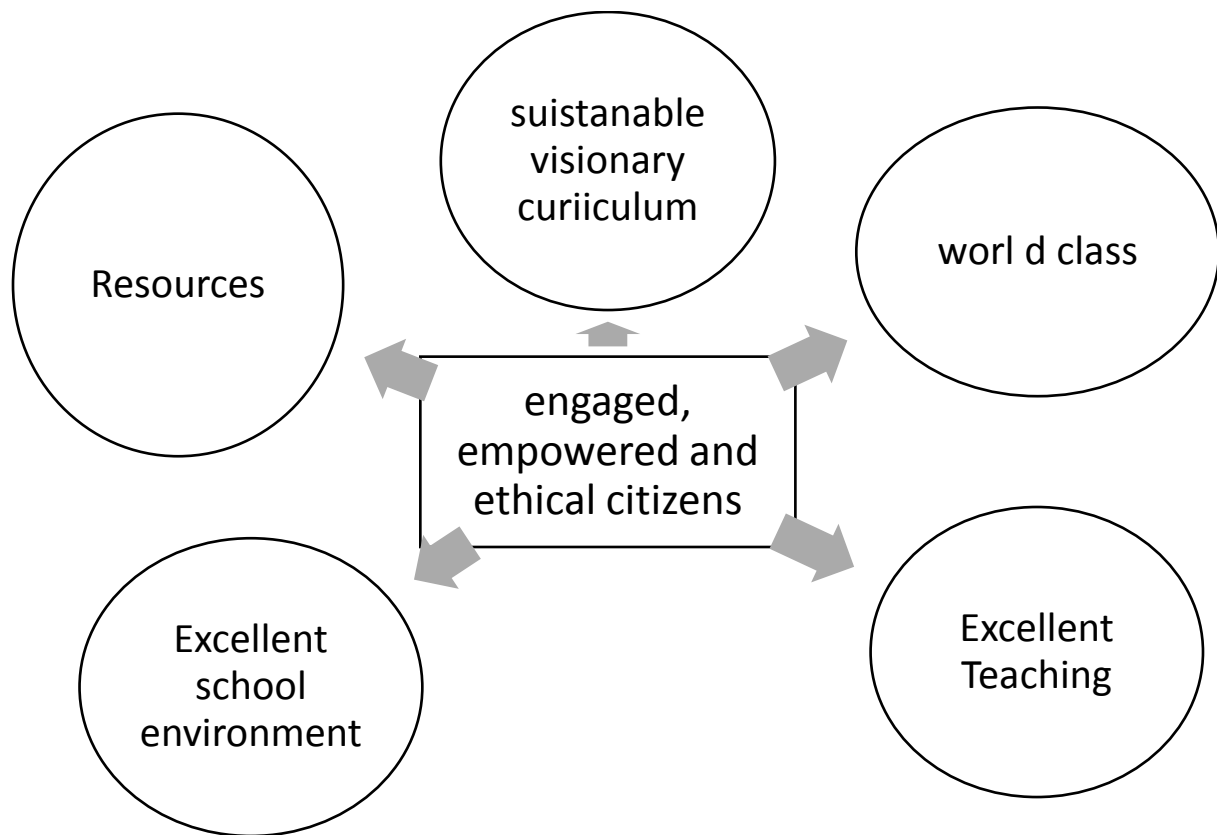
Enable every Kenyan to become an engaged empowered and an ethical citizen

Learners are facilitated to become useful members of the society

Mission

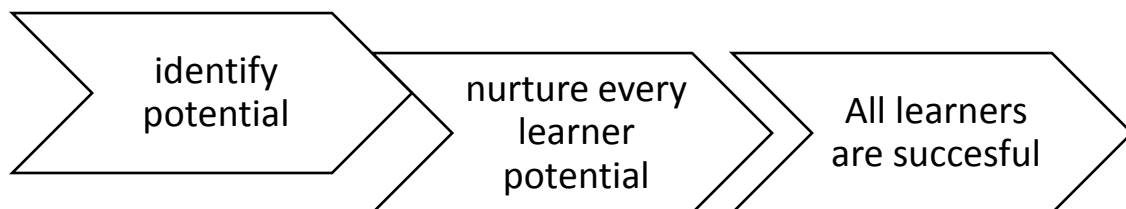
Nurturing every learners potential

All learners have abilities that can be nurtured for success



Curriculum reform mission

To nurture every learner potential



To nur
learner potential

ture every

National goals of education

Are statement that describe the aspirations of a country and the needs of the society that are to be met through education

Foster nationalism, patriotism and natural unity

Promote social economic, technological and industrial needs for national development

Promote individual development and self fulfillment

Promote sound moral and religious values

Promote social equity and responsibility

Promote respect for and development of Kenyans rich and various cultures

Promote international consciousness and fosters positive attitudes towards other nations

Promote positive attitudes towards good health and environment protection

Pillars of BECF

Values

Theoretical approaches

Guiding principals

Value bases Education (VBE) love , responsibility, respect, unity, peace, patriotism , social justice, integrity.

GUIDING PRINCIPLE OF BECF

The Framework is based on the following guiding principles:

1. Opportunity

In order to achieve the reforms' vision and mission, the curriculum provides learners with a variety of opportunities to enable them to identify their needs, talents and potential.

2. Excellence

Every learner will be nurtured to excel in their areas of greatest interest and ability. The Framework values excellence and competitiveness rather than raw competition for examination grades.

3. Diversity and Inclusion

Basic Education Curriculum Framework Nurturing Every Learner's Potential

Engaged, Empowered & Ethical Citizens

There are two dimensions in the guiding principle of diversity and inclusion. First, the Framework will guide learners to appreciate Kenya's diversity in terms of race, ethnicity, gender, language, culture, and religion. The second dimension relates to the fact that learners are different in terms of their learning needs and abilities and these differences need to be respected and valued within an inclusive learning environment.

4. Differentiated Curriculum and Learning

Differentiated curriculum and learning builds on the principle of diversity and inclusion. It ensures that the curriculum content and instructional approaches are appropriate for each learner. It provides space for teachers to adapt the curriculum to suit the learner. It does not demand that every learner learn the same content in the same way, in the same number of hours and at the same time.

5. Parental Empowerment and Engagement

Parents play a very important role in determining the success of a child's education. They have a shared responsibility with schools to provide an enabling environment that is conducive to learning and which motivates the child to achieve their full potential. The Framework will provide opportunities for schools to empower parents to contribute to the learning outcomes for their children and to be engaged at all tiers and levels of basic education.

6. Community Service Learning

Involving students in community service is a form of experiential education that enables students to apply their knowledge and skills in a different setting. Teachers then support students to analyse what they have learned by taking part in this activity and how it might be applied to their academic and personal development. Community service learning entails a balanced emphasis on both students' learning and addressing real needs in the community.

Core Competencies For Basic Education

Sessional Paper No. 2 of 2015 on „Reforming Education and Training in Kenya“ recommends a reformed curriculum that adopts a competency based approach

A competency based approach enables meaningful connections within and between subject areas through a focus on competencies.

Subjects and Subjects will continue to be taught and will be the vehicles through which the core competencies are developed over time.

Based on the Needs Assessment Study carried out by KICD, and the vision and mission of the BECF, the seven core competencies to be achieved by every learner in basic education are:

1. Communication and Collaboration
2. Self-efficacy
3. Critical Thinking and Problem Solving
4. Creativity and Imagination
5. Citizenship
6. Digital Literacy
7. Learning to Learn

The Framework seeks to develop these competencies so that all Kenyans can thrive in the 21st century.

Communication and Collaboration

Communication is the act of transferring information from one place to another, whether vocally, visually, or non-verbally. The discipline of communication focuses on how people use messages to generate meanings within and across various contexts, cultures, channels, and media. The discipline promotes the effective and ethical practice of human communication.

Collaboration is the process of two or more people or organizations working together to realize shared goals. Collaboration may require leadership, although this can be social within decentralized or egalitarian groups or teams that work collaboratively in relation to gaining greater resources, recognition and motivation. Strategies for effective communication enhance the attainment of greater collaboration among groups that ultimately increase the success of teams as they engage in collaborative problem solving.

Self-efficacy

Self-efficacy is a person's belief about his or her capabilities to perform tasks or assignments that can change and transform his or her life. It determines how the person feels, thinks, behaves and motivates themselves. Self-efficacy has the potential to determine four major processes namely cognitive, motivational, affective and selection processes.

Critical Thinking and Problem Solving

An important outcome of quality education is teaching learners how to think critically. The British Council (2015) identifies three types of thinking: reasoning, making judgements, and problem solving. It is possible for learners to reason in an uncritical way. When learners are empowered with critical thinking, they avoid being subjective, and use logic and evidence to arrive at conclusions.

Creativity and Imagination

Creativity and imagination refers to the ability to form new images and sensations in the mind, and to turn them into reality.

It is the ability to imagine things that are not real, to form pictures in the mind of things that one has not seen or experienced, and turn those pictures into real things.

It also refers to the act or power of forming mental images of things that are not present to the senses, or that are never wholly perceived in reality, and creating physical representations of those images.

Digital Literacy

Digital literacy can be described as having the knowledge, skills and behaviours which are necessary to effectively and safely use a wide range of digital content and devices. Such devices include mobile phones, smart phones, tablets, laptops and desktops among others. All these fall within the category of network enabled devices

THEORITICAL APPROACHES

A theory is an abstract general explanation of observations or a subject under study that can be relied upon to provide guidance for practice.

It attempts to predict behaviour or reach a reasonable and general set of explanations for an underlying issue. The Basic Education Curriculum Framework is underpinned by several theories.

Instructional Design Theory

This theory offers explicit guidance towards a new curriculum that explains how to help students learn and develop in the wake of emerging globalization

It offers guidance for fostering cognitive learning as "Clear information, in terms of goals, knowledge needed and performance expected; Thoughtful practice, in terms of opportunities for learners to engage in learning actively and reflectively; Informative feedback, in terms of clear and thorough counsel to learners, and; Strong intrinsic or extrinsic motivation."

Vygotsky's Social-Cultural Development Theory

Vygotsky's social-cultural theory emphasized that teaching and learning are highly social activities and that interactions with teachers, peers and instructional materials influence the cognitive and affective developments of learners (Kim and Baylor, 2006).

The theory argues that learning takes place when learners interact with each other, or have other social contact.

Learners negotiate meanings with people in the environment, and they achieve goals through interacting, both explicitly and implicitly, with the teacher, peers, materials, and atmosphere embedded in the context.

This theory underpins the basic education curriculum framework in terms of conceptualising and designing the necessary paradigm shifts that will facilitate creating rich learning environments which will stimulate all learners and help them to fulfil their potential.

Gardner's Multiple Intelligence Theory

Gardner's theory grew out of constructivism. It states that learners possess different kinds of minds and therefore learn, remember, perform, and understand in different ways.

He argues that we are all able to know the world through language, logical-mathematical analysis, spatial representation, musical thinking, use of the body to solve problems or to make things, an understanding of other individuals, and an understanding of ourselves.

Piaget's Cognitive Development Theory

Piaget's theory deals with how humans gradually come to acquire, construct, and use knowledge.

He looked at the impact a person's childhood had on their development, and the ways in which maturation affect a child's increasing capacity to understand their world.

Piaget asserted that children cannot undertake certain tasks until they are psychologically mature enough to do so.

According to this theory, there is progressive reorganization of mental processes resulting from biological maturation and environmental experience. It is important to note that children's thinking does not develop entirely smoothly, instead, there are certain points at which it "takes off" and moves into completely new areas and capabilities.

These points are in four stages of cognitive development:

Sensory motor (0-2 years),

Pre-operational (2-7 years),

Concrete operations (7-11 years) and

Formal operations (11 years and above).

This has been taken to mean that before these ages children are not capable (no matter how bright) of understanding things in certain ways, and has been used as the basis for scheduling the school curriculum.

Parents are encouraged to provide a rich, supportive environment for their child's natural propensity to grow and learn.

competency based curriculum has the following strengths:

1. Learner-focused

The focus should be less on the school and the system and more on the education and the learner. The curriculum should be responsive and relevant to learners and enable flexibility for teachers – who are designers of learning opportunities – to enable them to meet the diverse needs of the students.

2. Focus on Competencies

More focus should be directed to competencies and less on content. The goal should be the appropriate application of knowledge, and not necessarily just its acquisition. This kind of curriculum enables the creation of meaningful connections within and among subject areas by focusing on competencies.

Subjects/subjects/disciplines will continue to be taught and will be the vehicle through which literacy, numeracy and other competencies are developed over time.

3. Opportunities for Local Decision Making and Greater Depth of Study

Prescriptive curricula with limited flexibility should be avoided, as should programs that have fewer learning outcomes. The goal is to enable greater flexibility at the local level. Teachers will have autonomy in implementing the curriculum while teaching.

4. Balance Between Formative and Summative Assessment

Too much focus on summative assessment should be avoided. A range of assessment that focuses on the development of student learning outcomes, cross-curricular competencies, and literacy and numeracy should be adopted. *Basic Education Curriculum Framework Nurturing Every Learner's Potential Engaged, Empowered & Ethical Citizens*

5. Digitally Based

The design of the curriculum within a collaborative digital application enables it to be improved continuously and supports learning with flexible timing and pacing through a range of learning environments.

6. Collaborative and Co-development Models

Collaboration by all stakeholders in the curriculum design should be the norm. Co-creation of the curriculum with partners and stakeholders taps into local expertise to enhance its design and development. Content delivery will be flexible and exploratory.

7. Synchronous Development

Sequential development needs to be replaced with synchronous development. An integrated approach to develop programmes of study, assessments, and learning and teaching resources supports a common approach that encourages interdisciplinary learning.

Subjects

These are the subjects taught at various levels of the education. For example, English, Mathematics, Science and Technology. Subjects are the means through which the key competencies, values and knowledge will be taught across all levels of learning. However, they are not static and may change to address the ever changing needs of society.

Organization Structure of Basic Education

Basic Education will be organized into three (3) levels:

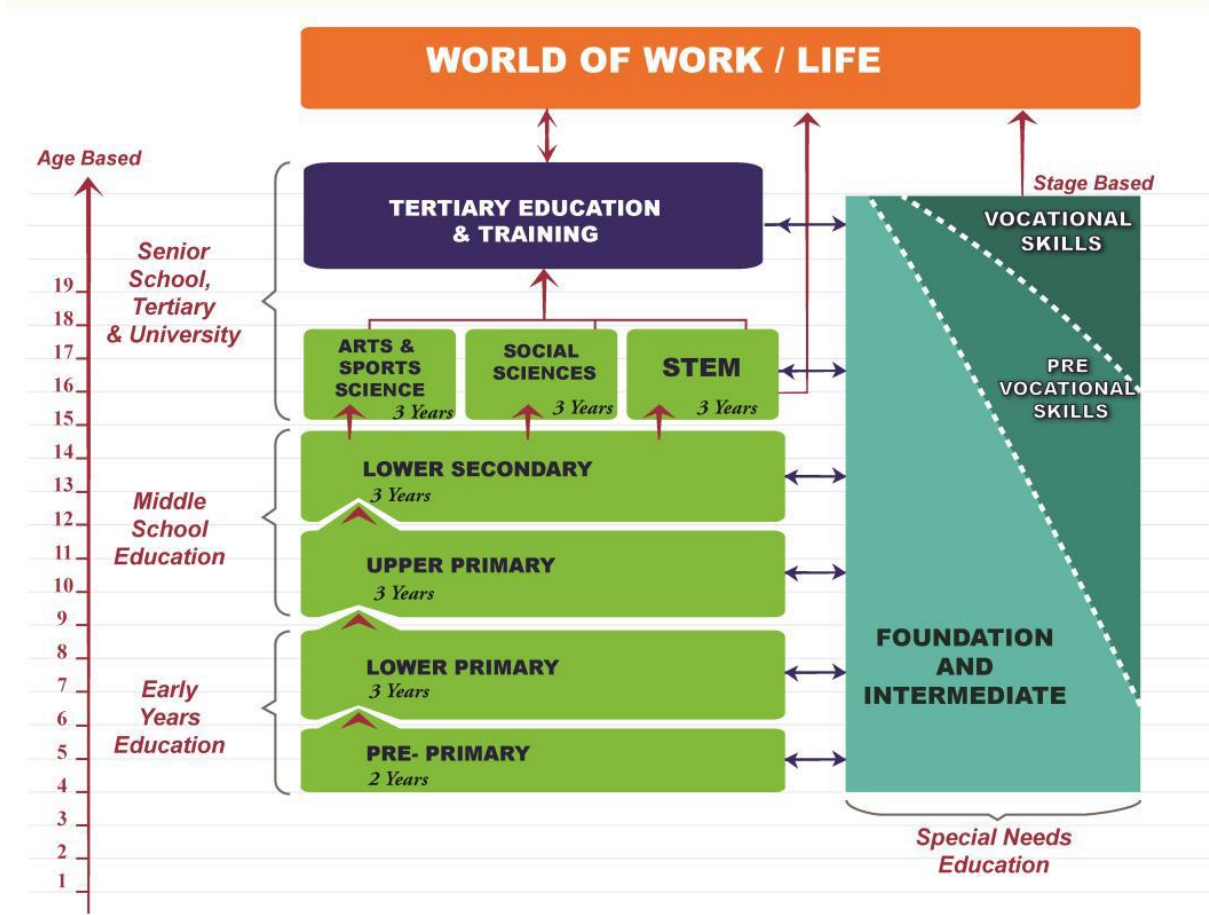
Early Years Education,

Middle School Education

Senior School.

Figure 2 presents a summary of the structural model. *Basic Education Curriculum Framework Nurturing Every Learner's Potential*

ORGANIZATION OF BASIC EDUCATION



9.4 TEACHER EDUCATION CURRICULUM FRAMEWORK.

Learning Outcomes

- o Analyse the teacher education curriculum framework in Kenya
- o Apply the teacher Education Curriculum Framework to the learning process in Kenya
- o Appreciate the need for teacher education curriculum framework to the current curriculum

Acronyms Abbreviations

AU	African Union
BECF	Basic Education Curriculum Framework
CBC	Competency Based Curriculum
CSL	Community Service Learning
CTS	Career and Technology Studies
ECDE	Early Childhood Development and Education
ESD	Education for Sustainable Development
ET	Engineering Technology
GCED	Global Citizenship Education
ICT	Information and Communication Technology
IEP	Individualized Educational Plan
KIE	Kenya Institute of Education
KICD	Kenya Institute of Curriculum Development
KNEC	Kenya National Examinations Council
KCBC	
LSE	Kenya Competence Based Curriculum
Life Skills Education	
LSV	Life Skills and Values
PCI	Pertinent and Contemporary Issues
SNE	Special Needs Education
SS	Senior School
STEM	Science, Technology, Engineering and Mathematics
TSC	Teachers' Service Commission
VTC	Vocational Training Centres

Continue with copying the hand written notes forwarded in whatsapp from page 1 title teacher education curriculum framework

9.6 TOPIC 14; CURRICULUM DESIGN

Learning Outcomes

- ☐ Discuss the attributes of curriculum design in curriculum development
- ☐ Apply the principles of curriculum design in curriculum development
- ☐ Discuss the components of curriculum design in Kenya
- ☐ Examine curriculum design from other countries
- ☐ Develop scope and sequence chart in curriculum design
- ☐ Discuss on importance of curriculum design.

Curriculum design

Is a term used to describe the purposeful, deliberate, and systematic organization of curriculum (instructional blocks) within a class or course.

It is a way for teachers to plan instruction. When teachers design curriculum, they identify what will be done, who will do it, and what schedule to follow.

N/B Curriculum design is not a one-step process; continuous improvement is a necessity..

Curriculum design then includes an arrangement of objectives, subject matter chosen, specific action plans for teaching, all forms-of instructional materials to be used, time schedules, activity descriptions and so forth. If one goes further and includes what pupils learn as part of curriculum, then components of evaluation also have to be added.

IMPORTANCE OF CURRICULUM DESIGN

1. It both creates and reflects culture and identity.

Curriculum reflects the national culture in which a school operates – different countries have different expectations of their students, even if teaching practices are similar. But it can also reflect and define the culture at the school level as well, from the specific needs of the neighbourhoods they serve to the topics where you want to differentiate yourself.

2. It keeps up with a changing world.

A good curriculum is never a one-and-done initiative. Revisiting it regularly allows you to not only review how things are going, but also make room for new topics that are relevant today. In fact, the World Economic Forum touts curriculum as a key player in helping educators keep up with the speed of changing trends, technology and skills students will need in the future.

3. It makes learning (and teaching) consistent.

Internal consistency happens when students inside your school can expect to walk away with the same set of skills, whether their teacher is a veteran in the profession or learning the ropes. Consistency should also stretch across a district, state or province, or country as well – a student in fifth grade should expect to graduate with the same foundational knowledge and skills, no matter where they learn.

4. It opens the doors for collaboration.

Having regular discussions about curriculum creates opportunities to get all stakeholders involved: teachers, administration, parents and communities. Getting input from everyone will help identify where you're strong and what gaps you need to address.

5. It saves schools money.

Textbooks aren't cheap, but they're common place because they're convenient – they provide a pre-built progression of knowledge that's hard to ignore for busy teachers. With a strong curriculum in place, schools can break away from an over-reliance on textbooks and take a more active and dynamic role in choosing better (and cheaper) texts.

6. It helps teachers align.

A good curriculum also connects teachers from across grade levels and subject areas to look at the big picture of student learning. Teachers can work together to plan a progression of topics that build off of ones that came before and connect across disciplines. The result? You reinforce knowledge over time and make sure that students are prepared for what's coming next.

7. It provides measurable targets.

Data matters. That's why a good curriculum sets measurable outcomes and tracks progress throughout the year. Teachers get a better view of what's happening in the classrooms, students know where they stand and parents

TYPES OF CURRICULUM DESIGN

There are three basic types of curriculum design:

- Subject-centered design
- Learner-centered design
- Problem-centered design

Subject-Centered Curriculum Design

Subject-centered curriculum design revolves around a particular subject matter or discipline. For example, a subject-centered curriculum may focus on math or biology. This type of curriculum design tends to focus on the subject rather than the individual.

Subject-centered curriculum design describes what needs to be studied and how it should be studied.

Learner-Centered Curriculum Design

In contrast, learner-centered curriculum design takes each individual's needs, interests, and goals into consideration. In other words, it acknowledges that students are not uniform and adjust to those student needs.

Learner-centered curriculum design is meant to empower learners and allow them to shape their education through choices.

Instructional plans in a learner-centered curriculum are differentiated, giving students the opportunity to choose assignments, learning experiences or activities. This can motivate students and help them stay engaged in the material that they are learning.

Problem-Centered Curriculum Design

Like learner-centered curriculum design, problem-centered curriculum design is also a form of student-centered design. Problem-centered curricula focus on teaching students how to look at a problem and come up with a solution to the problem.

Students are thus exposed to real-life issues, which helps them develop skills that are transferable to the real world. Problem-centered curriculum design increases the relevance of the curriculum and allows students to be creative and innovate as they are learning. The drawback to this form of curriculum design is that it does not always take learning styles into consideration.

Curriculum Design Tips

The following curriculum design tips can help educators manage each stage of the curriculum design process.

- Identify the needs of stakeholders (i.e., students) early on in the curriculum design process. This can be done through needs analysis, which involves the collection and analysis of data related to the learner. This data might include what learners already know and what they need to know to be proficient in a particular area or skill. It may also include information about learner perceptions, strengths, and weaknesses.
- Create a clear list of learning goals and outcomes. This will help you to focus on the intended purpose of the curriculum and allow you to plan instruction that can achieve the desired results. Learning goals are the things teachers want students to achieve in the course. Learning outcomes are the measurable knowledge, skills, and attitudes that students should have achieved in the course.
- Identify constraints that will impact your curriculum design. For example, time is a common constraint that must be considered. There are only so many hours, days, weeks or months in the term. If there isn't enough time to deliver all of the instruction that has been planned, it will impact learning outcomes.
- Consider creating a curriculum map (also known as a curriculum matrix) so that you can properly evaluate the sequence and coherence of instruction. Curriculum mapping provides visual diagrams or indexes of a curriculum. Analyzing a visual representation of the curriculum is a good way to quickly and easily identify potential gaps, redundancies or alignment issues in the sequencing of instruction. Curriculum maps can be created on paper or with software programs or online services designed specifically for this purpose.
- Identify the instructional methods that will be used throughout the course and consider how they will work with student learning styles. If the instructional methods are not conducive to the curriculum, the instructional design or the curriculum design will need to be altered accordingly.
- Establish evaluation methods that will be used at the end and during the school year to assess learners, instructors, and the curriculum. Evaluation will help you determine if the curriculum design is working or if it is failing. Examples of things that should be evaluated include the strengths and weaknesses of the curriculum and achievement rates related to learning outcomes. The most effective evaluation is ongoing and summative.

FEATURES/ATTRIBUTES OF A CURRICULUM

- ☐ The teacher - Quality Education requires quality teachers.

Good teachers bring a shining light into the learning environment.

They are ideal companions of the learners

These teachers should be given support with their continuing development in order to keep abreast with the changing demands of a learning society.

- ☐ The Learners - The learners are at the center stage in the educative process.

They are the most important factors in the learning environment.

There is no teaching without them

Teachers should understand and accept the learner's diverse background

Considering the domain of diversity of learners will allow the individual learner to develop his multiple intelligences at his own pace

Their needs should be addressed and be met that's why teachers are to provide learning opportunities and varied experiences.

□ Knowledge, Skills, Values • It should be remembered that what students learn will be obsolete in ten years, and half of what they need to know to succeed in work and in life has not yet been fully developed and will have to be learned as they go along in the future. • Calculators and typewriters are made obsolete by computers and the next generation will see these being replaced. And so, the value of the educational process lies not just in what they learn, but how they learn, and how good they will be in continuing to learn after they leave school.

□ Strategies and Methods • There is no best strategy that could work in a million of different student background and characteristics. • However, for teachers to teach effectively, they must use appropriate methodologies, approaches and strategies. • Teachers should select teaching methods, learning activities and instructional materials or resources appropriate to learners and aligned to objectives of the lesson. • Good teachers utilize information derived from assessment to improve teaching and learning and adopt a culture of excellence.

□ Performance • At the end of the teaching act, it is necessary to find out if they objectives set were accomplished. • In curriculum we call this the learning outcomes. • These learning outcomes indicate the performance of both the teachers and the learners. • Learning outcomes are the product performance of the learners as a result of teaching. • Performance is a feature of a curriculum that should be given emphasis.

Community Partners • Teaching is a collaborative undertaking. While teachers are the focal point in the learning process, they must draw upon the resources of their environment and of their partners to be effective. • Partnership is a means and not an end to be pursued in itself. • An absence of partnership often means a poor definition of education ends.

PRINCIPLES OF CURRICULUM DESIGN

The principles must be taken into account for all children and young people. The principles will help teachers and schools in their practice and as a basis for reviewing, evaluating and improving the learning and teaching. Although all should apply at any one stage, the principles will have different emphases as a child or young person learns and develops.

Learning should be designed on the basis of the following principle:

- Challenge and enjoyment
- Breadth
- Progression
- Depth
- Coherence
- Relevance
- Personalisation and choice.

Challenge and enjoyment Children and young people should find their learning challenging, engaging and motivating. The curriculum should encourage high aspirations and ambitions for all.. At every stage, children and young people should experience an appropriate level of challenge, to enable each individual to achieve his or her potential.

Breadth All children and young people should have the opportunities for a broad range of experiences. Their learning should be planned and organised so that they will learn and develop through a variety of contexts within both the classroom and other aspects of school life.

Progression Children and you people should experience continuous progression in their learning from 3 to 18 years. Each stage should build upon earlier knowledge and achievements. learners should be able to progress at a rate which is meets their individual needs and aptitudes.

Depth There should be opportunities for children and young people to develop their full capacity for different types of thinking and learning, exploring and achieving more advanced levels of understanding.

Coherence Children and young people's learning activities should combine to form a coherent experience. There should be clear links between different aspects of learning. Such links should be discussed with children and young people in order to bring different strands of learning together .

Relevance Children and young people should understand the purpose of their learning and related activities. They should see the value of what they are learning and its relevance to their lives, present and future.

COMPONENTS OF A CURRICULUM DESIGN

Any curriculum consists of several components: objectives, attitudes, time, students and teachers, needs analysis, classroom activities, materials, study skills, language skills, vocabulary, grammar and assessment.

Before setting up a program or course of study, these components should be determined and described in detail. In fact, these elements help to clarify various dimensions of the curriculum and consequently enhance its productivity.

☐ Objectives

Generally, objectives are one of the quint essential aspects of any course or program. Any curriculum usually determines its instructional objectives at the beginning of the course. These objectives should clearly elucidate the language elements or skills which the students might learn during the program.

☐ Attitudes

Certainly, students' attitudes determine whether or not they intend to learn a foreign or second language. Mainly, favourable attitudes towards the language and its speakers augment students' motivation and their learning rate .In fact, motivation for learning a second language is one of the realizations of positive attitudes towards the language.

☐ Time

One of the crucial factors which has tremendous effect on the learners' learning rate is the amount of time spent on teaching-learning activities in the classroom. Certainly, hours of instruction per week and month play an important role in language acquisition. The restricted amount of time could damage and decrease the quality and efficiency of the classroom teaching-learning activities.

☐ Needs Analysis

In order to devise a course and prepare materials and methods based on the students' and institution's objectives, we had better carry out needs analysis. Educational program should be based on an analysis of learner needs."

Needs analysis is the starting point which is usually done before, during and after the course in order to determine the course's outline, materials and resources.

☐ Students and Teachers

Students are certainly the main stakeholders in a course of study. Students have been marginalized and scant attention has been paid to them in designing and developing materials and methods. Generally, each classroom is made up of different types of students with "various personal characteristics and cultural background.

☐ Classroom Activities

The effective classroom activities and exercises can contribute to learning and make it enjoyable.

☐ Materials

Good materials not only can be taught straight forwardly but also can facilitate learning process argues that finding useful materials is difficult and their suitability "cannot be known in advance."Selecting the best and most appropriate materials which are available. Essentially, many teachers do not have time or are not provided with enough time to develop their own materials based on their students' needs and course objectives.

☐ Study Skills

Definitely, study skills could equip the students with some essential and necessary techniques in university. Study skills approach came to the fore because learners needed more than linguistic competence and language descriptions to achieve their goals.

CURRICULUM SCOPE AND SEQUENCE

Scope of the Curriculum

The scope is breadth and depth of a curriculum. Some curriculum are broad, in that they have many different ideas and subjects discussed, while not being very deep because nothing is discussed in detail. An example of this would be any kind of music or art appreciation class.

The scope refers to the areas of development addressed by the curriculum.

The sequence includes plans and materials for learning experiences to support and extend children's learning at various levels of development.

Education staff can use this resource to select and implement curriculum that includes an organized scope and sequence.

7 Strategies for Developing Your Own Curriculum as a New Teacher

Developing curriculum is a tough assignment for first year teachers. However, understanding what to expect and preparing ahead of time can be of great help. Here's a list of strategies for curriculum building.

Learning to Build Your Curriculum

General principles that you can use as a guideline for preparing your own curriculum. Here are seven principles to get you started.

Focus on the Students

When writing curriculum, it helps to remember that it's not about writing the best lesson plans or developing a perfect set of in-class projects and assignments. Instead, it's about meeting the needs of the students in a way that ensures the material is understood, maintained, and applied in and out of the classroom.

Ask for Help

Seek out seasoned teachers and ask for their input regarding your curriculum. If you don't have a mentor think about asking a fellow teacher if he or she would be willing to walk alongside you during the first year.

Choose a Supportive Programme

Most teachers will tell you that they don't plan their lessons or build curriculum on their own. In fact, oftentimes supportive computer software, online programs, or basic planning maps are used as a guide.

Avoid Prepackaged Curriculum

Packaged curriculum can be a great learning tool especially if you're looking for a hands-on sample to go by. However, it's not suggested that you use the curriculum as your set course of action.

Schedule Planning Time

Creating multiple lesson plans in order to build curriculum takes time. It's important to schedule in planning sessions and blocks of time to work on curriculum. Learning how to manage that time is also important.

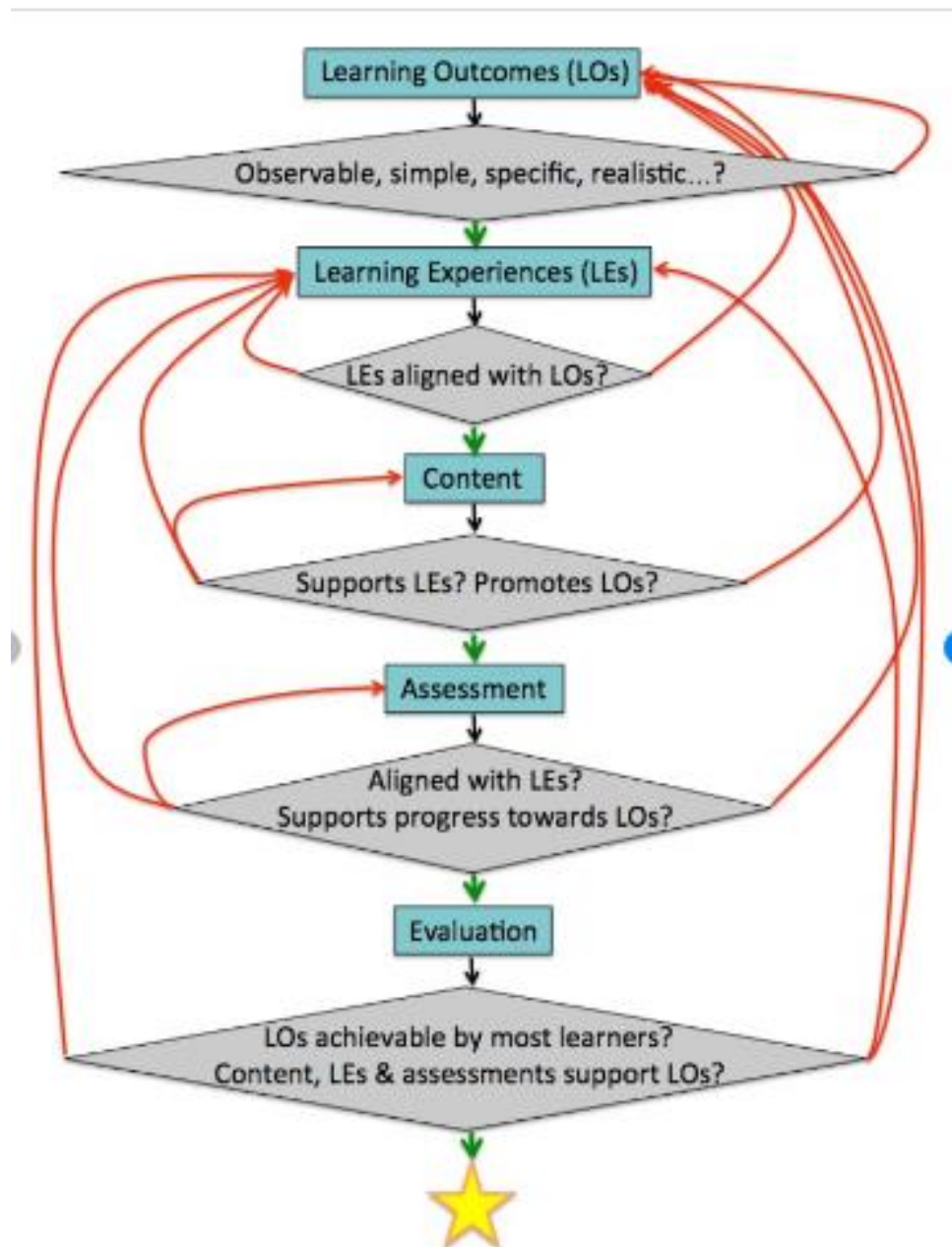
Remember U-Turns are Allowed

As a first year teacher developing curriculum for the first time, it's important to realize that it's not going to be perfect the initial time around. It may not even be where you want it to be the fourth or fifth time around and that's OK. Even when the finished product is ready for launch, there still may be a few bugs to work out.

Plan for Feedback and Assessments

Don't forget to build in assessments and time for feedback when developing your curriculum. You will need to be able to measure how well students are doing. Set aside time to engage students in conversations about the day's lesson and assignments.

DEVELOPING A SEQUENCE CHART IN CURRICULUM DESIGN



N/B The five phases of curriculum design and their dependencies.

For each phase (rectangles), the key questions that should be asked are shown (diamonds).

Where questions haven't been satisfied, that phase or previous phases or the LOs should be revisited (red arrows); once they have been satisfied, this feeds forward into the next phase(s) (green arrows)

When all questions are satisfied, the curriculum or instruction can be characterized, with concrete evidence, as successful (star).

INDIVIDUAL ASSIGNMENT

Compare and contrast curriculum designs from the following countries and reflect on Kenyan curriculum design .i.e

Uganda, Ghana , Asia Vs Kenya

9.7 VALUE BASED EDUCATION

LEARNING OUTCOME

- Examine the roles of values in society
- Examine the concept of VBE in learning
- Determine the principles of VBE in relation to the whole school approach (WSA)
- Demonstrate the application of VBE in real life.
- Outline and interpreted the core values identified in the Basic Education Curriculum Framework (BECF)

Values

Values are defined as standards that guide an individual on how to respond or behave in a given circumstance. Our values influence how we feel, act and make choices in life. Internationally, there has been a rise in the challenges and issues related with cultural integration, as well as ethnic and religious diversity.

Social values form an important part of the culture of the society.

Values account for the stability of social order. They provide the general guidelines for social conduct. Values such as fundamental rights, patriotism, respect for human dignity, rationality, sacrifice, individuality, equality, democracy etc. guide our behaviour in many ways.

Value Education

Means teaching and learning of ideals. The aim is for students to understand moral values and show their good behaviour and attitude towards society.

N/B The teaching of values will facilitate the achievement of the curriculum reforms' vision, particularly with respect to developing ethical citizens.

According to the Constitution of Kenya, 2010 The values stated in the Constitution include responsibility, respect, excellence, care and compassion, understanding and tolerance, honesty and trustworthiness, trust, and being ethical.

Functions of Values:

1. Values provide goals or ends for the members to aim for.
2. Values provide for stabilities and uniformities in group interaction. They hold the society together because they are shared in common. Some sociologists argue that shared values form the basis for social unity. They will therefore, have a sense of belonging to a social group. They will feel a part of the wider society.
3. Values bring legitimacy to the rules that govern specific activities. The rule are accepted as rules and followed mainly because they embody the values that most people accept.
4. Values help to bring about some kind of adjustment between different sets of rules. The people seek the same kinds of ends or goals in different field of their life. Hence, it is possible for them to modify the rules to help the pursuit of this end.

What is Values-based Education?

☐ Value based education (VBE) is an outcome anchored on culture, religion, morals, values and character formation for holistic development.

Values and attitudes are components for cognitive knowledge and are useful in attaining and maintaining peace across cultures.

☐ Values-based Education is an approach to teaching that works with values. It creates a strong learning environment that enhances academic achievement and develops students' social and relationship skills that last throughout their lives.

The positive learning environment is achieved through the positive values modelled by staff throughout the school.

Value Based Education (VBE) Include;

- Love
- Responsibility
- Respect
- Unity
- Peace
- Patriotism
- Social
- Justice
- Integrity

IMPORTANCE OF VALUE BASED EDUCATION

There is an inherent need and importance of value education in the present world:

- It helps in taking the right decisions in difficult situations thus improving decision-making abilities.
- It inculcates students with essential values like kindness, compassion and empathy.
- It awakens curiosity in children developing their values and interests. This further helps in skill development in students.
- It also fosters a sense of brotherhood and patriotism thus helping students become more open-minded and welcoming towards all cultures as well as religions.
- It provides a positive direction to a student's life as they are taught about the right values and ethics.
- It helps students find their true purpose towards serving society and doing their best to become a better version of themselves.
- With age comes a wide range of responsibilities. This can at times develop a sense of meaninglessness and can lead to a rise in mental health disorders, mid-career crisis and growing discontent with one's life. Value education aims to somewhat fill the void in people's lives.
- Helps people to be more convinced and committed to their goals and passions. This leads to the development of awareness which results in thoughtful and fulfilling decisions.

It instils a sense of 'meaning' behind what one is supposed to do and thus aids in personality development

PRINCIPLES OF VALUE BASED EDUCATION

Adoption of a whole school approach where the teaching and learning of positive values is seamless and reflected in the ethos of the school. Integration of four core values ---Respect, Tolerance, Equality and Peace---into the curriculum at all levels of education, including early learning and teacher training.

1. Integration of four core values ---Respect, Tolerance, Equality and Peace---into the curriculum at all levels of education, including early learning and teacher training.
2. Elimination of the culture of "mean score" and "certification" by:
 - o Equitable re-distribution of resources to bridge gaps between the few elite schools and vast majority of public schools.
 - o Comprehensive examinations reform that at a minimum scraps KCPE, replacing it with a system of assessment that moves away from the narrow focus on regurgitating facts to recognition of the learning needs of the whole child, and is age-appropriate, respects child rights and is values-friendly.
3. Enforcement of policies and laws incorporating the four core values and reviewing school rules and regulations to align with the values and principles stipulated in the Kenyan Constitution.
4. Intensification of parental education and community sensitization through partnership with civil society and faith-based organizations on Value-based Education and with an emphasis on the core values.

A whole school approach.

Is cohesive and collaborative action by a school community that is being strategically and continually constructed to improve student learning, behavior and wellbeing?

CORE VALUES IN BASIC EDUCATION CURRICULUM FRAMEWORK (BECF)

Based on the Needs Assessment Study carried out by KICD, and the vision and mission of the BECF, the seven core competencies to be achieved by every learner in basic education are:

1. Communication and Collaboration
2. Self-efficacy
3. Critical Thinking and Problem Solving
4. Creativity and Imagination
5. Citizenship
6. Digital Literacy
7. Learning to Learn

The Framework seeks to develop these competencies so that all Kenyans can thrive in the 21st century.

Communication and Collaboration

Communication is the act of transferring information from one place to another, whether vocally, visually, or non-verbally. The discipline of communication focuses on how people use messages to generate meanings within and across various contexts, cultures, channels, and media. The discipline promotes the effective and ethical practice of human communication.

Collaboration is the process of two or more people or organizations working together to realize shared goals. Collaboration may require leadership, although this can be social within decentralized or teams that work collaboratively in relation to gaining greater resources.

Self-efficacy

Self-efficacy is a person's belief about his or her capabilities to perform tasks or assignments that can change and transform his or her life. It determines how the person feels, thinks, behaves and motivates themselves. Self-efficacy has the potential to determine four major processes namely cognitive, motivational, affective and selection processes.

Critical Thinking and Problem Solving

An important outcome of quality education is teaching learners how to think critically. The British Council (2015) identifies three types of thinking: reasoning, making judgements, and problem solving. It is possible for learners to reason in an uncritical way

Creativity and Imagination

Creativity and imagination refers to the ability to form new images and sensations in the mind, and to turn them into reality.

It is the ability to imagine things that are not real, to form pictures in the mind of things that one has not seen or experienced, and turn those pictures into real things.

Learning to Learn

Learning is a continuous process that begins at birth and continues until death; it is the process through which we use our experience to deal with new situations and to develop relationships.

Digital Literacy can be described as having the knowledge, skills and behaviours which are necessary to effectively and safely use a wide range of digital content and devices. Such devices include mobile phones, smart phones, tablets, laptops and desktops among others. .

Citizenship

Is the state of being vested with the rights, privileges, and duties of a citizen. It creates a sense of belonging and attachment to one's nation. A sense of citizenship helps to equip young people to deal with situations of conflict and controversy knowledgeably and tolerantly.

WHAT IS THE IMPORTANCE OF MORAL VALUES IN THE LEARNING PROCESS?

WHAT ARE MORAL VALUES?

Moral values are the key components of a person's character. They are personality traits guiding people to make decisions and judgements according to their own sense of what is right and wrong, based on collective and individual experiences.

Moral values are essential in whatever stage of life we are in, the values that most adults have are the ones that were instilled in us during our childhood. They form the fundamentals in any student's life.

It is said that students are the future nation, and this future of our country depends greatly upon the values imparted to them during their student life. Moral values pave the path for all their decisions in life, as without these values, children do not have any guidance and their life may seem directionless. In order to be accepted and respected by society, parents and caregivers should make sure of imbibing these strong moral values in children as a lifestyle itself.

N/B We believe that moral values such as kindness, courage, humility, honesty, truthfulness, integrity, respect, hard-work, tolerance, compassion, empathy, and inclusivity are extremely important to be instilled from an early age, to build a child's character, as it forms the very core of their being and becomes the foundation of their moral beliefs throughout their lives..

Therefore, we will look at the importance of moral values, which are the basic values that reflect integrity and humanity in a student's life, in this blog.

IMPORTANCE OF MORAL VALUES IN A LEARNING PROCESS

A student's life is full of challenges, changes, growth, and evolution. A growing child imbibes a lot of lessons throughout their student years, most of which stick with them for the

rest of their lives. Students with high ethical values become accountable, responsible, and honest people, while those without moral values to abide by becoming a threat to themselves and society.

Importance to teaching moral values to students from a young age

1. It Helps In Strong Character Building:

Being aware of values from an early age helps in developing the child's character and forming the very core of their being. A strong character development happens because of the virtues and values that children learn during their childhood, which remains as a sound foundation for their moral beliefs in the future.

2. It Helps In Distinguishing Right From Wrong:

Moral values serve as a guide in differentiating between right and wrong, right from the beginning. While we see almost every day that students at times can contribute to undesirable behaviour like bullying or cheating in exams etc

3. Boost Self-Confidence And Positivity:

When children do something good, they tend to feel good about themselves, which further motivates them, while also giving them an ample amount of self-confidence.

This self-affirmation of being able to give back or help others, in turn, helps in boosting their self-confidence along with helping them develop positive feelings and positive relationships in their lives.

4. Shapes Attitudes, Beliefs And Determines Their Adult Behaviour:

Today's students are the ones that will become tomorrow's doctors, business people, engineers, scientists, politicians, police, etc., and hold the most important positions that will either uplift or let down societies, communities, countries, and the world at large.

Moral values should be properly implemented right from a young age because the virtues you teach your children today will determine how they behave as adults.

5. Negative Peer Influence:

This peer pressure plays a significant role in most of their decisions, from something as simple as the choice of clothing to the way of life.

Good moral values act as a moral compass that helps children stay off the negative influences from their peers, social media, or, in general, as they grow into their teenage years or even adulthood. According to some health sites, children that feel good about themselves will resist negative peer pressure better.

6. Serves As A Support In Tough Situations:

Moral values give children perspective, make them strong, and transpire firm beliefs in them, which makes them work hard and have courage when they are facing challenges in life..

7. Helps Them In Their Relationships With Others:

While children are taught responsibility from a young age, they figure out their role and purpose in society. It is with the understanding that the world does not revolve around them and that they have a duty to follow towards their fellow human beings that will teach them to put the emotions and needs of others before themselves and see others' perspectives.

Teaching values such as sharing, compassion, cooperation, acceptance, equality, generosity, and justice are extremely important for children to make everyday decisions in an unprejudiced and empathetic manner. From something as simple as showing courage to complain about a bully at school to showing perseverance in achieving goals at work will all help the entire society.

N/B Moral values are the building blocks of personality that help in building the better character of people.

It is absolutely important to teach moral values in students because it is then that they take their first steps towards life, and it matters that they do it right.

These moral values shape their attitudes, beliefs, and ideas and help them develop into undeterred and morally strong individuals.

9.8 PARENTAL EMPOWERMENT AND ENGAGEMENT(PE&E)

Learning Outcome

- ☐ Distinguish parental empowerment from parental engagement
- ☐ Discuss the role of PE and E in learning.
- ☐ Examine the pillars of PE and E
- ☐ Explore and analyse the strategies of PE and E.
- ☐ Evaluate the role of stakeholders in PE and E.
- ☐ Examine the legal and policy framework for PE & E
- ☐ Discuss how parents can be empowered and engaged to contribute to the learning
- ☐ Appreciate the role of parental empowerment in child's learning and value addition

DEFINITION OF TERMS

Community: A group of individuals who share norms, beliefs, values and aspirations.

Education: The act or process of acquiring desirable knowledge, skills, attitudes and values.

Empowerment: Building the capacity of parents to actively nurture learner's potential by enhancing their knowledge, skills, attitudes and practices.

Engagement: Provision of activities and opportunities that foster positive parent-learner connections, thereby enhancing learner's competencies.

Parent: A person who brings up and cares for another, for example, biological parents, guardians or care-givers who oversee the growth and development of a child.

Parental Empowerment

Is a process or activity that is undertaken to enhance the capacity of parents so that they can actively nurture the potential of their children in and out of school. However, optimal learning can best be achieved through a multi-dimensional approach involving all members of school community i.e learners, parents, teachers, support staff, Board of Management (BoM) and the Parent Association (PA).

Parental Engagement

Is the active participation of parents in their children's learning and holistic development in collaboration with teachers and other stakeholders.

Learning experiences at home should take place to reinforce what is learnt in school.

These connections can be maintained through various activities and provision of opportunities to parents to render services.

Some strategies have been proposed that parents will engage in towards the education of their children.

ROLES OF PARENTAL EMPOWERMENT AND ENGAGEMENT IN LEARNING

Empowered and engaged parents have a positive impact on the overall development of the learner in areas such as

- ☐ Acquisition and nurturing of moral and religious values
- ☐ Improved health and nutrition
- ☐ Increased enrolment
- ☐ Identification of appropriate career choice.
- ☐ Understands the child's school calendar
- ☐ Visits school on certain occasions
- ☐ Is concerned with the child's welfare
- ☐ Does follow-ups in the child's school
- ☐ Provides for the child's needs
- ☐ Ensures safety and security of the child
- ☐ Is ethical and a good role model
- ☐ Provides career guidance to the child
- ☐ Disciplines the child
- ☐ Spends quality time with the child
- ☐ Supports education activities of the learning institution of his/her child
- ☐ Respects the child
- ☐ Identifies and nurtures the child's talents

PILLARS OF PARENTAL EMPOWERMENT AND ENGAGEMENT

Pillars of PE&E.

The Guidelines on PE&E are premised on two pillars namely: Empowerment and Engagement.

☐ Parental Empowerment

Empowerment refers to the enhancement of knowledge, skills, attitudes and practices of an individual or a group in pursuit of a mutual goal.

It is the process of becoming stronger and more confident in performing one's role.

Therefore, parental empowerment is the process or activity that is undertaken to enhance the capacity of parents so that they can actively nurture the potential of their children in and out of school.

However, optimal learning can best be achieved through a multi-dimensional approach involving all members of a school community.

These members include learners, parents, teachers, support staff, Board of Management (Bo M) and the Parent Association (PA).

This approach will create parental awareness through advocacy on their responsibilities in the learning of their children.

☐ Parental Engagement

Parental engagement is the active participation of parents in their children's learning and holistic development in collaboration with teachers and other stakeholders.

Learning experiences at home should take place to reinforce what is learnt in school.

These connections can be maintained through various activities and provision of opportunities to parents to render services.

N/B Parents, teachers, BoM, school management and the community need to be empowered in order to be engaged. Capacity building of all stakeholders is critical to ensure active parental involvement in school activities.

This will help nurture the potential of learners in and out of school.

STRATEGIES OF PARENTAL EMPOWERMENT AND ENGAGEMENT

1. Strategies for Parental Empowerment

Empowerment will be implemented through the following strategies:

- a) Training
- b) Advocacy
- c) Resourcing

a) Training

This is a structured process geared towards the acquisition of skills, concepts, change of attitude and advancement of knowledge.

Training boosts the performance of an individual or a group of people with a common objective or goal. For this reason, parents, teachers and other members of the school community will be trained on their roles in nurturing learners' potential.

Immediate Outcomes

Training as a strategy will seek to achieve the following outcomes.

1. Informed and strengthened engagement amongst members of the school community especially parents
2. Enhanced capacity of the Parents Association (PA) to effectively implement Guidelines on PE&E.
3. Improved knowledge, attitudes and skills on parents' roles and responsibilities in the education of their children.
4. Improved collaboration between teachers and parents in achieving learning outcomes.

Activities to be undertaken to realize the immediate training outcomes

1. Orient the BoM and school management on the importance of PE&E and their role in the implementation of the Guidelines.
2. Train PAs on PE&E facilitation skills at school and county level.
3. Train parents on their roles and responsibility in boosting the achievement of desired learning outcomes.
4. Sensitize teachers on the need for teacher-parent partnership in enhancing the attainment of learning outcomes.
5. Sensitize the support staff on the critical role of parents in supporting the achievement of learning outcomes.

b) Advocacy

Various activities will be undertaken to support parents in promoting a safe and supportive environment for learners in school.

Immediate Outcomes

Advocacy as an empowerment strategy is aimed at achieving the following outcomes:

1. Increased awareness on the importance of parental participation
2. Increased enrolment, retention and transition of learners in schools
3. Increased resource base for learning
4. Improved family support systems that are pro-education
5. Informed communication amongst stakeholders

Activities to be undertaken to realize the immediate outcomes

The following are suggested activities that can be undertaken by the school and the community to support the attainment of the stated outcomes on advocacy. These are:

1. Establish an advocacy committee to address issues that affect education
2. PA to lobby for resources from relevant stakeholders to support school improvement plans
3. PA to lobby relevant government agencies for pro-family life legislation and policy formulation
4. Organize PE&E conferences at different levels where key stakeholders can be empowered
5. Establish a communication office or center for stakeholder engagement and feedback

c) Resourcing

This refers to the processes and actions that will facilitate availability of facilities, materials, finances, human resources and other assets necessary for effective learning.

Immediate Outcomes

1. Increased parents' participation in resource mobilization for school activities
2. Enhanced resource-base at school level

3. Enhanced accountability and efficiency in school resource utilization

Activities to be undertaken to realize the Immediate resourcing Outcomes

1. Mobilize communities to pool resources for common use in learning institutions
2. Undertake a resource gap analysis and formulate an implementable action plan
3. Create appropriate working relationships through linkages, partnerships and referrals where necessary
4. Monitor resource utilization and management within the school community for accountability purposes.

2. Strategies of Parental Engagement

Parental engagement is the active participation of parents in their children's learning and holistic development in collaboration with teachers and other stakeholders.

Learning experiences at home should take place to reinforce what is learnt in school. These connections can be maintained through various activities and provision of opportunities to parents to render services.

N/B Parents, teachers, BoMs, school management and the community need to be empowered in order to be engaged. Capacity building of all stakeholders is critical to ensure active parental involvement in school activities.

This will help nurture the potential of learners in and out of school.

Parental engagement therefore will be enhanced through the following strategies:

1. Participatory decision making
2. Communication and collaboration
3. Learning and development process
4. Resourcing, volunteering and linkages.

1. Participatory Decision Making

Parents should be involved in the formulation of decisions affecting the learner's overall wellbeing within and outside the school.

Immediate outcome

Enhanced parental participation and ownership of decisions affecting children's education.

Activities to be undertaken to realize the immediate outcomes

1. Participate in BoM and PA forums, attending meetings, offering their candidature, voting and sharing ideas on how to develop the school.
2. Take part in discussions concerning their children's observed behaviour. This is aimed at helping parents participate in the identification of their children's natural abilities, academic capabilities, career guidance and choices.
3. Monitor children's growth and development as well as the identification of learning difficulties/disabilities where they may occur.
4. Proactively participate in decisions concerning the general wellbeing of their children in and out of school.

2. Communication and Collaboration

Parents are key stakeholders in the school community. Timely and effective communication enables parents to collaborate with the school towards the improvement of their children's well-being.

Parents should be regularly updated and provided with opportunities to respond on matters concerning learning and general issues on education.

Immediate Outcome

Enhanced mutual understanding and timely action leading to better performance and the learner's general wellbeing.

Immediate Outcome Activities

Communication and collaboration between the school and parents can be enhanced through the following activities:

1. Teacher – parent collaboration to develop action plan for each learner.
2. Prompt and coordinated action between the school and the parents on issues such as discipline, emerging issues such as illnesses or bereavement.
3. Peaceful and prompt resolution of conflicts.
4. Timely sharing of the MoE and school calendars of events and programme of activities with parents to encourage parental involvement.
5. Parents engaging with their children on school assignments such as homework and projects.

3. Development and Learning Process

Quality learning should facilitate holistic growth and development. As learners grow, they negotiate various developmental milestones in the following six dimensions: physical, social, intellectual, emotional, moral and spiritual. Parents should be actively involved in monitoring and providing support to their children as they negotiate the developmental milestones.

For instance, adolescents require the assurance of parental love and support throughout their schooling.

Parents should create enabling environments to nurture values and develop character in their children.

Immediate Outcome

Enhanced parental engagement in partnership with the teacher in the child's developmental process and learning to realize their full potential.

Activities to be undertaken to realize the immediate outcome

1. Introduce and nurture values in their child to facilitate moral and social development.
2. Involve the child in play and age-appropriate chores for their physical, intellectual, spiritual, social and emotional development.
3. Attend and actively participate in school activities and meetings. These activities may include academic clinics, music and drama festivals, symposiums, games and sports activities.
4. Take children to places of worship and encourage them to share what they learn with their teachers. Parents should also provide continuous moral and spiritual guidance to their children.
5. Provide adequate resources and an enabling environment for children to grow, learn and acquire age-appropriate life and selfskills.
6. Involve the child in safe and age-appropriate family and community activities that enhance learning.

4. Resourcing, Volunteering and Linkages

Schools require various kinds of resources to function effectively and efficiently. These resources include human, physical and financial.

Parents are an important resource that the school should tap into to improve the wellbeing of the school community.

They can support school activities and development through volunteerism, provision of own resources, relevant linkages and networking for the school.

Immediate Outcome

Cost effective access to goods and services through enhanced linkages with parents and other service and resource providers

Activities to be undertaken to realize immediate outcome

1. Participate in identifying, quantifying and prioritizing the resource needs of the school both in cash and in kind.
2. Encourage and facilitate their children to volunteer their services in planned school activities if the activities are age, gender and culturally appropriate.
3. Provide opportunities for parents to volunteer their services and resources to the schools.
4. Create useful linkages with relevant government and nongovernment agencies to provide resources as may be required.
5. Participate in fundraising to improve the school facilities and services.
6. Provide their children with simple classroom aids or materials as requested by the teacher

LEGAL AND POLICY FRAMEWORK FOR PE&E

PE &E Guidelines are anchored in the existing legal frameworks and policies, which include

a) The Convention on the Rights of the Child (CRC -1990)

The Convention lays the foundational principles from which all rights are derived. Helping children to understand their rights will require parents to guide them to make choices with knowledge of accompanying consequences.

These principles require children to respect the rights of others, especially their parents. Rights of children should be respected by all educational stakeholders.

b) The African Charter on the Rights and Welfare of the Child (1999)

The Children's Charter originated because the member states of the African Union believed that the CRC missed important socio-cultural and economic realities particular to Africa. It emphasizes the need to include African cultural values and experiences when dealing with the rights of the child.

c) The Constitution of Kenya (CoK 2010)

The Constitution of Kenya in Article 26 recognizes the dignity and sanctity of life by acknowledging the right to life. Article 45 then goes on to acknowledge the family as a fundamental unit of society and the necessary basis of social order.

Article 53 is premised on the provisions of the CRC and the African charter.

It states that a child is entitled to:

- A name, nationality, basic education, nutrition, shelter and healthcare.
- Protection from abuse, neglect, harmful cultural practices, all forms of violence as well as exploitative labour.
- Provision from both parents whether married or not.

d) Basic Education Act, 2013

According to the Basic Education Act, 2013, Article 31 (3) indicates that a parent or guardian has a right to participate in the character development of their children. This fosters good habits, acceptable values and behaviour for effective living as an individual member of the society.

e) The Sessional Paper No.1 of 2019 on Reforming Education and Training

The Sessional paper states that a child's school life should be enriched to enable him or her experience and cope better with life.

f) Basic Education Curriculum Framework (BECF -2016)

The BECF captures the aspirations of most Kenyans in relation to the provision of Basic and Teacher Education in the country.

The BECF has identified PE&E as one of the critical components of the competency based curriculum.

The Guidelines on PE&E have been developed to actualize this aspiration.

The Guidelines will support the realization of learning outcomes by enhancing the capacity of curriculum implementers. Indeed, teachers require the right skills to enable them create and strengthen partnerships with parents.

HOW PARENTS CAN BE EMPOWERED & ENGAGED TO CONTRIBUTE TO LEARNING

Sensitize Parents on:

- The importance and highlights of the PE&E programme
- Good parenting practices that support children's learning at home and at school

Train parents on:

- Train parents on the highlights of the Competency Based Curriculum
- Train parents on their role in supporting achievement of learning outcomes.eg how to get information from a report card, how to help with homework, how to monitor.

Parenting skills:

Role modeling, hygiene and nutrition of children; positive discipline, life skills, drugs and substance use, human sexuality, progressive cultural practices, dealing with emerging trends among others.
responsible parent.

N/B

INDIVIDUAL ASSIGNMENT

- ☐ Evaluate the role of stakeholders in parental empowerment and engagement

ROLE OF STAKEHOLDERS IN PE&E.

A stakeholder in education is anyone who has an interest in the success of a school or school system. They are the parties that are either directly or indirectly affected by the success of an education system. This includes government officials, school board members, administrators, and teachers.

ROLES

9.9 COMMUNITY SERVICE LEARNING (CSL)

Learning Outcome;

- ☐ Examine the concept of CSL in enhancing learning
- ☐ Distinguish between community service and community service learning
- ☐ Discuss the rationale of CSL in the learning
- ☐ Explore the principles of CSL in the learning process
- ☐ Analyse ways of engaging the community in CLS activities
- ☐ Examine CSL at the various levels of learning
- ☐ Appreciate the importance CSL in the learning process

COMMUNITY SERVICE LEARNING

It combines classroom learning and community service, giving students the opportunity to work with community members to solve local problems.

This integration of theory and practice enables learners to experience, reflect, and learn from the projects they undertake.

CSL creates rich learning experiences, and delivers benefits to the community. At a personal level, global research shows it develops learners' personal values, critical thinking, problem-solving, imagination and creativity.

Project Based Learning in CSL

Project Based Learning is a research-backed pedagogical approach where learners solve real-world problems through planning and executing a project in partnership with the community and under the supervision of the teacher.

DISTINGUISH BETWEEN COMMUNITY SERVICE AND CSL

Community service is unpaid work performed by a person or group of people for the benefit and betterment of their community without any form of compensation. Community service can be distinct from volunteering, since it is not always performed on a voluntary basis and may be compulsory. While community service learning combines classroom learning and community service, giving students the opportunity to work with community members to solve local problems. Community service learning is intended to equip learners with skills to enable them use their talents to turn around problems into solutions.

RATIONALE FOR CSL IN THE LEARNING

1.CSL was a timely idea whose conceptualisation had deviated from past practices where education had concentrated on theory and not practical learning. it exposed the learners to more practical learning to translate what is in their minds into hands-on skills.

2. That CSL is a positive reinforcement in the curriculum. It does not only focus on knowledge capture but also gives students relevant skills.

3.It add value to the students as the practical component led to opportunities for development of talents.

4. That CSL had impacted on learners' motivational levels because it encourages them to take personal initiatives and plan to make the environment better.

5.Is a good program for the learners who are not very endowed in academic performance. It offers them a chance to acquire and diversify skills and talents.

6. CSL equips learners with knowledge and skills that will enable them to solve problems in the community.
7. That CSL is interesting and learner-centred and is the only hands-on subject that enables learners to apply what they have learnt in solving problems facing their communities
8. CSL helps learners to love their own school and become more responsible and self-driven
9. That recycling of waste products had been enhanced.

PRINCIPLES OF CSL IN THE LEARNING PROCESS

1: Academic Credit Is for Learning, Not for Service

Academic credit is not awarded for doing service or for the quality of the service, but rather for the student's demonstration of academic and civic learning.

2: Establish Learning Objectives

It is a service-learning maxim that one cannot develop a quality service-learning course without first setting very explicit learning objectives. This principle is foundational to service-learning.

3: Establish Criteria for the Selection of Service Placements

Requiring students to serve in any community-based organization as part of a service-learning course is tantamount to requiring students to read any book as part of a traditional course.

4: Provide Educationally-Sound Learning Strategies To Harvest Community Learning and Realize Course Learning Objectives

Requiring service-learning students to merely record their service activities and hours, as their journal assignment is tantamount to requiring students in engineering to log their activities and hours in the lab.

5: Prepare Students for Learning from the Community

Most students lack experience with both extracting and making meaning from experience and in merging it with other academic and civic course learning strategies. Therefore, even an exemplary reflection journal assignment will yield, without sufficient support, uneven responses.

6: Minimize the Distinction between the Students' Community Learning Role and Classroom Learning Role

Classrooms and communities are very different learning contexts. Each requires students to assume a different learner role. The solution is to shape the learning environments so that students assume similar learning roles in both contexts.

7: Rethink the Faculty Instructional Role

If faculty encourage students' active learning in the classroom, what would be a commitment and consistent change in one's teaching role? Commensurate with the proceeding principle's recommendation for an active student learning posture, this principle advocates that service-learning teachers, too, rethink their role.

8: Maximize the Community Responsibility Orientation of the Course

One of the necessary conditions of a service-learning course is purposeful civic learning. Designing classroom norms and learning strategies that not only enhance academic learning but also encourage civic learning are essential to purposeful academic learning.

COMMUNITY ENGAGEMENT

Community engagement pedagogies, often called “service learning,” are ones that combine learning goals and community service in ways that can enhance both student growth and the common good.

Ways of engaging the community in CSL

Community engagement: 6 tips to engage your community

Community engagement depends on the benefits for the members constituting it. Here are some techniques to start community engagement and more important, to keep it going.

1. Get people to tell stories.

People love to share themselves with others. If you become a conduit for this type of communication between people, you will capture the emotions of your audience like no other. You get bonus points if you provide a large audience for the true performers in your ranks.

2. Use images.

Images are better able to convey emotion. Make sure that the images that you publish have as much emotional impact as you can put into them.

3. Be a responsive community engagement leader.

Make sure that you respond to the people that take the time to respond to you. There is nothing more emotionally satisfying than receiving encouragement and understanding from a leader of a social circle.

4. Make sure that your contributions stay grounded

It is easy to become irrelevant once your network begins to grow a little. You may start to think that you must cater to a wider spectrum by glossing up your image. What ends up happening is that you try too hard and people begin to see through you.

5. Community engagement starts with understanding and adding value.

If you always come from the mindset of making someone else’s day better, you will get great emotional responses from your audience. No matter if your posts are entertaining in nature or serious in scope, if they are relevant and precise, you will always get a great response.

6. Fall back on entertainment and education.

If you cannot find something with the appropriate gravitas for the day, you can always fall back on entertaining and educational content. It is sometimes good to break up the monotony with a light-hearted story that everyone can get into.

COMMUNITY SERVICE LEARNING AT VARIOUS LEVELS OF LEARNING

Developing CSL in Kenya The Kenyan competency-based curriculum (CBC) is being designed to develop skills and apply these in real-life situations.

SENIOR SCHOOL

Conceptualising CSL in Kenya, The new senior school learning area will be organised into six strands.

1. Citizenship: This motivates and educates young people to become thoughtful, responsible, patriotic, accountable, active citizens who participate in public life, continuously identifying community needs and solutions, and engaging to conserve the environment.

2. Life Skills: These span learners’ ability to make decisions, set goals, solve problems, and cope with stress. They equip learners to manage interpersonal relationships with empathy, resist peer pressure, become assertive and develop leadership skills

3. Communication: These abilities equip learners to contribute to teams and community groups through the media of the information age.

4. Entrepreneurship: This strand equips learners to manage small businesses that solve poverty. It spans resources, business ideas, business opportunities, market gaps and social entrepreneurship.

5. Financial Literacy: Financial literacy helps learners raise funds and invest funds in learners’ business projects.

Learners will be able to join or start community groups (such as savings SACCOs), understanding methods of saving, and saving goals.

6. Research: This enables learners to start practicing research skills while still at a young age. Learners will be able to analyse information, identify the root cause of issues, identify effective solutions, report on their findings and make recommendations.

INDIVIDUAL ASSIGNMENT

Research on CSL at Early Years and Middle School Education.

9.10 COMMUNITY SERVICE LEARNING INTERGRATION STRATEGY.

Learning Outcome;

- o Analyse CSL activities in curriculum designs of primary level.
- o Develop a CSL activity that will incorporate all skills from different learning areas.
- o Apply project based learning in carrying out the identified CSL activity
- o Administer assessment tools for the CSL activity carried out.
- o Generate a report on the CSL activity carried out
- o Appreciate project based learning strategy for CSL integration

ASSIGNMENT

This assignment is a project based learning which will be carried out in TWO GROUPS I.E OPTION A and OPTION B

Guidelines on how to carry out the task

- a) Study different curriculum designs to identify a CSL activity that addresses challenges in the community.
- b) Develop a CSL activity will incorporate skills from different learning areas.
- c) Implement a CSL project in the community (school) using project based learning approach.
- d) Carry out assessment for the project done
- e) Write a report on the CSL activity carried out.
- f) Organize an exhibition for the CSL projects to the community.